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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\A150NRC14EB002SYNGAS COMP 2nd Stage Intercooler.pvdb

Date of Analysis : Sep 5, 2026 4:41pm

PV Elite 28, January 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 1095.000 mm. ID
 Vessel Design Length, Tangent to Tangent 13523.67 mm.
 Specified Datum Line Distance 0.00 mm.
 Shell Side Design Temperature 200 °C
 Channel Side Design Temperature 85 °C
 Shell Side Design Pressure 191.000 bars
 Channel Side Design Pressure 147.000 bars
 Shop Shell Side Test Pressure 251.822 bars
 Shop Channel Side Test Pressure 202.134 bars
 Wind Design Code ASCE/SEI 7-22
 Earthquake Design Code ASCE/SEI 7-22

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Flange	SA-350 LF2	1	...	K03011	+IT
Nozzle	SA-350 LF2	1	...	K03011	+IT
Nozzle Flg	SA-105	...	...	K03504	
Nozzle Flg	SA-350 LF2	1	...	K03011	+IT
Tubes	SA-179	...	...	K01200	
Tubesheet	SA-350 LF2	1	...	K03011	+IT
Flg Bolting	SA-193 B7	...	64 < t <= 100	G41400	
Hrz Bolting	SA-36	...	...	K02600	
Saddles	SA-516 70	...	...	K02700	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (bars & mm.):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
CH. COVER-1	147.000	0.00	160.000	3.0000	N/A	No
CH. FLG-1	147.094	0.00	165.000	3.0000	N/A	No
CH. SHELL-1	147.094	0.00	165.000	3.0000	N/A	No
MAIN SHELL	191.094	0.00	199.000	3.0000	N/A	No
CH. SHELL-2	147.094	0.00	165.000	3.0000	N/A	No
CH. FLG-2	147.094	0.00	165.000	3.0000	N/A	No
CH. COVER-2	147.000	0.00	160.000	3.0000	N/A	No

Liquid Level: 955.00 mm. Dens.: 0.001 kg./cm³ Sp. Gr.: 1.000

Element Types and Properties:

Element Type	"To" Elev mm.	Element Length mm.	Nominal Thickness mm.	Finished Thickness mm.	Reqd Thk Internal mm.	Reqd Thk External mm.	Long Eff	Circ Eff
Body Flg	293.2	290.0	...	290.0	277.9	...	1.00	1.00
Body Flg	714.2	421.0	...	310.0	301.5	...	1.00	1.00
Cylinder	2014.2	1300.0	74.0	74.0	65.7	...	1.00	1.00
Cylinder	11308.2	9094.0	90.0	90.0	86.1	...	1.00	1.00
Cylinder	12808.2	1300.0	70.0	74.0	65.7	...	1.00	1.00



Body Flg	13233.7	421.0	...	310.0	301.5	...	1.00	1.00
Body Flg	13523.7	290.0	...	290.0	277.9	...	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
TS OUT NOZ. T2	WRC-297	0.877	Shell	Passed
SH. IN NOZ. S1	WRC-297	0.949	Shell	Passed
SH. OUT NOZ. S	WRC-297	0.949	Shell	Passed
TS. IN NOZ. T1	WRC-297	0.877	Shell	Passed

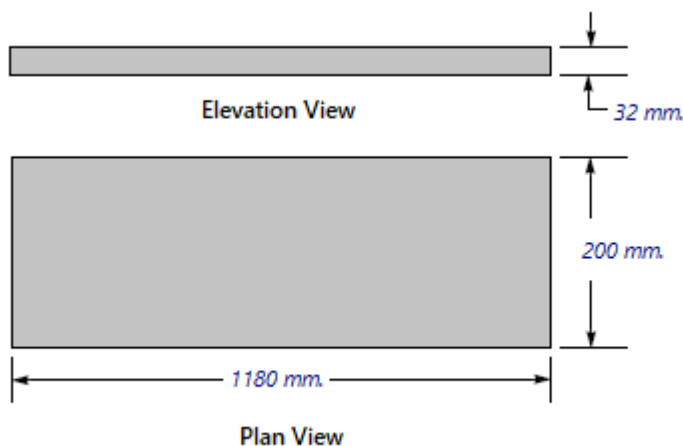
Foundation/Support Summary

Saddle Parameters:

Saddle Width	160.000 mm.
Saddle Bearing Angle	120.000 deg.
Centerline Dimension	1300.000 mm.
Wear Pad Width	520.000 mm.
Wear Pad Thickness	16.000 mm.
Wear Pad Bearing Angle	132.000 deg.
Distance from Saddle to Tangent	1900.000 mm.

Baseplate Length	1180.000 mm.
Baseplate Thickness	32.000 mm.
Baseplate Width	200.000 mm.
Number of Ribs (including outside ribs)	4
Rib Thickness	16.000 mm.
Web Thickness	16.000 mm.
Height of Center Web	306.000 mm.
Number of Bolts in Baseplate	3

Baseplate Sketch



Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	61216.12 Kg
Maximum Transverse Saddle Shear Load	4307.20 Kg
Maximum Longitudinal Saddle Shear Load	8614.40 Kg

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	46252.89 Kg
Maximum Transverse Saddle Shear Load	323.43 Kg
Maximum Longitudinal Saddle Shear Load	108.02 Kg

Weights:

Fabricated - Bare w/o removable internals	70357.5 kg.
Shop Test - Fabricated + water (full)	89897.7 kg.



Shipping	- Fab. + rem. intls.+ shipping app.	70646.3	kg.
Erected	- Fab. + rem. intls.+ insul. (etc)	79331.3	kg.
Empty	- Fab. + intls. + details + wghts.	79331.3	kg.
Operating	- Empty + operating liquid (no ca)	89937.9	kg.

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Nozzle Calculation Summary (bars & mm.):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
TS OUT NOZ. T2	156	...	...	OK	26.47	OK	Passed
DRAIN NOZ. D1	165	...	...	OK	6.42	OK	No Calc[*]
SH. IN NOZ. S1	196	...	...	OK	49.46	OK	Passed
SH. OUT NOZ. S2	196	...	...	OK	49.49	OK	Passed
TS. IN NOZ. T1	155	...	...	OK	26.48	OK	Passed
VENT NOZ. V1	165	...	...	OK	6.42	OK	No Calc[*]

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 196 Nozzle : SH. OUT NOZ. S2 [Shellside]
 Minimum MAWP Nozzles : 155 Nozzle : TS. IN NOZ. T1 [Tubeside]

[*] - This was a small opening and the areas were not computed.

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm.	Layout Angle deg	Dia. Limit mm.
30	TS OUT NOZ. T2	1264.175	90.000	824.800
30	DRAIN NOZ. D1	1314.175	-90.000	202.325
40	SH. IN NOZ. S1	3114.176	90.000	1231.200
40	SH. OUT NOZ. S2	10408.176	-90.000	1231.200
50	TS. IN NOZ. T1	12258.175	-90.000	824.800
50	VENT NOZ. V1	12108.176	90.000	202.325

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
DRAIN NOZ. D1	2.000 in	160	LWN	2.375	8.738	...	...	374.83	1500
VENT NOZ. V1	2.000 in	160	LWN	2.375	8.738	...	...	374.83	1500
TS OUT NOZ. T2	16.000 in	Actual	RTJWn	19.188	40.490	...	...	468.99	1500
TS. IN NOZ. T1	16.000 in	Actual	RTJWn	19.188	40.490	...	...	468.99	1500
SH. IN NOZ. S1	24.000 in	Actual	RTJWn	28.688	59.540	...	...	592.85	1500
SH. OUT NOZ. S2	24.000 in	Actual	RTJWn	28.688	59.540	...	...	592.85	1500

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm.):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
DRAIN NOZ.	SA-350 LF2	60.000	10.000	...	...	...
VENT NOZ. V	SA-350 LF2	8.738	10.000	...	...	...
TS OUT NOZ.	SA-350 LF2	60.000	10.000	...	...	...
TS. IN NOZ.	SA-350 LF2	60.000	10.000	...	...	...
SH. IN NOZ.	SA-350 LF2	80.000	10.000	...	...	...
SH. OUT NOZ	SA-350 LF2	80.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm.	Layout Angle deg	Proj Outside mm.	Proj Inside mm.	Installed in Component
DRAIN NOZ. D1	1314.175	-90.0	300.00	0.00	CH. SHELL-1
VENT NOZ. V1	12108.176	90.0	300.00	0.00	CH. SHELL-2
TS OUT NOZ. T2	1264.175	90.0	300.00	0.00	CH. SHELL-1
TS. IN NOZ. T1	12258.175	-90.0	300.00	0.00	CH. SHELL-2
SH. IN NOZ. S1	3114.176	90.0	200.00	0.00	MAIN SHELL
SH. OUT NOZ. S2	10408.176	-90.0	200.00	0.00	MAIN SHELL

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	BLIND FLANGE: 290.0mm. THK X 1638.0mm. OD	SA-350 LF2
2	WELD NECK FLANGE: 310.0mm. THK X 1638.0mm. OD	SA-350 LF2
2	CYLINDER: 74.0mm. THK X 1095.0mm. ID X 1300.0mm.	SA-516 70
1	CYLINDER: 90.0mm. THK X 1095.0mm. ID X 9094.0mm.	SA-516 70
18	BAFFLES: 1105.0mm DIA X 8.0mm THK	SA-516 70
2	TUBESHEET: 1275.0mm X 200.0mm THK	SA-350 LF2
468	TUBES: 9494.0mm X 25.4mm DIA X 3.4mm THK	SA-179
2	SADDLE: 160.0mm X 120 DEG	SA-516 70
1	INSULATION: 9094mm X 40mm THK	HOT INSULATION
2	CLASS 1500 GR 1.1, 2.0" BLIND FLANGE(S)	SA-350 LF2
1	GASKET: 1127.0mm. OD X 1101.0mm. ID	...
48	BODY FLANGE BOLTS: 90.0mm. DIA	SA-193 B7
96	NUTS FOR BODY FLANGE BOLTS: 90.0mm. DIA	...
1	GASKET: 1133.0mm. OD X 1101.0mm. ID	...
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm.	E*	PWHT reqd
Notes								
MAIN SHELL [8]	B	29	28		0.970	90.000	1.00	Yes
FIXED SADDLE [24]	B	-31	-32		0.970	16.000	1.00	No
SLIDING SADDL [24]	B	-31	-32		0.970	16.000	1.00	No
SH. IN NOZ. S1 [1]	B	29	27		0.955	90.000	1.00	Yes
Nozzle Flg [5]	!	-46	-60					
SH. OUT NOZ. S [1]	B	29	27		0.956	90.000	1.00	Yes
Nozzle Flg [5]	!	-46	-59					
Nozl. Bolting [22]		-48			1.000	64.000	1.00	No
Tubesheet: SS [13]	B	13	10		0.960	90.000	1.00	No
Warmest MDMT:		29	28					
CH. COVER-1 [11]	!	-46	-46		0.950	72.500	1.00	No
CH. FLG-1 [11]	!	-46	-46		0.902	74.000	1.00	Yes
CH. FLG-2 [11]	!	-46	-46		0.902	74.000	1.00	Yes
CH. COVER-2 [11]	!	-46	-46		0.950	72.500	1.00	No
CH. SHELL-1 [8]	B	25	22		0.937	74.000	1.00	Yes
CH. SHELL-2 [8]	B	25	22		0.937	74.000	1.00	Yes
TS OUT NOZ. T2 [1]	B	25	19		0.882	74.000	1.00	Yes
Nozzle Flg [5]	A	-18	-42					
DRAIN NOZ. D1 [1]	B	-29	-48	-29	0.599	7.645	1.00	No
Nozzle Flg [5]	!	-46	-46					
TS. IN NOZ. T1 [1]	B	25	19		0.883	74.000	1.00	Yes
Nozzle Flg [5]	!	-46	-70					
VENT NOZ. V1 [1]	B	-29	-48	-29	0.598	7.645	1.00	No
Nozzle Flg [5]	!	-46	-46					
Nozl. Bolting [22]		-48			1.000	64.000	1.00	No
Tubesheet: CS [14]	B	9	-5		0.748	74.000	1.00	No
Bolting [21]		-40						
Bolting [21]		-40						
Warmest MDMT:		25	22					
Exchanger Side	Computed MDMT		Required MDMT		Pass/Fail			
Shell	28.0		-28.9		Fail			
Channel/Tube	22.0		-28.9		Fail			

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld



UG-84(b)(2) was not considered.
UCS-66(g) was not considered.
UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.
UCS-66(i) applies to impact tested materials not by specification and
UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (Kgfcmm2mm):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm.
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0689	bars
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm.
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm.
Thickness	inches	25.4000	mm.

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	191	bars
Channel Side Design Pressure	147	bars
Shell Side Design Temperature	200.0	°C
Channel Side Design Temperature	85.0	°C
Radiography, Shell Side	None	
Radiography, Channel Side	None	
Service Type, Shell Side	None	
Service Type, Channel Side	None	
MDMT (CET), Shell Side	-28.9	°C
MDMT (CET), Tube Side	-28.9	°C
User defined MAWP, Shell Side	0	bars
User defined MAWP, Channel Side	0	bars
User defined MAPnc, Shell Side	0	bars
User defined MAPnc, Channel Side	0	bars
User defined Test Pres., Shell Side	0	bars
User defined Test Pres., Channel Side	0	bars

Projection of Nozzle from Vessel Top	0	mm.
Projection of Nozzle from Vessel Bottom	0	mm.
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS



Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-22
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	[V] 180 Km/hr
Surface Roughness Category	D: Flat, unobstructed
Type of Surface	Moderately Smooth
Base Elevation	300 mm.
Percent Wind for Hydrotest	33.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment	H or Hh 0 mm.
Distance Upwind of Crest	Lh 0 mm.
Distance from Crest to the Vessel	x 0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.

Seismic Design Code	ASCE/SEI 7-22
Importance Factor	1.500
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio	z/h 0.000
Amplification Factor	Ap 0.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Component resonance duct. fact.	[CAR] 0.000
Component strength factor	[Rpo] 0.000
Overstrength factor	[Omega0] 0.000
Response Acceleration Factor	[SMS] 0.326
Response Acceleration Factor	[SM1] 0.139
Building Importance Factor	[Ie] 1.000
Site Latitude	0.000 deg
Site Longitude	0.000 deg

Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N

Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No



Use the MAWP to compute the MDMT Yes
 For thickness ratios <= 0.35, MDMT will be -155F (-104C) Yes
 For PWHT & P1 Materials the MDMT can be < -55F (-48C) No

 Using Metric Material Databases, ASME II D Yes
 Calculate B31.3 type stress for Nozzles with Loads Yes
 Reduce the MDMT due to lower membrane stress Yes
 Consider Longitudinal Stress in MDMT Calculations Yes

Complete Listing of Vessel Elements and Details:

Element#	1/7
Element From Node	10
Element To Node	20
Element Type	Flange
Description	CH. COVER-1
Distance "FROM" to "TO"	290 mm.
Flange Outside Diameter	1638 mm.
Element Thickness	290 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	0 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	147 bars
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 bars
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-350 LF2 +IT
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	225 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm.
Yield Stress, Operating	229.57 N./mm ²
UCS-66 Chart Curve Designation	Impact Tested
External Pressure Chart Name	CS-2
UNS Number	K03011
Class / Thickness / Grade	1 ::
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element#	2/7
Element From Node	20
Element To Node	30
Element Type	Flange
Description	CH. FLG-1
Distance "FROM" to "TO"	421 mm.
Flange Inside Diameter	1095 mm.
Element Thickness	310 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	0 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	147 bars
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 bars
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-350 LF2 +IT
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element From Node	20
Detail Type	Liquid
Detail ID	Liquid: 20
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	955 mm.
Liquid Density	0.0009984 kg./cm ³



Element#	3/7
Element From Node	30
Element To Node	40
Element Type	Cylinder
Description	CH. SHELL-1
Distance "FROM" to "TO"	1300 mm.
Inside Diameter	1095 mm.
Element Thickness	74 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	74 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	147 bars
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 bars
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	234 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm.
Yield Stress, Operating	242 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	30
Detail Type	Liquid
Detail ID	Liquid: 30
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	955 mm.
Liquid Density	0.0009984 kg./cm ³

Element From Node	30
Detail Type	Nozzle
Detail ID	TS OUT NOZ. T2
Dist. from "FROM" Node / Offset dist	550 mm.
Nozzle Diameter	16 in.
Nozzle Schedule	None
Nozzle Class	1500
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	893.94 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-350 LF2 +IT

Element From Node	30
Detail Type	Nozzle
Detail ID	DRAIN NOZ. D1
Dist. from "FROM" Node / Offset dist	600 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	160
Nozzle Class	1500
Layout Angle	-90.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	0 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-350 LF2 +IT



Element From Node	40
Element To Node	50
Element Type	Cylinder
Description	MAIN SHELL
Distance "FROM" to "TO"	9094 mm.
Inside Diameter	1095 mm.
Element Thickness	90 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	90 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	191 bars
Design Temperature Internal Pressure	200 °C
Design External Pressure	0 bars
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	40
Detail Type	Saddle
Detail ID	FIXED SADDLE
Dist. from "FROM" Node / Offset dist	1900 mm.
Width of Saddle	160 mm.
Height of Saddle at Bottom	1300 mm.
Saddle Contact Angle	120.0
Height of Composite Ring Stiffener	0 mm.
Width of Wear Plate	520 mm.
Thickness of Wear Plate	16 mm.
Contact Angle, Wear Plate (degrees)	132.0
Friction coefficient	0.60000002
Moment Factor	3.0
Dimension E at base (optional)	0 mm.
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 mm.

Element From Node	40
Detail Type	Saddle
Detail ID	SLIDING SADDLE
Dist. from "FROM" Node / Offset dist	7194 mm.
Width of Saddle	160 mm.
Height of Saddle at Bottom	1300 mm.
Saddle Contact Angle	120.0
Height of Composite Ring Stiffener	0 mm.
Width of Wear Plate	520 mm.
Thickness of Wear Plate	16 mm.
Contact Angle, Wear Plate (degrees)	132.0
Friction coefficient	0.1
Moment Factor	3.0
Dimension E at base (optional)	0 mm.
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 mm.

Element From Node	40
Detail Type	Liquid
Detail ID	Liquid: 40
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	955 mm.
Liquid Density	0.0009984 kg./cm³

Element From Node	40
Detail Type	Insulation
Detail ID	HOT
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Insulation	9094 mm.
Thickness of Insulation	40 mm.
Density	0.000192 kg./cm³



Element From Node 40
 Detail Type Nozzle
 Detail ID SH. IN NOZ. S1
 Dist. from "FROM" Node / Offset dist 900 mm.
 Nozzle Diameter 24 in.
 Nozzle Schedule None
 Nozzle Class 1500
 Layout Angle 90.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 2389.6 Kg
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-350 LF2 +IT

Element From Node 40
 Detail Type Nozzle
 Detail ID SH. OUT NOZ. S2
 Dist. from "FROM" Node / Offset dist 8194 mm.
 Nozzle Diameter 24 in.
 Nozzle Schedule None
 Nozzle Class 1500
 Layout Angle -90.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 2389.6 Kg
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-350 LF2 +IT

Element From Node 40
 Detail Type Weight
 Detail ID datasheet weigh
 Dist. from "FROM" Node / Offset dist 4547 mm.
 Miscellaneous Weight 8685 Kg
 Offset from Element Centerline 0 mm.

Element# 5/7
 Element From Node 50
 Element To Node 60
 Element Type Cylinder
 Description CH. SHELL-2
 Distance "FROM" to "TO" 1300 mm.
 Inside Diameter 1095 mm.
 Element Thickness 74 mm.
 Internal Corrosion Allowance 3 mm.
 Nominal Thickness 70 mm.
 External Corrosion Allowance 0 mm.
 Design Internal Pressure 147 bars
 Design Temperature Internal Pressure 85 °C
 Design External Pressure 0 bars
 Design Temperature External Pressure 85 °C
 Effective Diameter Multiplier 1.2
 Material Name SA-516 70
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Weld is pre-Heated No

Element From Node 50
 Detail Type Liquid
 Detail ID Liquid: 50
 Dist. from "FROM" Node / Offset dist 0 mm.
 Height/Length of Liquid 955 mm.
 Liquid Density 0.0009984 kg./cm³

Element From Node 50
 Detail Type Nozzle
 Detail ID TS. IN NOZ. T1
 Dist. from "FROM" Node / Offset dist 750 mm.
 Nozzle Diameter 16 in.
 Nozzle Schedule None



Nozzle Class	1500
Layout Angle	-90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	907.02 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-350 LF2 +IT

Element From Node	50
Detail Type	Nozzle
Detail ID	VENT NOZ. V1
Dist. from "FROM" Node / Offset dist	600 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	160
Nozzle Class	1500
Layout Angle	90.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	16.353 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-350 LF2 +IT

Element#	6/7
Element From Node	60
Element To Node	70
Element Type	Flange
Description	CH. FLG-2
Distance "FROM" to "TO"	421 mm.
Flange Inside Diameter	1095 mm.
Element Thickness	310 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	0 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	147 bars
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 bars
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-350 LF2 +IT
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	225 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm.
Yield Stress, Operating	229.57 N./mm ²
UCS-66 Chart Curve Designation	Impact Tested
External Pressure Chart Name	CS-2
UNS Number	K03011
Class / Thickness / Grade	1 ::
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element From Node	60
Detail Type	Liquid
Detail ID	Liquid: 60
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	955 mm.
Liquid Density	0.0009984 kg./cm ³

Element#	7/7
Element From Node	70
Element To Node	80
Element Type	Flange
Description	CH. COVER-2
Distance "FROM" to "TO"	290 mm.
Flange Outside Diameter	1638 mm.
Element Thickness	290 mm.



Internal Corrosion Allowance	3	mm.
Nominal Thickness	0	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	147	bars
Design Temperature Internal Pressure	85	°C
Design External Pressure	0	bars
Design Temperature External Pressure	85	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-350 LF2	+IT
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	

XY Coordinate Calculations:

From	To	X (Horiz.) mm.	Y (Vert.) mm.	DX (Horiz.) mm.	DY (Vert.) mm.
CH. COVER-1		293.175	...	290	...
CH. FLG-1		714.175	...	421	...
CH. SHELL-1		2014.17	...	1300	...
MAIN SHELL		11308.2	...	9094	...
CH. SHELL-2		12808.2	...	1300	...
CH. FLG-2		13233.7	...	421	...
CH. COVER-2		13523.7	...	290	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Flange Input Data Values

Description: CH. COVER-1 :

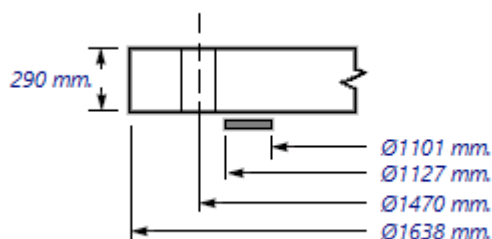
CH. COVER-1

Flange Type		Blind
Design Pressure	P	147.00 bars
Design Temperature		85 °C
Internal Corrosion Allowance	ci	3.0000 mm.
External Corrosion Allowance	ce	0.0000 mm.
Use Corrosion Allowance in Thickness Calcs.		Yes
Flange Outside Diameter	A	1638.000 mm.
Flange Thickness	t	290.0000 mm.
Flange Material		SA-350 LF2 +IT
Flange Material UNS number		K03011
Flange Allowable Stress At Temperature	Sfo	138.00 N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00 N./mm ²
Bolt Material		SA-193 B7
Bolt Allowable Stress At Temperature	Sb	159.00 N./mm ²
Bolt Allowable Stress At Ambient	Sa	159.00 N./mm ²

Warning:

It seems as if the selected bolt material does not match the bolt diameter range allowed per the material specification. Please check the material selection for the given nominal bolt diameter.

Perimeter along the Center of the Bolts	L	4618.141 mm.
Diameter of Bolt Circle	C	1470.000 mm.
Nominal Bolt Diameter	a	90.0000 mm.
Type of Threads		TEMA Metric
Number of Bolts		24
Flange Face Outside Diameter	Fod	1130.000 mm.
Flange Face Inside Diameter	Fid	0.000 mm.
Flange Facing Sketch		Sketch 1a
Gasket Outside Diameter	Go	1127.000 mm.
Gasket Inside Diameter	Gi	1101.000 mm.
Gasket Factor	m	3.0000
Gasket Design Seating Stress	y	68.95 N./mm ²
Gasket Thickness	tg	3.1750 mm.
Raised face depth	Fd	0.0000 mm.



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Gasket Contact Width,	$N = (G_o - G_i)/2$	13.000 mm.
Basic Gasket Width,	$b_o = N/2$	6.500 mm.
Effective Gasket Width,	$b = C_b * \sqrt{b_o}$	6.425 mm.
Gasket Reaction Diameter,	$G = G_o - 2 * b$	1114.151 mm.

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 1114^2 \cdot 147$$

$$= 1461476.500 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.42 \cdot \pi \cdot 1114 \cdot 3 \cdot 147$$

$$= 202256.391 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(1461476 + 202256 + 0, 0)$$

$$= 1663732.875 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 68.9 \cdot 6.42 \cdot \pi \cdot 1114 + 0 \cdot 0 \cdot 0$$

$$= 158103.953 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G)/Sb, Wg/Sa)$$

$$= \max((1663732 + 0 + 4 \cdot 0/1114)/159, 158104/159)$$

$$= 1026.179 \text{ cm}^2$$

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	1026.179	1268.901	
Radial Distance between Bolts and Edge:	84.000	84.000	
Circ. Spacing between the Bolts:	181.500	191.874	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi (Go + Gi))$$

$$= 1269 \cdot 159 / (68.9 \cdot \pi (1127 + 1101))$$

$$= 41.804 \text{ mm. [Note: Exceeds actual gasket width, 13.000]}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa(Am + Ab) / 2$$

$$= 159 (1026 + 1269)/2$$

$$= 1860494.00 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo$$

$$= 1663732.88 \text{ Kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1470 - 1114)/2$$

$$= 177.9246 \text{ mm.}$$

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$= 1.9 \cdot Wm1 \cdot hG \cdot Bsc / (t^2 \cdot G) + C \cdot Z \cdot Peq \cdot G^2 / t^2$$

$$= 1.9 \cdot 1663732 \cdot 178 \cdot 1 / (287^2 \cdot 1114) + 0.3 \cdot 1 \cdot 147 \cdot 1114^2 / 287^2$$

$$= 126.57 \text{ N./mm}^2$$

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:

$$= 1.9 \cdot W \cdot hG \cdot Bsc / (t^2 \cdot G)$$

$$= 1.9 \cdot 1860494 \cdot 178 \cdot 1 / (287^2 \cdot 1114)$$

$$= 67.21 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= Wm1 / Ab$$

$$= 1663732/1269$$



= 128.58 N./mm²

Bolt Stress, Seating [BSa]:

= (Wm2 / Ab)
 = (158104/1269)
 = 12.22 N./mm²

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Tangential Flange	126.57	138.00	67.21 138.00
Bolting	128.58	159.00	12.22 159.00

Reqd. Blind Flange Thickness at Center 277.857 mm.
 Reqd. Blind Flange Thickness at Gasket 203.293 mm.

Estimated M.A.W.P. (Operating) 160 bars
 Estimated Finished Weight of Flange at given Thk. 4736.4 kg.
 Estimated Unfinished Weight of Forging at given Thk 4736.4 kg.

Minimum Design Metal Temperature Results:

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Tangential Flange	123.96	138.00	65.83 138.00
Bolting	128.58	159.00	12.22 159.00

Minimum Req'd Thickness, ambient, no Corrosion: 274.879 mm.

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Flange Input Data Values

Description: CH. FLG-1 :

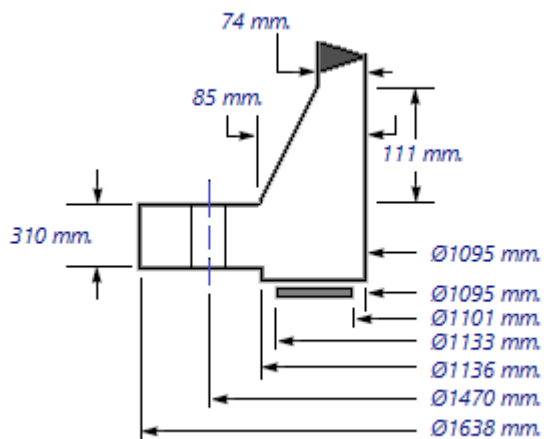
CH. FLG-1

Flange Type	Integral Weld Neck		
Design Pressure	P	147.09	bars
Design Temperature		85	°C
Internal Corrosion Allowance	ci	3.0000	mm.
External Corrosion Allowance	ce	0.0000	mm.
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	1095.000	mm.
Flange Outside Diameter	A	1638.000	mm.
Flange Thickness	t	310.0000	mm.
Thickness of Hub at Small End	go	74.0000	mm.
Thickness of Hub at Large End	gl	85.0000	mm.
Length of Hub	h	111.0000	mm.
Flange Material	SA-350 LF2 +IT		
Flange Material UNS number	K03011		
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material	SA-193 B7		
Bolt Allowable Stress At Temperature	Sb	159.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	159.00	N./mm ²

Warning:

It seems as if the selected bolt material does not match the bolt diameter range allowed per the material specification. Please check the material selection for the given nominal bolt diameter.

Diameter of Bolt Circle	C	1470.000	mm.
Nominal Bolt Diameter	a	90.0000	mm.
Type of Threads	TEMA Metric		
Number of Bolts	24		
Flange Face Outside Diameter	Fod	1136.000	mm.
Flange Face Inside Diameter	Fid	1095.000	mm.
Flange Facing Sketch	Sketch 1a		
Gasket Outside Diameter	Go	1133.000	mm.
Gasket Inside Diameter	Gi	1101.000	mm.
Gasket Factor	m	3.7500	
Gasket Design Seating Stress	y	62.06	N./mm ²
Gasket Thickness	tg	4.5000	mm.
Raised face depth	Fd	0.0000	mm.





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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (147 \cdot (1095/2 + 3)) / (138 \cdot 1 - 0.6 \cdot 147) + Ca$$

$$= 65.6917 \text{ mm.}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 71) / (550 + 0.6 \cdot 71)$$

$$= 165.187 \text{ bars}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 74) / (548 + 0.6 \cdot 74)$$

$$= 172.516 \text{ bars}$$

Corroded Flange ID,	Bcor = B+2*ci	1101.000 mm.
Corroded Large Hub,	g1Cor = g1-ci	82.000 mm.
Corroded Small Hub,	g0Cor = go-ci	71.000 mm.
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	102.500 mm.
Gasket Contact Width,	N = (Go - Gi)/2	16.000 mm.
Basic Gasket Width,	bo = N/2	8.000 mm.
Effective Gasket Width,	b = Cb*sqrt(bo)	7.127 mm.
Gasket Reaction Diameter,	G = Go - 2 * b	1118.745 mm.

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 1119^2 \cdot 147$$

$$= 1474492.500 \text{ Kg}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 7.13 \cdot \pi \cdot 1119 \cdot 3.75 \cdot 147$$

$$= 281815.219 \text{ Kg}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 1101^2 \cdot 147$$

$$= 1428087.875 \text{ Kg}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 1474492 - 1428087$$

$$= 46404.641 \text{ Kg}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(1474492 + 281815 + 0, 0)$$

$$= 1756307.750 \text{ Kg}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 62.1 \cdot 7.13 \cdot \pi \cdot 1119 + 0 \cdot 0$$

$$= 158511.547 \text{ Kg}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((1756307 + 0 + 4 \cdot 0 / 1119) / 159, 158512 / 159)$$

$$= 1083.278 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 90 + 6 \cdot 310 / (3.75 + 0.5)$$

$$= 617.647 \text{ mm.}$$

Actual Circumferential Bolt Spacing [Bs]:

$$\begin{aligned}
 &= C \cdot \sin(\pi / n) \\
 &= 1470 \cdot \sin(\pi / 24) \\
 &= 191.874 \text{ mm.}
 \end{aligned}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$\begin{aligned}
 &= \max(\sqrt{Bs / (2a + t)}, 1) \\
 &= \max(\sqrt{192 / (2 \cdot 90 + 310)}, 1) \\
 &= 1.0000
 \end{aligned}$$

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	1083.278	1268.901	
Radial Distance between Hub and Bolts:	101.500	102.500	
Radial Distance between Bolts and Edge:	84.000	84.000	
Circ. Spacing between the Bolts:	181.500	191.874	617.647

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$\begin{aligned}
 &= Ab \cdot Sa / (y \cdot \pi(Go + Gi)) \\
 &= 1269 \cdot 159 / (62.1 \cdot \pi(1133 + 1101)) \\
 &= 46.324 \text{ mm. [Note: Exceeds actual gasket width, 16.000]}
 \end{aligned}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$\begin{aligned}
 &= Sa \cdot Ab \\
 &= 159 \cdot 1269 \\
 &= 2057255.12 \text{ Kgf}
 \end{aligned}$$

Gasket Load for the Operating Condition [HG]:

$$\begin{aligned}
 &= Wo - H \\
 &= 1756307 - 1474492 \\
 &= 281815.22 \text{ Kgf}
 \end{aligned}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$\begin{aligned}
 &= (C - G) / 2 \\
 &= (1470 - 1119) / 2 \\
 &= 175.6274 \text{ mm.}
 \end{aligned}$$

Distance to Face Pressure Reaction [ht]:

$$\begin{aligned}
 &= 0.5((C - B) / 2 + hg) \\
 &= 0.5((1470 - 1101) / 2 + 176) \\
 &= 180.0637 \text{ mm.}
 \end{aligned}$$

Distance to End Pressure Reaction [hd]:

$$\begin{aligned}
 &= (C - B - g1) / 2 \\
 &= (1470 - 1101 - 82) / 2 \\
 &= 143.5000 \text{ mm.}
 \end{aligned}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	1428088.	143.5000	1.0000	204935.
Face Pressure, Mt	46405.	180.0637	1.0000	8356.
Gasket Load, Mg	281815.	175.6274	1.0000	49495.
Gasket Seating, Matm	2057255.	175.6274	1.0000	361318.

Total Moment for Operation, Mo 262786. Kg-m.
 Total Moment for Gasket seating, Mg 361318. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 279.591 mm.
 Hub Ratio, h/h0 = HL / H0 0.397
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.155

Flange Factors for Integral Flange:

Factor F per 4.16.5	0.890
Factor V per 4.16.5	0.456
Factor f per 4.16.5	1.000

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 1638/1101$$

$$= 1.488$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169(K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169(1.49^2 \cdot \log_{10}(1.49))) / (1.49^2 - 1)) / (1.49 - 1)$$

$$= 5.059$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.49^2(1 + 8.55246 \cdot \log_{10}(1.49)) - 1) / ((1.0472 + 1.9448 \cdot 1.49^2) \cdot (1.49 - 1))$$

$$= 1.716$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$$

$$= (1.49^2(1 + 8.55246 \cdot \log_{10}(1.49)) - 1) / (1.36136(1.49^2 - 1) \cdot (1.49 - 1))$$

$$= 5.560$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.49^2 + 1) / (1.49^2 - 1)$$

$$= 2.648$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V$$

$$= 5.56 \cdot 71^2 \cdot 280 / 0.46$$

$$= 17181050.000 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.89/280$$

$$= 0.003 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3/d$$

$$= (310 \cdot 0.0032 + 1) / 1.72 + 310^3/17181050$$

$$= 2.892$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g_1^2)$$

$$= (1 \cdot 262786/1101) / (2.89 \cdot 82^2)$$

$$= 120.39 \text{ N./mm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g_1^2)$$

$$= (1 \cdot 361318/1101) / (2.89 \cdot 82^2)$$

$$= 165.52 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (2.32 \cdot 262786/1101) / (2.89 \cdot 310^2)$$

$$= 19.50 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (2.32 \cdot 361318/1101) / (2.89 \cdot 310^2)$$

$$= 26.82 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$



$$= (5.06 \cdot 262786 / (310^2 \cdot 1101)) - 2.65 \cdot 19.5$$

$$= 71.58 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot \text{Matm} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRa}$$

$$= (5.06 \cdot 361318 / (310^2 \cdot 1101)) - 2.65 \cdot 26.8$$

$$= 98.42 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (\text{SHo} + \max(\text{SRO}, \text{STo})) / 2$$

$$= (120 + \max(19.5, 71.6)) / 2$$

$$= 95.98 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (\text{SHA} + \max(\text{SRa}, \text{STa})) / 2$$

$$= (166 + \max(26.8, 98.4)) / 2$$

$$= 131.97 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= \text{Wm1} / \text{Ab}$$

$$= 1756307 / 1269$$

$$= 135.74 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (\text{Wm2} / \text{Ab})$$

$$= (158512 / 1269)$$

$$= 12.25 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	120.39	207.00	165.52	207.00
Radial Flange	19.50	138.00	26.82	138.00
Tangential Flange	71.58	138.00	98.42	138.00
Maximum Average	95.98	138.00	131.97	138.00
Bolting	135.74	159.00	12.25	159.00

Minimum Required Flange Thickness 301.473 mm.

Estimated M.A.W.P. (Operating) 165 bars
 Estimated Finished Weight of Flange at given Thk. 3052.8 kg.
 Estimated Unfinished Weight of Forging at given Thk 3803.1 kg.



Coefficients for f and $f2$

	f	$f2$	$f3$
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

$$= \max(1; F1num / F1den)$$

$$= \max(1; 0.74/1.47)$$

$$= 1.0000$$

Hub Stress Correction Factor [f2]:

$$= \max(0.25; fn2) \text{ if } h/h0 < 0.35, \text{ else } \max(0.25; fn3)$$

$$= \max(0.25; 0.32/1.54) \text{ if } h/h0 < 0.35, \text{ else } \max(0.25; 0.17/0.77)$$

$$= 0.2500$$

In this case, the hub ratio $h/h0$, 0.3970 was ≥ 0.35 .

PV Elite iterates on the assembly bolt load stress, Sf_{max} , until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.0 N./mm ²
Minimum permissible bolt stress	Sbmin	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sfmax	436.0 N./mm ²
Target assembly gasket stress	SgT	207.0 N./mm ²
Maximum permissible gasket stress	Sgmax	380.0 N./mm ²
Minimum gasket operating stress	Sgmin-O	97.0 N./mm ²
Minimum gasket seating stress	Sgmin-S	140.0 N./mm ²
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	52.8709 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:



Flange Material Ambient Yield Stress	Sya	250.0 N./mm ²
Flange Material Operating Yield Stress	Syo	229.6 N./mm ²

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (1133^2 - 1101^2)$$

$$= 561.5 \text{ cm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Ab)$$

$$= 207 \cdot 561 / (24 \cdot 52.9)$$

$$= 91.6 \text{ N./mm}^2$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(91.6; 507)$$

$$= 91.6 \text{ N./mm}^2$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(91.6; 290)$$

$$= 290.0 \text{ N./mm}^2$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(290; 436)$$

$$= 290.0 \text{ N./mm}^2$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$\therefore Sbsel \geq Sgmin-S(Ag / (nb \cdot Abs))$$

$$\therefore 290 \geq 140 (561 / (24 \cdot 52.9))$$

$$\therefore 290.0 \geq 61.9 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$\therefore Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$\therefore 290 \geq (97 \cdot 561 + \pi/4 \cdot 147 \cdot 1101^2) / (0.7 \cdot 52.9 \cdot 24)$$

$$\therefore 290.0 \geq 219.0 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$\therefore Sbsel \leq Sgmax(Ag / (nb \cdot Abs))$$

$$\therefore 290 \leq 380 (561 / (24 \cdot 52.9))$$

$$\therefore 290.0 \leq 168.1 \text{ N./mm}^2 \text{ Check Fails}$$

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 361318 / 1.9807 \cdot 0.46 / (2.74 \cdot 201334 \cdot 71^2 \cdot 280 \cdot 0.3)$$

$$= 0.361 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 262786 / 1.9807 \cdot 0.46 / (2.74 \cdot 197675 \cdot 71^2 \cdot 280 \cdot 0.3)$$

$$= 0.268 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 361318 / 1.9807 \cdot 0.46 / (2.89 \cdot 201334 \cdot 71^2 \cdot 280 \cdot 0.3)$$

$$= 0.342 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 262786 / 1.9807 \cdot 0.46 / (2.89 \cdot 197675 \cdot 71^2 \cdot 280 \cdot 0.3)$$

$$= 0.253 \text{ (should be } \leq 1)$$

Minimum Design Metal Temperature Results:

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.



Impact Test Temperature provided per Specification -46 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed	
Longitudinal Hub	119.57	207.00	162.68	207.00
Radial Flange	20.62	138.00	28.06	138.00
Tangential Flange	69.64	138.00	94.75	138.00
Maximum Average	94.60	138.00	128.71	138.00
Bolting	135.74	159.00	12.25	159.00

Minimum Req'd Thickness, ambient, no Corrosion: 296.342 mm.

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Flange Input Data Values

Description: CH. FLG-2 :

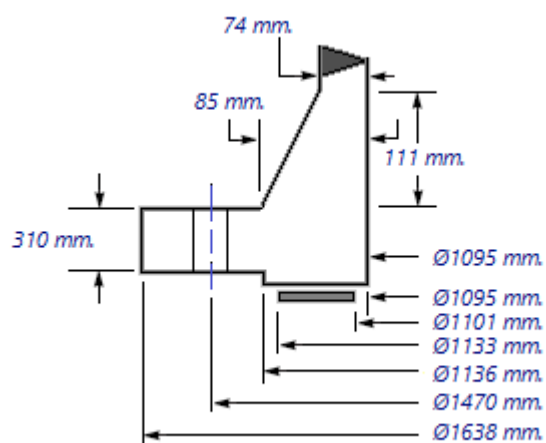
CH. FLG-2

Flange Type	Integral Weld Neck		
Design Pressure	P	147.09	bars
Design Temperature		85	°C
Internal Corrosion Allowance	ci	3.0000	mm.
External Corrosion Allowance	ce	0.0000	mm.
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	1095.000	mm.
Flange Outside Diameter	A	1638.000	mm.
Flange Thickness	t	310.0000	mm.
Thickness of Hub at Small End	go	74.0000	mm.
Thickness of Hub at Large End	gl	85.0000	mm.
Length of Hub	h	111.0000	mm.
Flange Material	SA-350 LF2 +IT		
Flange Material UNS number	K03011		
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material	SA-193 B7		
Bolt Allowable Stress At Temperature	Sb	159.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	159.00	N./mm ²

Warning:

It seems as if the selected bolt material does not match the bolt diameter range allowed per the material specification. Please check the material selection for the given nominal bolt diameter.

Diameter of Bolt Circle	C	1470.000	mm.
Nominal Bolt Diameter	a	90.0000	mm.
Type of Threads	TEMA Metric		
Number of Bolts	24		
Flange Face Outside Diameter	Fod	1136.000	mm.
Flange Face Inside Diameter	Fid	1095.000	mm.
Flange Facing Sketch	Sketch 1a		
Gasket Outside Diameter	Go	1133.000	mm.
Gasket Inside Diameter	Gi	1101.000	mm.
Gasket Factor	m	3.7500	
Gasket Design Seating Stress	y	62.06	N./mm ²
Gasket Thickness	tg	4.5000	mm.
Raised face depth	Fd	0.0000	mm.



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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (147 \cdot (1095/2 + 3)) / (138 \cdot 1 - 0.6 \cdot 147) + Ca$$

$$= 65.6917 \text{ mm.}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 71) / (550 + 0.6 \cdot 71)$$

$$= 165.187 \text{ bars}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 74) / (548 + 0.6 \cdot 74)$$

$$= 172.516 \text{ bars}$$

Corroded Flange ID,	Bcor = B+2*ci	1101.000 mm.
Corroded Large Hub,	g1Cor = g1-ci	82.000 mm.
Corroded Small Hub,	g0Cor = go-ci	71.000 mm.
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	102.500 mm.
Gasket Contact Width,	N = (Go - Gi)/2	16.000 mm.
Basic Gasket Width,	bo = N/2	8.000 mm.
Effective Gasket Width,	b = Cb*sqrt(bo)	7.127 mm.
Gasket Reaction Diameter,	G = Go - 2 * b	1118.745 mm.

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 1119^2 \cdot 147$$

$$= 1474492.500 \text{ Kg}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 7.13 \cdot \pi \cdot 1119 \cdot 3.75 \cdot 147$$

$$= 281815.219 \text{ Kg}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 1101^2 \cdot 147$$

$$= 1428087.875 \text{ Kg}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 1474492 - 1428087$$

$$= 46404.641 \text{ Kg}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(1474492 + 281815 + 0, 0)$$

$$= 1756307.750 \text{ Kg}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 62.1 \cdot 7.13 \cdot \pi \cdot 1119 + 0 \cdot 0$$

$$= 158511.547 \text{ Kg}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((1756307 + 0 + 4 \cdot 0 / 1119) / 159, 158512 / 159)$$

$$= 1083.278 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 90 + 6 \cdot 310 / (3.75 + 0.5)$$

$$= 617.647 \text{ mm.}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 1470 \cdot \sin(\pi / 24)$$

$$= 191.874 \text{ mm.}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs / (2a + t)}, 1)$$

$$= \max(\sqrt{192 / (2 \cdot 90 + 310)}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	1083.278	1268.901	
Radial Distance between Hub and Bolts:	101.500	102.500	
Radial Distance between Bolts and Edge:	84.000	84.000	
Circ. Spacing between the Bolts:	181.500	191.874	617.647

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 1269 \cdot 159 / (62.1 \cdot \pi(1133 + 1101))$$

$$= 46.324 \text{ mm. [Note: Exceeds actual gasket width, 16.000]}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 159 \cdot 1269$$

$$= 2057255.12 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 1756307 - 1474492$$

$$= 281815.22 \text{ Kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1470 - 1119) / 2$$

$$= 175.6274 \text{ mm.}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((1470 - 1101) / 2 + 176)$$

$$= 180.0637 \text{ mm.}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (1470 - 1101 - 82) / 2$$

$$= 143.5000 \text{ mm.}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	1428088.	143.5000	1.0000	204935.
Face Pressure, Mt	46405.	180.0637	1.0000	8356.
Gasket Load, Mg	281815.	175.6274	1.0000	49495.
Gasket Seating, Matm	2057255.	175.6274	1.0000	361318.

Total Moment for Operation, Mo 262786. Kg-m.
 Total Moment for Gasket seating, Mg 361318. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 279.591 mm.
 Hub Ratio, h/h0 = HL / H0 0.397
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.155

Flange Factors for Integral Flange:

Factor F per 4.16.5	0.890
Factor V per 4.16.5	0.456
Factor f per 4.16.5	1.000

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 1638/1101$$

$$= 1.488$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169(K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169(1.49^2 \cdot \log_{10}(1.49))) / (1.49^2 - 1)) / (1.49 - 1)$$

$$= 5.059$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.49^2(1 + 8.55246 \cdot \log_{10}(1.49)) - 1) / ((1.0472 + 1.9448 \cdot 1.49^2) \cdot (1.49 - 1))$$

$$= 1.716$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$$

$$= (1.49^2(1 + 8.55246 \cdot \log_{10}(1.49)) - 1) / (1.36136(1.49^2 - 1) \cdot (1.49 - 1))$$

$$= 5.560$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.49^2 + 1) / (1.49^2 - 1)$$

$$= 2.648$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V$$

$$= 5.56 \cdot 71^2 \cdot 280 / 0.46$$

$$= 17181050.000 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.89/280$$

$$= 0.003 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3/d$$

$$= (310 \cdot 0.0032 + 1) / 1.72 + 310^3/17181050$$

$$= 2.892$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g_1^2)$$

$$= (1 \cdot 262786/1101) / (2.89 \cdot 82^2)$$

$$= 120.39 \text{ N./mm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g_1^2)$$

$$= (1 \cdot 361318/1101) / (2.89 \cdot 82^2)$$

$$= 165.52 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (2.32 \cdot 262786/1101) / (2.89 \cdot 310^2)$$

$$= 19.50 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (2.32 \cdot 361318/1101) / (2.89 \cdot 310^2)$$

$$= 26.82 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$



$$= (5.06 \cdot 262786 / (310^2 \cdot 1101)) - 2.65 \cdot 19.5$$

$$= 71.58 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot \text{Matm} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRa}$$

$$= (5.06 \cdot 361318 / (310^2 \cdot 1101)) - 2.65 \cdot 26.8$$

$$= 98.42 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (\text{SHo} + \max(\text{SRO}, \text{STo})) / 2$$

$$= (120 + \max(19.5, 71.6)) / 2$$

$$= 95.98 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (\text{SHA} + \max(\text{SRa}, \text{STa})) / 2$$

$$= (166 + \max(26.8, 98.4)) / 2$$

$$= 131.97 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= \text{Wm1} / \text{Ab}$$

$$= 1756307 / 1269$$

$$= 135.74 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (\text{Wm2} / \text{Ab})$$

$$= (158512 / 1269)$$

$$= 12.25 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	120.39	207.00	165.52	207.00
Radial Flange	19.50	138.00	26.82	138.00
Tangential Flange	71.58	138.00	98.42	138.00
Maximum Average	95.98	138.00	131.97	138.00
Bolting	135.74	159.00	12.25	159.00

Minimum Required Flange Thickness 301.473 mm.

Estimated M.A.W.P. (Operating) 165 bars
 Estimated Finished Weight of Flange at given Thk. 3052.8 kg.
 Estimated Unfinished Weight of Forging at given Thk 3803.1 kg.

Coefficients for f and $f2$

	f	$f2$	$f3$
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

$$= \max(1; F1num / F1den)$$

$$= \max(1; 0.74/1.47)$$

$$= 1.0000$$

Hub Stress Correction Factor [f2]:

$$= \max(0.25; fn2) \text{ if } h/h0 < 0.35, \text{ else } \max(0.25; fn3)$$

$$= \max(0.25; 0.32/1.54) \text{ if } h/h0 < 0.35, \text{ else } \max(0.25; 0.17/0.77)$$

$$= 0.2500$$

In this case, the hub ratio $h/h0$, 0.3970 was ≥ 0.35 .

PV Elite iterates on the assembly bolt load stress, $Sfmax$, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.0 N./mm ²
Minimum permissible bolt stress	Sbmin	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sfmax	436.0 N./mm ²
Target assembly gasket stress	SgT	207.0 N./mm ²
Maximum permissible gasket stress	Sgmax	380.0 N./mm ²
Minimum gasket operating stress	Sgmin-O	97.0 N./mm ²
Minimum gasket seating stress	Sgmin-S	140.0 N./mm ²
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	52.8709 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:



Flange Material Ambient Yield Stress	Sya	250.0 N./mm ²
Flange Material Operating Yield Stress	Syo	229.6 N./mm ²

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (1133^2 - 1101^2)$$

$$= 561.5 \text{ cm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Ab)$$

$$= 207 \cdot 561 / (24 \cdot 52.9)$$

$$= 91.6 \text{ N./mm}^2$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(91.6; 507)$$

$$= 91.6 \text{ N./mm}^2$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(91.6; 290)$$

$$= 290.0 \text{ N./mm}^2$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(290; 436)$$

$$= 290.0 \text{ N./mm}^2$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$\therefore Sbsel \geq Sgmin-S(Ag / (nb \cdot Abs))$$

$$\therefore 290 \geq 140 (561 / (24 \cdot 52.9))$$

$$\therefore 290.0 \geq 61.9 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$\therefore Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$\therefore 290 \geq (97 \cdot 561 + \pi/4 \cdot 147 \cdot 1101^2) / (0.7 \cdot 52.9 \cdot 24)$$

$$\therefore 290.0 \geq 219.0 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$\therefore Sbsel \leq Sgmax(Ag / (nb \cdot Abs))$$

$$\therefore 290 \leq 380 (561 / (24 \cdot 52.9))$$

$$\therefore 290.0 \leq 168.1 \text{ N./mm}^2 \text{ Check Fails}$$

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 361318 / 1 \cdot 9807 \cdot 0.46 / (2.74 \cdot 201334 \cdot 71^2 \cdot 280 \cdot 0.3)$$

$$= 0.361 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 262786 / 1 \cdot 9807 \cdot 0.46 / (2.74 \cdot 197675 \cdot 71^2 \cdot 280 \cdot 0.3)$$

$$= 0.268 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 361318 / 1 \cdot 9807 \cdot 0.46 / (2.89 \cdot 201334 \cdot 71^2 \cdot 280 \cdot 0.3)$$

$$= 0.342 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 262786 / 1 \cdot 9807 \cdot 0.46 / (2.89 \cdot 197675 \cdot 71^2 \cdot 280 \cdot 0.3)$$

$$= 0.253 \text{ (should be } \leq 1)$$

Minimum Design Metal Temperature Results:

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.



Impact Test Temperature provided per Specification -46 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed	
Longitudinal Hub	119.57	207.00	162.68	207.00
Radial Flange	20.62	138.00	28.06	138.00
Tangential Flange	69.64	138.00	94.75	138.00
Maximum Average	94.60	138.00	128.71	138.00
Bolting	135.74	159.00	12.25	159.00

Minimum Req'd Thickness, ambient, no Corrosion: 296.342 mm.

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Flange Input Data Values

Description: CH. COVER-2 :

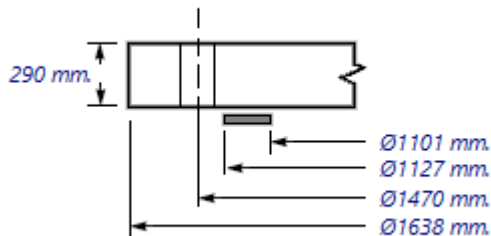
CH. COVER-2

Flange Type		Blind
Design Pressure	P	147.00 bars
Design Temperature		85 °C
Internal Corrosion Allowance	ci	3.0000 mm.
External Corrosion Allowance	ce	0.0000 mm.
Use Corrosion Allowance in Thickness Calcs.		Yes
Flange Outside Diameter	A	1638.000 mm.
Flange Thickness	t	290.0000 mm.
Flange Material		SA-350 LF2 +IT
Flange Material UNS number		K03011
Flange Allowable Stress At Temperature	Sfo	138.00 N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00 N./mm ²
Bolt Material		SA-193 B7
Bolt Allowable Stress At Temperature	Sb	159.00 N./mm ²
Bolt Allowable Stress At Ambient	Sa	159.00 N./mm ²

Warning:

It seems as if the selected bolt material does not match the bolt diameter range allowed per the material specification. Please check the material selection for the given nominal bolt diameter.

Perimeter along the Center of the Bolts	L	4618.141 mm.
Diameter of Bolt Circle	C	1470.000 mm.
Nominal Bolt Diameter	a	90.0000 mm.
Type of Threads		TEMA Metric
Number of Bolts		24
Flange Face Outside Diameter	Fod	1130.000 mm.
Flange Face Inside Diameter	Fid	0.000 mm.
Flange Facing Sketch		Sketch 1a
Gasket Outside Diameter	Go	1127.000 mm.
Gasket Inside Diameter	Gi	1101.000 mm.
Gasket Factor	m	3.0000
Gasket Design Seating Stress	y	68.95 N./mm ²
Gasket Thickness	tg	3.1750 mm.
Raised face depth	Fd	0.0000 mm.



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Gasket Contact Width,	$N = (G_o - G_i)/2$	13.000 mm.
Basic Gasket Width,	$b_o = N/2$	6.500 mm.
Effective Gasket Width,	$b = C_b * \sqrt{b_o}$	6.425 mm.
Gasket Reaction Diameter,	$G = G_o - 2 * b$	1114.151 mm.

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 1114^2 \cdot 147$$

$$= 1461476.500 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.42 \cdot \pi \cdot 1114 \cdot 3 \cdot 147$$

$$= 202256.391 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(1461476 + 202256 + 0, 0)$$

$$= 1663732.875 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 68.9 \cdot 6.42 \cdot \pi \cdot 1114 + 0 \cdot 0 \cdot 0$$

$$= 158103.953 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G)/Sb, Wg/Sa)$$

$$= \max((1663732 + 0 + 4 \cdot 0/1114)/159, 158104/159)$$

$$= 1026.179 \text{ cm}^2$$

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	1026.179	1268.901	
Radial Distance between Bolts and Edge:	84.000	84.000	
Circ. Spacing between the Bolts:	181.500	191.874	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi (Go + Gi))$$

$$= 1269 \cdot 159 / (68.9 \cdot \pi (1127 + 1101))$$

$$= 41.804 \text{ mm. [Note: Exceeds actual gasket width, 13.000]}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 159 \cdot 1269$$

$$= 2057255.12 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo$$

$$= 1663732.88 \text{ Kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1470 - 1114) / 2$$

$$= 177.9246 \text{ mm.}$$

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$= 1.9 \cdot Wm1 \cdot hG \cdot Bsc / (t^2 \cdot G) + C \cdot Z \cdot Peq \cdot G^2 / t^2$$

$$= 1.9 \cdot 1663732 \cdot 178 \cdot 1 / (287^2 \cdot 1114) + 0.3 \cdot 1 \cdot 147 \cdot 1114^2 / 287^2$$

$$= 126.57 \text{ N./mm}^2$$

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:

$$= 1.9 \cdot W \cdot hG \cdot Bsc / (t^2 \cdot G)$$

$$= 1.9 \cdot 2057255 \cdot 178 \cdot 1 / (287^2 \cdot 1114)$$

$$= 74.32 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= Wm1 / Ab$$

$$= 1663732 / 1269$$



= 128.58 N./mm²

Bolt Stress, Seating [BSa]:

= (Wm2 / Ab)
 = (158104/1269)
 = 12.22 N./mm²

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Tangential Flange	126.57	138.00	74.32 138.00
Bolting	128.58	159.00	12.22 159.00

Reqd. Blind Flange Thickness at Center 277.857 mm.
 Reqd. Blind Flange Thickness at Gasket 213.618 mm.

Estimated M.A.W.P. (Operating) 160 bars
 Estimated Finished Weight of Flange at given Thk. 4736.4 kg.
 Estimated Unfinished Weight of Forging at given Thk 4736.4 kg.

Minimum Design Metal Temperature Results:

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Note: UCS-66(b)-(c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Tangential Flange	123.96	138.00	72.79 138.00
Bolting	128.58	159.00	12.22 159.00

Minimum Req'd Thickness, ambient, no Corrosion: 274.879 mm.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd bars	Nominal Thickness mm.	Total Corr Allowance mm.	Element Diameter mm.	Allowable Stress(SE) N./mm ²	Element#
CH. COVER-1		147	...	3	1638	138	1
CH. FLG-1		147.09	...	3	1095	138	2
CH. SHELL-1		147.09	74	3	1095	138	3
MAIN SHELL		191.09	90	3	1095	138	4
CH. SHELL-2		147.09	70	3	1095	138	5
CH. FLG-2		147.09	...	3	1095	138	6
CH. COVER-2		147	...	3	1638	138	7

Element Required Thickness and MAWP :

From	To	Design Pressure bars	M.A.W.P. Corroded bars	M.A.P. New & Cold bars	Minimum Thickness mm.	Required Thickness mm.	Element#
CH. COVER-1		147	160	164	290	277.857	1
CH. FLG-1		147	165	172	310	301.473	2
CH. SHELL-1		147	165	173	74	65.6918	3
MAIN SHELL		191	199	206	90	86.1442	4
CH. SHELL-2		147	165	173	74	65.6918	5
CH. FLG-2		147	165	172	310	301.473	6
CH. COVER-2		147	160	164	290	277.857	7

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

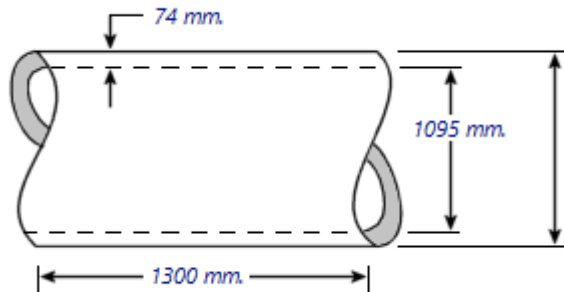
Shell Side MAWP = 194 bars
 Shell Side MAPnc = 199 bars
 Channel Side MAWP = 155 bars
 Channel Side MAPnc = 164 bars

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

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Internal Pressure Results for: CH. SHELL-1

Cylindrical Shell Design Information:					
Design Pressure	P	147.094 bars	Int. Design Temperature	85 °C	
Material		SA-516 70	Ext. Design Temperature	85 °C	
Length		1300.000 mm.	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	1095.000 mm.	Actual Stress	122.9 N./mm ²	
Spec. Min. Thk t _e , or s _a		74.0000 mm.	Spec. Nominal Thk.	74.000 mm.	
Surface Area		50765.0 cm ²	SG of Contents	1.000	
Element Volume		1224445.1 cm ³	Weight of Contents	1132.5 Kg	
Empty Weight		2738.2 Kg	Operating Weight	3870.7 Kg	
			Comp. All. Ope. Stress	147.3 N./mm ²	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (147 \cdot 550) / (138 \cdot 1 - 0.6 \cdot 147)$$

$$= 62.6918 + 3.0000 = 65.6918 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.094 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 71) / (550 + 0.6 \cdot 71)$$

$$= 165 - 0.094 = 165 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 74) / (548 + 0.6 \cdot 74)$$

$$= 173 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (147 \cdot (550 + 0.6 \cdot 71)) / (1 \cdot 71)$$

$$= 122.882 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 6.330 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

Minimum Design Metal Temperature Results:



Govrn. thk, tg = 74, tr = 66.5, c = 3 mm., E* = 1
 Thickness Ratio = $tr * E* / (tg - c) = 0.94$, Temp. Reduction = 3 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 25 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 22 °C

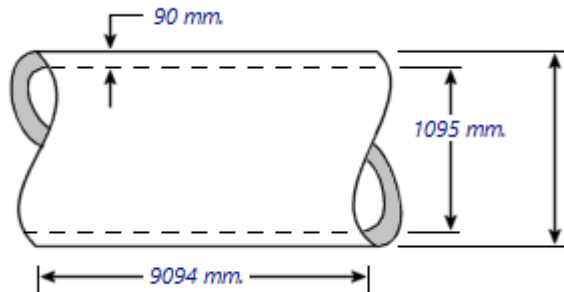
Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	155.487 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-29 °C
Governing Thk.	74.000 mm.	Basic MDMT	25 °C
Stress/Thk. Ratio	0.937	Temperature Difference	3 °C
UCS-66(c) red.	No	Computed Min. Temp.	22 °C
		Joint Efficiency E*	100 %

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Internal Pressure Calculation Results:

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Internal Pressure Results for: MAIN SHELL

Cylindrical Shell Design Information:					
Design Pressure	P	191.094 bars	Int. Design Temperature		200 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		9094.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	1095.000 mm.	Actual Stress		132.4 N./mm ²
Spec. Min. Thk t,e, or sa		90.0000 mm.	Spec. Nominal Thk.		90.000 mm.
Surface Area		364263.0 cm ²	SG of Contents		1.000
Element Volume		6408535.0 cm ³	Weight of Contents		6404.6 Kgf
Empty Weight		23615.4 Kgf	Operating Weight		30020.0 Kgf
			Comp. All. Ope. Stress		141.4 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (191 \cdot 550) / (138 \cdot 1 - 0.6 \cdot 191)$$

$$= 83.1442 + 3.0000 = 86.1442 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.094 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 87) / (550 + 0.6 \cdot 87)$$

$$= 199 - 0.094 = 199 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 90) / (548 + 0.6 \cdot 90)$$

$$= 206 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (191 \cdot (550 + 0.6 \cdot 87)) / (1 \cdot 87)$$

$$= 132.390 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * tnom / Rf * (1 - Rf / Ro)) 7.595 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

Minimum Design Metal Temperature Results:



Govrn. thk, tg = 90, tr = 84.4, c = 3 mm., E* = 1
 Thickness Ratio = $tr * E* / (tg - c) = 0.97$, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 28 °C

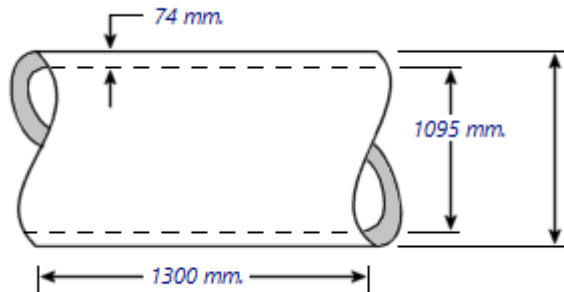
Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	193.709 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-29 °C
Governing Thk.	90.000 mm.	Basic MDMT	29 °C
Stress/Thk. Ratio	0.970	Temperature Difference	2 °C
UCS-66(c) red.	No	Computed Min. Temp.	28 °C
		Joint Efficiency E*	100 %

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Internal Pressure Calculation Results:

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Internal Pressure Results for: CH. SHELL-2

Cylindrical Shell Design Information:					
Design Pressure	P	147.094 bars	Int. Design Temperature	85 °C	
Material		SA-516 70	Ext. Design Temperature	85 °C	
Length		1300.000 mm.	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	1095.000 mm.	Actual Stress	122.9 N./mm ²	
Spec. Min. Thk t,e, or sa		74.0000 mm.	Spec. Nominal Thk.	70.000 mm.	
Surface Area		50765.0 cm ²	SG of Contents	1.000	
Element Volume		1224445.1 cm ³	Weight of Contents	1132.5 Kg	
Empty Weight		2738.2 Kg	Operating Weight	3870.7 Kg	
			Comp. All. Ope. Stress	141.4 N./mm ²	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (147 \cdot 550) / (138 \cdot 1 - 0.6 \cdot 147)$$

$$= 62.6918 + 3.0000 = 65.6918 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.094 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 71) / (550 + 0.6 \cdot 71)$$

$$= 165 - 0.094 = 165 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 74) / (548 + 0.6 \cdot 74)$$

$$= 173 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (147 \cdot (550 + 0.6 \cdot 71)) / (1 \cdot 71)$$

$$= 122.882 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * tnom / Rf * (1 - Rf / Ro)) 6.330 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

Minimum Design Metal Temperature Results:



Govrn. thk, tg = 74, tr = 66.5, c = 3 mm., E* = 1
 Thickness Ratio = $tr * E* / (tg - c) = 0.94$, Temp. Reduction = 3 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 25 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 22 °C

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	155.487 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-29 °C
Governing Thk.	74.000 mm.	Basic MDMT	25 °C
Stress/Thk. Ratio	0.937	Temperature Difference	3 °C
UCS-66(c) red.	No	Computed Min. Temp.	22 °C
		Joint Efficiency E*	100 %

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Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	251.822 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	248.300 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	258.507 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	213.080 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	273.130 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	273.130 bars
App 27-4	= M.A.W.P.	193.709 bars

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	202.134 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	191.100 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	212.577 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	171.036 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	210.210 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	210.210 bars
App 27-4	= M.A.W.P.	155.487 bars

Horizontal Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
CH. SHELL-1	169.0	234.0	0.722	202.24
MAIN SHELL	174.5	234.0	0.746	251.93
CH. SHELL-2	169.0	234.0	0.722	202.24

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
TS OUT NOZ. T2	Nozzle	138.00	138.00	1.000
DRAIN NOZ. D1	Nozzle	138.00	138.00	1.000
SH. IN NOZ. S1	Nozzle	138.00	138.00	1.000
SH. OUT NOZ. S2	Nozzle	138.00	138.00	1.000
TS. IN NOZ. T1	Nozzle	138.00	138.00	1.000
VENT NOZ. V1	Nozzle	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
CH. COVER-1	138.00	138.00	1.000
CH. FLG-1	138.00	138.00	1.000
CH. SHELL-1	138.00	138.00	1.000
MAIN SHELL	138.00	138.00	1.000
CH. SHELL-2	138.00	138.00	1.000
CH. FLG-2	138.00	138.00	1.000
CH. COVER-2	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	92.40	92.40	1.000



Tubesheet Material	138.00	138.00	1.000
--------------------	--------	--------	-------

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
TS OUT NOZ. T2	123.38	225.00	0.548
DRAIN NOZ. D1	98.23	225.00	0.437
SH. IN NOZ. S1	152.27	225.00	0.677
SH. OUT NOZ. S2	152.27	225.00	0.677
TS. IN NOZ. T1	123.38	225.00	0.548
VENT NOZ. V1	98.23	225.00	0.437

Elements Suitable for Test Pressure.

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MDMT Calculations for Stiffening Rings:

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Input Echo, Tubesheet Number 1, Description: FIXED TS

Shell Data:

Main Shell Description: MAIN SHELL

Shell Test Pressure		251.82 bars
Shell Thickness	ts	90.0000 mm.
Shell Internal Corrosion Allowance	cas	0.0000 mm.
Shell External Corrosion Allowance	casext	0.0000 mm.
Inside Diameter of Shell	Ds	1095.000 mm.
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: CH. SHELL-1

Channel Type:		Cylinder
Channel Test Pressure		202.13 bars
Channel Thickness	tc	74.0000 mm.
Channel Internal Corrosion Allowance	cac	0.0000 mm.
Channel External Corrosion Allowance	cacext	0.0000 mm.
Inside Diameter of Channel	Dc	1095.000 mm.
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	468
Tube Wall Thickness	et	3.4100 mm.
Tube Outside Diameter	D	25.4000 mm.
Total Straight Tube Length	Lt	9494.00 mm.
Straight Tube Length (bet. inner tubsht faces) L		9094.00 mm.
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-179
Tube Material UNS Number		K01200
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		92.40 N./mm ²
Tube Allowable Stress At Ambient		92.40 N./mm ²
Tube Yield Stress At design Temperature	Syt	180.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	40.3900 mm.
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	3.0000 mm.
Groove Weld Leg	ag	3.0000 mm.
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	506.085 mm.
Largest Center-to-Center Tube Distance	Ul	0.0000 mm.
Length of Expanded Portion of Tube	ltx	51.0000 mm.
Tube-side pass partition groove depth	hg	0.0000 mm.
Area of Tube Layout	Ap	5976.1 cm ²
Perimeter of Tube Layout	Cp	4005.8 mm.

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf A



Tubesheet Design Metal Temperature T 21.11 °C
 Tubesheet Material SA-350 LF2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number K03011
 Tubesheet Allowable Stress at Temperature S 138.00 N./mm²
 Tubesheet Allowable Stress at Ambient Tt 138.00 N./mm²
 Thickness of Tubesheet h 200.0000 mm.
 Tubesheet Corr. Allowance (Shell side) Cats 0.0000 mm.
 Tubesheet Corr. Allowance (Channel side) Catc 0.0000 mm.
 Tubesheet Outside Diameter A 1275.000 mm.

Area of the Untubed Lanes AL 0.0 cm²

Additional Data for Fixed/Floating Tubesheet Exchangers:

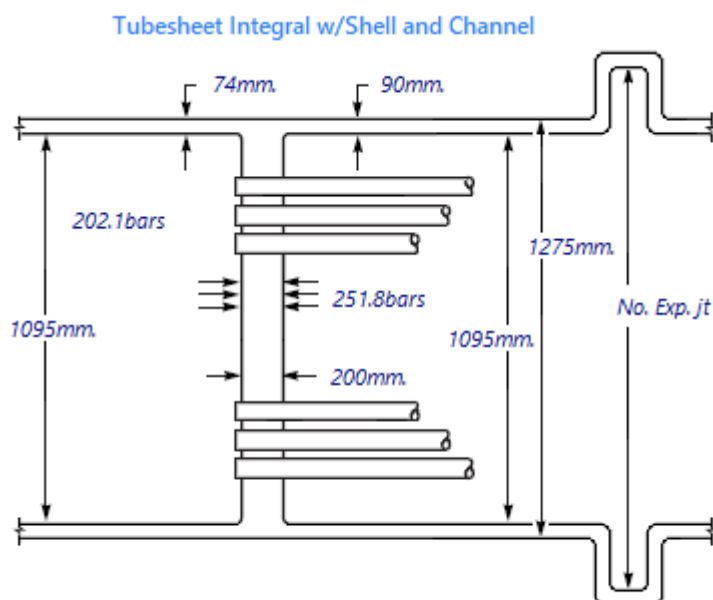
Unsupported Tube Span under consideration l 1608.000 mm.
 Tube End condition corresponding to Span (l) k 1.00

Ignore Radial Thermal Exp. effects (4.18.8/4.18.9) Yes
 Perform Differential Pressure Design No
 Run Multiple Load Cases Yes

Mean Shell Metal Temp. along Shell len. Tsm 21.11 °C
 Mean Tube Metal Temp. along Tube length Ttm 21.11 °C
 Junction Stress Reduction option Perform Plastic Calculation

Tubesheet Integral with Both
 Tubesheet Extended as Flange No

Is Exchanger in Creep range (skip EP, Use 3S for Sps) No



Configuration a:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
 Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len 21.1 °C 0.0000115191 /°C
 Elastic Mod. at Design Temperature 21.1 °C 0.20270E+09 KPa.



Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
 Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
 Tubes - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
 TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.

Please insure these properties are consistent with the type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (202 \cdot (25.4/2 - 0)) / (92.4 \cdot 1 + 0.4 \cdot 202)$
 $= 2.5549 + 0.0000 = 2.5549 \text{ mm.}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-1 at 21.11 °C
 Elastic Modulus for Material 199943392.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
3.4100	25.40	9094.00	7.45	50.0000	0.0198260	95.84

EMAWP = $(2.167 / (D/T) - 0.0833) \cdot B = 199 \text{ bars}$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
4.0567	25.40	9094.00	6.26	50.0000	0.0280591	95.84

EMAWP = $(2.167 / (D/T) - 0.0833) \cdot B = 252 \text{ bars}$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	4.0567 mm.
Allowable Internal Pressure at Corroded thickness	277.93 bars
Required Internal Design Pressure	202.13 bars
Allowable External Pressure at Corroded thickness	198.98 bars
Required External Design Pressure	251.82 bars
Required Thickness due to Shell Side pressure	4.0567 mm.

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME Fixed Tubesheet Calculations for Configuration a:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.8.4 Step 1:

Tube expansion depth ratio [p]:

$$= l_{tx} / h \text{ (modified for corrosion if present)}$$

$$= 51/200$$

$$= 0.2550 \text{ (must be } 0 \leq p \leq 1 \text{)}$$

Effective tube hole diameter [d*]:

$$= \max(dt - 2 \cdot tt(EtT/E) \cdot (StT/S) \cdot (p); dt - 2 \cdot tt)$$

$$= \max(25.4 - 2 \cdot 3.41 (202701232/202701232) \cdot (92.4/138) \cdot (0.25); 25.4 - 2 \cdot 3.41)$$

$$= 24.2356 \text{ mm.}$$

Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot r_o + dt$$

$$= 2 \cdot 506 + 25.4$$

$$= 1037.570 \text{ mm.}$$

Determine the basic ligament efficiency for shear [μ]:

$$= (p - dt) / p$$

$$= (40.4 - 25.4) / 40.4$$

$$= 0.371$$

Equivalent outer tube limit radius [ao]:

$$= Do / 2$$

$$= 1038/2$$

$$= 518.7850 \text{ mm.}$$

Effective tube pitch [p*]:

$$= p / \sqrt{ (1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot Do \cdot p) / (\pi \cdot Do^2)) }$$

$$= 40.4 / \sqrt{ (1 - 4 \cdot \min(0 \cdot 100 ; 4 \cdot 1038 \cdot 40.4) / (\pi \cdot 1038^2)) }$$

$$= 40.3900 \text{ mm.}$$

Effective ligament efficiency for bending [μ*]:

$$= (p* - d*) / p*$$

$$= (40.4 - 24.2) / 40.4$$

$$= 0.4000$$

Ratio [ps]:

$$= as / ao$$

$$= 548/519$$

$$= 1.0554$$

Ratio [pc]:

$$= ac / ao$$

$$= 548/519$$

$$= 1.0554$$

Compute parameter [xt]:

$$= 1 - Nt((dt - 2 \cdot tt) / (2 \cdot ao))^2$$

$$= 1 - 468 ((25.4 - 2 \cdot 3.41) / (2 \cdot 519))^2$$

$$= 0.8499$$

Determine parameter [xs]:

$$= 1 - Nt(dt / (2 \cdot ao))^2$$

$$= 1 - 468 (25.4 / (2 \cdot 519))^2$$

$$= 0.7195$$

Determine the value [h'g]:

$$= \max((hg - CATC); 0) \text{ (For pressure only cases)}$$

$$= \max((0 - 0); 0)$$

$$= 0.000 \text{ mm.}$$

ASME Section VIII-2 Part 4.18.8.4 Step 2:

Axial Shell Stiffness [Ks]:

$$= \pi \cdot ts(D_s + ts) Es / L$$

$$= \pi \cdot 90 (1095 + 90) 202701232 / 9094$$

$$= 761568.0625 \text{ Kg/mm.}$$

Axial Tube Stiffness [Kt]:

$$= \pi \cdot t (D_t - t) E_t / L$$

$$= \pi \cdot 3.41 (25.4 - 3.41) 202701232 / 9094$$

$$= 535.4605 \text{ Kg/mm.}$$

Stiffness Factor [Ks,t]:

$$= K_s / (N_t \cdot K_t)$$

$$= 761568 / (468 \cdot 535)$$

$$= 3.03903$$

Rigidity Ratio [J]:

$$= 1 / (1 + K_s / K_j)$$

$$= 1 / (1 + 761568 / 0)$$

$$= 1.0000 (= 1 \text{ if No Exp. Jt.})$$

Shell coefficient [β_s]:

$$= ((12(1 - \nu_s^2))^{1/4} / \sqrt{(D_s + t_s)t_s})$$

$$= ((12(1 - 0.3^2))^{1/4} / \sqrt{(1095 + 90)90})$$

$$= 0.0056 \text{ 1/mm.}$$

Shell coefficient [ks]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - \nu_s^2))$$

$$= 0.0056 \cdot 202701232 \cdot 90^3 / (6(1 - 0.3^2))$$

$$= 1506490752.0000 \text{ bars} \cdot \text{mm.}^2$$

Shell coefficient [λ_s]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(1095 + 90) \cdot 1506490752 / 200^3 \cdot (1 + 200 \cdot 0.0056 + (200 \cdot 0.0056)^2 / 2)$$

$$= 3659170.7500 \text{ bars}$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2(D_s + t_s)))$$

$$= 1095^2 / (4 \cdot 202701232 \cdot 90) \cdot (1 - 1095 \cdot 0.3 / (2(1095 + 90)))$$

$$= 0.14153\text{E-}01 \text{ mm./N./mm}^2$$

Channel coefficient [β_c]:

$$= ((12(1 - \nu_c^2))^{1/4} / \sqrt{(D_c + t_c)t_c})$$

$$= ((12(1 - 0.3^2))^{1/4} / \sqrt{(1095 + 74)74})$$

$$= 0.0062 \text{ 1/mm.}$$

Channel coefficient [kc]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_c^2))$$

$$= 0.0062 \cdot 202701232 \cdot 74^3 / (6(1 - 0.3^2))$$

$$= 929803968.0000 \text{ bars} \cdot \text{mm.}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (Cm\Delta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot Cq\Delta)$$

$$= 6(1095 + 74 / \cos(0)) \cdot 929803968 / 200^3 \cdot (1 + 200 \cdot 0.0062 \cdot (1 + 1) / 2 + (200 \cdot 0.0062)^2 / 2 \cdot 1)$$

$$= 2445722.0 \text{ bars}$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot \nu_c / (2(D_c + t_c)))$$

$$= 1095^2 / (4 \cdot 202701232 \cdot 74) \cdot (1 - 1095 \cdot 0.3 / (2(1095 + 74)))$$

$$= 0.17175\text{E-}01 \text{ mm./N./mm}^2$$

Channel coefficient [zetac]:

$$zetac = 0$$

ASME Section VIII-2 Part 4.18.8.4 Step 3:

$$h/p = 4.951721 ; \mu^* = 0.399961 ; E^* = 83784328 \text{ KPa.}$$

$$E^*/E = 0.413339 ; \nu^* = 0.310763$$

Note: As h/p (4.952) is > 2, data values for h/p = 2 were used.

Tube Bundle Stiffness Factor [Xa]:

$$= \left(\frac{24(1 - \nu^2) N t \cdot E t \cdot t t (d t - t t) a o^2}{E \cdot L \cdot H^3} \right)^{1/4}$$

$$= \left(\frac{24(1 - 0.31^2) 468 \cdot 202701232 \cdot 3.41}{(25.4 - 3.41) 519^2 / (83784328 \cdot 9094 \cdot 200^3)} \right)^{1/4}$$

$$= 1.6154$$

Values from Table 4.18.3

$$Z_d = 0.347565 ; Z_v = 0.178075 ; Z_m = 0.724769$$

$$Z_a = 0.577057E+00 ; Z_w = 0.178075$$

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Diameter ratio [K]:

$$= A / D_o$$

$$= 1275/1038$$

$$= 1.2288$$

Compute coefficient [F]:

$$= (1 - \nu) / E \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.31) / 83784328 \cdot (3659170 + 2445722 + 202701232 \cdot \ln(1.23))$$

$$= 5.3657$$

Compute parameter [ϕ]:

$$= (1 + \nu) F$$

$$= (1 + 0.31) 5.37$$

$$= 7.0331$$

Compute parameter [Q1]:

$$= (\rho_s - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m)$$

$$= (1.06 - 1 - 7.03 \cdot 0.18) / (1 + 7.03 \cdot 0.72)$$

$$= -0.196325$$

Compute Parameter [Qz1]:

$$= (Z_d + Q1 \cdot Z_w) / 2 \cdot X_a^4$$

$$= (0.35 + -0.2 \cdot 0.18) / 2 \cdot 1.62^4$$

$$= 1.0644$$

Compute parameter [Qz2]:

$$= (Z_v + Q1 \cdot Z_m) / 2 \cdot X_a^4$$

$$= (0.18 + -0.2 \cdot 0.72) / 2 \cdot 1.62^4$$

$$= 0.1218$$

Compute parameter [U]:

$$= (Z_w + (\rho_s - 1) Z_m) X_a^4 / (1 + \phi \cdot Z_m)$$

$$= (0.18 + (1.06 - 1) 0.72) 1.62^4 / (1 + 7.03 \cdot 0.72)$$

$$= 0.2437$$

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor [yb]:

$$= 0 \text{ (config a)}$$

Compute Parameter [y]:

$$= 0.000 \text{ mm. (For Pressure only cases)}$$

Calculate parameter [ws]:

$$= \rho_s \cdot k_s \cdot \beta_s \cdot \delta S (1 + h \cdot \beta_s)$$

$$= 1.06 \cdot 1506490752 \cdot 0.0056 \cdot 0.014152893 (1 + 200 \cdot 0.0056)$$

$$= 26470.7578 \text{ mm.}^2$$

Calculate parameter [$\omega \cdot s$]:

$$= a o^2 (\rho_s^2 - 1) \cdot (\rho_s - 1) / 4 - \omega_s$$

$$= 519^2 (1.06^2 - 1) \cdot (1.06 - 1) / 4 - 26471$$

$$= -26047.0723 \text{ mm.}^2$$

Parameter [ωc]:

$$= \rho_c \cdot k_c [\delta C (C_m \Theta + \beta C + C_q \Delta \cdot h \cdot \beta C^2) - \zeta c (C_m \Theta + h \cdot C_q \Theta \cdot h \cdot \beta C / 2)]$$

$$= 1.06 \cdot 929803968 [0.017175065 (1 + 0.0062 \cdot 1 \cdot 200) -$$

$$0.000 (1/200 + 1 \cdot 0.0062/2))$$

$$= 23293.8086 \text{ mm.}^2$$

Parameter [$\omega \cdot c$]:

$$= a_o^2 [(p_c^2 + 1) \cdot (p_c - 1)/4 - (p_s - 1)/2] - \omega c$$

$$= 519^2 [(1.06^2 + 1) \cdot (1.06 - 1)/4 - (1.06 - 1)/2] - 23294$$

$$= -22870.1211 \text{ mm.}^2$$

Pressure [$P \cdot S$]:

$$= 0 \text{ For Pressure only cases or Configurations d,e,f,A,B,C,D}$$

Pressure [$P \cdot C$]:

$$= 0 \text{ For Pressure only cases or Configurations b,c,d,B,C,D}$$

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Pressure [$P's$]:

$$= P_s \{ x_s + 2(1 - x_s) v_t + [2/K_{st} \cdot (D_s/D_o)^2] v_s - [(p_s^2 - 1)/(J \cdot K_{st})] - [(1 - J)/(2J \cdot K_{st})][(D_j^2 - (D_s)^2)/D_o^2] \}$$

$$= 252 \{ 0.72 + 2(1 - 0.72)0.3 + [2/3.04 \cdot (1095/1038)^2]0.3 - [(1.06^2 - 1)/(1 \cdot 3.04)] - [(1 - 1)/(2 \cdot 1 \cdot 3.04)][(0^2 - (1095)^2)/1038^2] \}$$

$$= 269.5179 \text{ bars}$$

Pressure [$P't$]:

$$= [x_t + 2(1 - x_t) v_t + 1/(J \cdot K_{st})] P_t$$

$$= [0.85 + 2(1 - 0.85)0.3 + 1/(1 \cdot 3.04)] 202$$

$$= 256.5121 \text{ bars}$$

Pressure [P_v]:

$$= N_t \cdot K_t \cdot \gamma / (\pi \cdot a_o^2)$$

$$= 468 \cdot 535 \cdot 0 / (3.14 \cdot 519^2)$$

$$= 0.0000 \text{ bars}$$

Pressure [P_w]:

$$= -\gamma_b \cdot U \cdot W \cdot / (2 \cdot \pi \cdot a_o^2)$$

$$= -0 \cdot 0.24 \cdot 0 / (2 \cdot \pi \cdot 519^2)$$

$$= -0.0000 \text{ bars}$$

Calculate the pressure [P_{Prim}]:

$$= - (U/a_o^2) \cdot (\omega \cdot s \cdot P_s - \omega \cdot c \cdot P_t)$$

$$= - (0.24/519^2) \cdot (-26047 \cdot 252 - -22870 \cdot 202)$$

$$= 1.7533 \text{ bars}$$

Calculate the pressure [P_w]:

$$= (U/a_o^2) \cdot (\omega \cdot s \cdot P_s - \omega \cdot c \cdot P_t)$$

$$= (0.24/519^2) \cdot (26471 \cdot 0 - 23294 \cdot 0)$$

$$= 0.0000 \text{ bars}$$

Determine the effective pressure [P_e]:

$$= J \cdot K_{st} / (1 + J \cdot K_{st} (Q_{z1} + (p_s - 1) Q_{z2})) \cdot (P's - P't + P_v + P_w + P_w + P_{\text{Prim}})$$

$$= 0.1000E+01 \cdot 3.04 / (1 + 1 \cdot 3.04 (1.06 + (1.06 - 1)0.12)) \cdot (270 - 257 + 0 + 0 + -0 + 1.75)$$

$$= 10.5406 \text{ bars}$$

ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor [Q_2]:

$$= [((\omega \cdot s \cdot P_s - \omega \cdot c \cdot P_t) - (\omega \cdot s \cdot P_s - \omega \cdot c \cdot P_t)) + W \cdot \gamma_b / (2 \cdot \pi)] / (1 + \phi \cdot Z_m)$$

$$= [((-26047 \cdot 252 - -22870 \cdot 202) - (26471 \cdot 0 - 23294 \cdot 0)) \cdot 0.01 + 0 \cdot 0 / (2 \cdot \pi)] / (1 + 7.03 \cdot 0.72)$$

$$= -3238.5342 \text{ Kg}$$

Calculate factor [Q_3]:

$$= Q_1 + 2 \cdot Q_2 / (P_e \cdot a_o^2)$$

$$= -0.2 + 2 \cdot -3239 / (10.5 \cdot 519^2)$$

$$= -0.42022$$

Fm Value from Table 4.18.3 = 0.210110

Tubesheet Bending Stress - original thickness [σ]:

$$= (1.5 \cdot F_m / \mu) \cdot (2 \cdot a_o / (H - h'g))^2 \cdot P_e$$

$$= (1.5 \cdot 0.21 / 0.4) \cdot (2 \cdot 519 / (200 - 0))^2 \cdot 10.5$$

$$= 22.3555 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 \cdot \beta T \cdot S_y = 1.5 \cdot 0.95 \cdot 250 = 356 \text{ N./mm}^2, 4.18.233$$

Tubesheet Bending Stress - final thickness [σ f]:

$$= (1.5 \cdot F_m / \mu) \cdot (2 \cdot a_o / (h - h'g))^2 \cdot P_e$$

$$= (1.5 \cdot 0.015 / 0.4) \cdot (2 \cdot 519 / (5.11 - 0))^2 \cdot 1.53$$

$$= 356.2330 \text{ N./mm}^2$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$= h + C_{ats} + C_{atc}$$

$$= 5.11 + 0 + 0$$

$$= 5.1103 \text{ mm.}$$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [τ limit]:

$$= (2 \cdot \mu \cdot h / a_o) \cdot (0.8 \cdot \beta T \cdot S_y)$$

$$= (2 \cdot 0.37 \cdot 200 / 519) \cdot 190$$

$$= 54.3683 \text{ N./mm}^2$$

Shear Stress is not required to be computed; [P_e] <= τ limit

Tubesheet Shear Stress is probably low, use the following req. thk:

$$\text{Tubesheet thickness (Incl. Corr.)} = 3.8100 \text{ mm.}$$

$$\text{Tubesheet Shear Stress} = 13.0920 \text{ N./mm}^2$$

Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd] :

$$= \max(H_{reqB}; H_{reqS})$$

$$= \max(5.11; 3.81)$$

$$= 5.1103 \text{ mm.}$$

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The Ftmin and Ftmax Coefficients from Table 4.18.4:

$$F_{tmin} = 0.93, F_{tmax} = 1.03$$

First Extreme Tube Axial Stress from among all the tubes [σ t1]:

$$= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmin}) / (X_t - X_s)$$

$$= ((252 \cdot 0.72 - 202 \cdot 0.85) - (10.5) \cdot 0.93) /$$

$$(0.85 - 0.72)$$

$$= -0.3012 \text{ N./mm}^2$$

Second Extreme value of Tube Axial Stress from among all the tubes [σ t2]:

$$= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmax}) / (X_t - X_s)$$

$$= ((252 \cdot 0.72 - 202 \cdot 0.85) - (10.5) \cdot 1.03) /$$

$$(0.85 - 0.72)$$

$$= -1.1594 \text{ N./mm}^2$$

Maximum Tube Axial Stress [σ t,max]:

$$= \max(|\sigma_{t,1}|; |\sigma_{t,2}|) = 1.16 \text{ N./mm}^2$$

Allowable Tube Stress [σ t,A]:

$$= \beta T \cdot S_{y,t} = 170.9969 \text{ N./mm}^2$$

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:

$$= \max((3.25 - 0.25(Z_d + Q_3 \cdot Z_w)X_a^4); 1.25)$$

$$= \max((3.25 - 0.25(0.35 + -0.42 \cdot 0.18)1.62^4); 1.25)$$

$$= 2.0000 \text{ (Should be } \leq 2)$$

Determine the factor [rt]:

$$= (\sqrt{(dt^2 + (dt - 2 \cdot tt)^2)})/4$$

$$= (\sqrt{(25.4^2 + (25.4 - 2 \cdot 3.41)^2)})/4$$

$$= 7.8676 \text{ mm.}$$

Determine the factor [Ct]:

$$= \sqrt{(2 \cdot \pi^2 \cdot Et/Syt)}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 202701232/180)}$$

$$= 157.1179$$

Determine the factor [Ft]:

$$= k \cdot L/r$$

$$= 1 \cdot 1608/7.87$$

$$= 204.3835$$

Buckling Allowable Stress [Stb]:

$$= (1/Fs) \cdot (\pi^2 \cdot Et/Ft^2) \text{ (Ct} \leq \text{Ft)}$$

$$= (1/2) \cdot (\pi^2 \cdot 202701232/204^2)$$

$$= 170.997 \text{ N./mm}^2 \text{ (Never greater than Sot)}$$

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:

$$= \sigma_{t,max} \cdot \text{Tube Area}$$

$$= 1.16 \cdot 2.36$$

$$= 27.85 \text{ Kg}$$

Tube Metal Area [At]:

$$= \pi (do - t) t$$

$$= \pi (25.4 - 3.41) 3.41$$

$$= 2.356 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (do - t) S$$

$$= \pi \cdot 3.41 (25.4 - 3.41) 171$$

$$= 4107.609 \text{ Kg}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot ag (do + 0.67 \cdot ag) Sw \text{ (but not } > \text{ Ft)}$$

$$= 0.85 \cdot \pi \cdot 3 (25.4 + 0.67 \cdot 3) 171$$

$$= 3828.7654 \text{ Kg}$$

Weld Strength Factor [fw]:

$$= S / Sw \text{ where } Sw = \min(S \text{ and } St)$$

$$= 171/171$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot af (do + 0.67 \cdot af) Sw \text{ (but not } > \text{ Ft)}$$

$$= 0.55 \cdot \pi \cdot 3 (25.4 + 0.67 \cdot 3) 171$$

$$= 2477.4363 \text{ Kg}$$

Minimum Length of Weld [ar]:

$$= 2(\sqrt{((0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t)fw \cdot fd)} - 0.75 \cdot do)$$

$$= 2(\sqrt{((0.75 \cdot 25.4)^2 + 1.07 \cdot 3.41 \cdot (25.4 - 3.41)1 \cdot 1)} - 0.75 \cdot 25.4)$$

$$= 4.002 \text{ mm.}$$

Minimum Required Fillet Weld Leg [afr]: 2.0460 mm.

Minimum Required Groove Weld Leg [agr]: 2.0460 mm.

Tube-Tubesheet Jt allowable, 4108 is >= tube strength 4108 Kg

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:
 $= \beta T \cdot S_{y,s} = 0.95 \cdot 260 = 370 \text{ N./mm}^2$

Axial Membrane Stress in Shell $[\sigma_s, m]$:
 $= a o^2 / ((D_s + t_s) t_s) \cdot [P_e + (p_s^2 - 1)(P_s - P_t)] + a s^2 \cdot P_t / ((D_s + t_s) t_s)$
 $= 519^2 / ((1095 + 90) 90) \cdot [10.5 + (1.06^2 - 1)(252 - 202)] + 548^2 \cdot 202 / ((1095 + 90) 90)$
 $= 60.9028 \text{ N./mm}^2$

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Note:
 For a given Shell thickness of 90 mm., the minimum Shell length adjacent to the tubesheet should be 565.068 mm.

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	22.4 <=	356.2 N./mm ²	Pass
Tubesheet Shear Stress	2.1 <=	190.0 N./mm ²	Pass
Maximum Tube Stress	1.2 <=	171.0 N./mm ²	Pass
Minimum Tube Stress (Buckling)	-1.2 <=	-171.0 N./mm ²	Pass
Maximum Force on any one Tube	27.8 <=	4107.6 Kg	Pass
Axial Membrane Stress in Shell	60.9 <=	370.5 N./mm ²	Pass

Thickness Results for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm.)	Required	Actual	P/F
Tubesheet Thickness :	5.110	200.000	Ok
Tube-Tubesheet Fillet Weld Leg :	2.046	3.000	Ok
Tube-Tubesheet Groove Weld Leg :	2.046	3.000	Ok

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Stresses				Case Type	Pass/Fail
		Bend	Allwd	Shear	Allwd			
D1uc	152.129 ...	213	356	37	190	Ps+Pt-Th	D1	Ok
D2uc	159.373 ...	235	356	40	190	Ps+Pt-Th	D2	Ok
D3uc	5.110 ...	22	356	2	190	Ps+Pt-Th	D3	Ok
Max:	159.3729 ... mm.	0.660	0.208	(Str. Ratio)				

Load Case Definitions:

[Ps & Pt]:
 Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[(+)-Th]:
 With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:

Operating Load Cases using the Maximum and Minimum Operating Pressures and Operating Temperatures

Shell Axial Membrane Stress Summary:

---- Shell Stresses ----					:	---- Shell Band Stress ----					:	Pass
Case#	Ten	Allwd	Cmp	Allwd	:	Ten	Allwd	Cmp	Allwd	:	Fail	
D1uc	4	370	...	...	:	...	...	...	...	:		Ok
D2uc	57	370	...	...	:	...	...	...	...	:		Ok
D3uc	61	370	...	...	:	...	...	...	...	:		Ok
Max RATIO	0.164		...		:	...		...		:		

Tube, Shell and Channel Stress Summary:

---- Tube Stresses ----					Tube Loads		Shell Stress		Channel Stress		Pass
Case#	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	Fail
D1uc	18	170	...	-170	423	4107	...	370	...	370	Ok
D2uc	18	170	-18	-170	430	4107	...	370	...	370	Ok
D3uc	1	170	-1	-170	28	4107	...	370	...	370	Ok
Max RATIO	0.105		0.105		0.105		...		...		

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm.)	Required	Actual	P/F
Tubesheet thickness :	159.373	200.000	Ok
Tube thickness :	4.057	3.410	Fail
Tube-tubesheet fillet weld leg :	2.046	3.000	Ok
Tube-tubesheet groove weld leg :	2.046	3.000	Ok

Min Shell length of thk, (90.000) adj. to tubesheet: 565.068 mm.

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Extends by : +.000 mm.
 Due to Pressure Shell Compresses by : -1.168 mm.
 Due to Pressure + Thermal Shell Compresses by : -1.168 mm.

Summary of Exchanger Pressure and Thermal Displacements: (mm.)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.1665	...	...	...	...	...
2	-1.1678	...	...	...	...	...
3	-1.0012	...	...	...	...	...
4	...	...	...	...	...	...
5	...	...	...	...	...	...
6	...	...	...	...	...	...
7	...	...	...	...	...	...
8	...	...	...	...	...	...

External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm.	Outside Diameter mm.	Corroded Thickness mm.	Factor A	Factor B N./mm ²
10	20	No Calc	...	287	No Calc	No Calc
20	30	No Calc	...	307	No Calc	No Calc
30	Tsht	1299.97	1243	71	0.018602	122.731
Tsht	40	No Calc	...	71	No Calc	No Calc
40	Tsht	9094.01	1275	87	0.0054322	120.315
Tsht	50	No Calc	...	87	No Calc	No Calc
50	60	1300.02	1243	71	0.018601	122.731
60	70	No Calc	...	307	No Calc	No Calc
70	80	No Calc	...	287	No Calc	No Calc

External Pressure Calculations:

From	To	External Actual T. mm.	External Required T. mm.	External Design Pressure bars	External M.A.W.P. bars
10	20	290	157.546	...	No Calc
20	30	310	301.473	...	No Calc
30	Tsht	74	No Calc	...	93.5
Tsht	40	74	No Calc	...	No Calc
40	Tsht	90	No Calc	...	109
Tsht	50	90	No Calc	...	No Calc
50	60	74	No Calc	...	93.5
60	70	310	301.473	...	No Calc
70	80	290	213.618	...	No Calc

Minimum

93.5

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm.	Allowable Length Bet. Stiffeners mm.	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	No Calc	No Calc	No Calc	No Calc
30	Tsht	1299.97	No Calc	No Calc	No Calc
Tsht	40	No Calc	No Calc	No Calc	No Calc
40	Tsht	9094.01	No Calc	No Calc	No Calc
Tsht	50	No Calc	No Calc	No Calc	No Calc
50	60	1300.02	No Calc	No Calc	No Calc
60	70	No Calc	No Calc	No Calc	No Calc
70	80	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm ³	Corroded Metal Wgt. kg.	Corroded ID Volume cm ³	Extra due Misc % kg.
10	20	4736.37	...	4736.37	...	473.637
20	30	3052.77	399740	3019	400889	305.277
30	40	2738.23	1224445	2633.96	1237900	273.823
40	50	23615.4	6408535	22886	6502660	2361.54
50	60	2738.23	1224445	2633.96	1237900	273.823
60	70	3052.77	399740	3019	400889	305.277
70	80	4736.37	...	4736.37	...	473.637
Total		44670	9656907.00	43664	9780240.00	4467



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm.	Y Offset, Dtl. Cent. mm.	Z Offset Dtl. Cent. mm.	Description
20	Liqd	369.709	210.5	70	...	Liquid: 20
30	Liqd	1132.46	650	70	...	Liquid: 30
30	Nozl	983.335	550	750.7	...	TS OUT NOZ. T2
30	Nozl	30.0494	600	-568.925	...	DRAIN NOZ. D1
40	Sadl	375.078	1900	870.75	...	FIXED SADDLE
40	Sadl	375.078	7194	870.75	...	SLIDING SADDLE
40	Liqd	6404.62	4547	70	...	Liquid: 40
40	Insl	288.869	4547	...	...	HOT
40	Nozl	2628.6	900	852.3	...	SH. IN NOZ. S1
40	Nozl	2628.6	8194	-852.3	...	SH. OUT NOZ. S2
40	Wght	8685	4547	...	...	datasheet weigh
50	Liqd	1132.46	650	70	...	Liquid: 50
50	Nozl	997.718	750	-750.7	...	TS. IN NOZ. T1
50	Nozl	17.9887	600	568.925	...	VENT NOZ. V1
60	Liqd	369.709	210.5	70	...	Liquid: 60
30	FTsh	1772.67	1400	...	...	FIXED TS
30	Tube	8923.75	6047	...	...	Tubes
30	RTsh	1772.67	10694	...	...	Tubesheet
40	Baf1	39.7098	1010	...	...	Baffles
40	Baf1	39.7098	1457	...	...	Baffles
40	Baf1	39.7098	1904	...	...	Baffles
40	Baf1	39.7098	2351	...	...	Baffles
40	Baf1	39.7098	2798	...	...	Baffles
40	Baf1	39.7098	3245	...	...	Baffles
40	Baf1	39.7098	3692	...	...	Baffles
40	Baf1	39.7098	4139	...	...	Baffles
40	Baf1	39.7098	4586	...	...	Baffles
40	Baf1	39.7098	5033	...	...	Baffles
40	Baf1	39.7098	5480	...	...	Baffles
40	Baf1	39.7098	5927	...	...	Baffles
40	Baf1	39.7098	6374	...	...	Baffles
40	Baf1	39.7098	6821	...	...	Baffles
40	Baf1	39.7098	7268	...	...	Baffles
40	Baf1	39.7098	7715	...	...	Baffles
40	Baf1	39.7098	8162	...	...	Baffles
40	Baf1	39.7098	8609	...	...	Baffles

Total Weight of Each Detail Type:

Saddles 750.2
 Liquid 9409.0



Insulation	288.9
Nozzles	7286.3
Weights	8685.0
Exchanger Components	13183.9
Liquid in Tubes	1197.6

Sum of the Detail Weights 40800.8 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	49137.1	49137.1	49137.1	49137.1	49137.1	49137.1
Saddles	750.2	750.2	750.2	750.2	750.2	750.2
Nozzles	7286.3	7286.3	7286.3	7286.3	7286.3	7286.3
Exchanger	13183.9	13183.9	13183.9	13183.9	13183.9	13183.9
Insulation	...	...	288.9	288.9	288.9	288.9
Empty Weights	...	...	...	8685.0	8685.0	...
Test Weights	...	8685.0	...	...	...	...
Ope Weights	...	...	...	...	...	8685.0
Ope. Liquid	...	...	...	...	...	9409.0
Tube Ope Lqd	...	...	...	...	...	1197.6
Test Liquid	...	9651.0	...	...	...	...
Tube Tst Lqd	...	1204.2	...	...	...	...
Totals	70357.5	89897.7	70646.3	79331.3	79331.3	89937.9

Field Installation Options:

Miscellaneous Weight Percent: 10.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	70357.5 kg.
Shop Test Wt.	- Fabricated weight + water (full)	89897.7 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	70646.3 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	79331.3 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	79331.3 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	89937.9 kg.
Oper. Wt. + CA	- Corr wt. + operating liquid	88831.9 kg.

Exchanger Tube Data

Volume of Exchanger tubes :	1204908.9 cm ³
Weight of Ope Liq in tubes :	1197.6 kg.
Weight of Water in tubes :	1204.2 kg.

The Corroded Weight and thickness are used in the Horizontal Vessel Analysis (Ope Case) and Earthquake Load Calculations.

Field Test weight is computed in the corroded condition.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	35995.8	...
20	30	32019.1	14482.6
30	40	50765	44720.6
40	50	364263	312838
50	60	50765	44720.6
60	70	32019.1	14482.6
70	80	35995.8	...

Total 601822.7 431243.9 cm²



Total	647.8	464.2 ft ²
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Nozzle Flange MAWP Results: (bars & °C)

Nozzle Description	Flange Ope.	Rating Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	- - - UG-44	Max Pressure 50%	DNV
TS OUT NOZ. T2	238.12	255.10	85	1500	GR 1.1	127.787	81.773	38.123	29.397
DRAIN NOZ. D1	238.12	255.10	85	1500	GR 1.1	...	...	...	...
SH. IN NOZ. S1	219.10	255.10	200	1500	GR 1.1	85.192	99.640	19.101	19.101
SH. OUT NOZ. S	219.10	255.10	200	1500	GR 1.1	84.052	00.779	19.101	19.101
TS. IN NOZ. T1	238.12	255.10	85	1500	GR 1.1	127.787	81.773	38.123	29.397
VENT NOZ. V1	238.12	255.10	85	1500	GR 1.1	...	...	...	...

Shellside Flange Rating

Lowest Flange Pressure Rating was (Ope)[ShellSide]: 199.640 bars
 Lowest Flange Pressure Rating was (Amb)[ShellSide]: 255.100 bars

Channelside Flange Rating

Lowest Flange Pressure Rating was (Ope)[TubeSide]: 181.773 bars
 Lowest Flange Pressure Rating was (Amb)[TubeSide]: 255.100 bars

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm.	Fm Value	Resolved Moment ME Kg-m.	Axial Force FE KgF	MAWP PD bars	Maximum Pressure Pmax bars	Pass Fail
TS OUT NOZ. T2	431.800	0.30	19191	13053	155.487	181.773	Pass
SH. IN NOZ. S1	647.700	0.30	43180	19580	193.709	199.640	Pass
SH. OUT NOZ. S2	647.700	0.30	42560	19580	193.709	200.779	Pass
TS. IN NOZ. T1	431.800	0.30	19191	13053	155.487	181.773	Pass

Evaluating Flange on Nozzle [TS OUT NOZ. T2]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(19191) + 4 \cdot \text{abs}(13053) \cdot 431.800 \leq \pi \cdot 431.800^3 \cdot ((238.123 - 155.487) + .300 \cdot 238.123)$$

$$= 0.32960E+06 \leq 0.39740E+06 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle TS OUT NOZ. T2 [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(238.123, 0.300 \cdot 238.123 + 238.123 - 329602.2 / (\pi \cdot 431.800^3))$$

$$= \min(238.123, 181.773)$$

$$= 181.773 \text{ bars}$$

Evaluating Flange on Nozzle [SH. IN NOZ. S1]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(43180) + 4 \cdot \text{abs}(19580) \cdot 647.700 \leq \pi \cdot 647.700^3 \cdot ((219.101 - 193.709) + .300 \cdot 219.101)$$

$$= 0.74161E+06 \leq 0.79323E+06 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle SH. IN NOZ. S1 [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(219.101, 0.300 \cdot 219.101 + 219.101 - 741606.8 / (\pi \cdot 647.700^3))$$

$$= \min(219.101, 199.640)$$

$$= 199.640 \text{ bars}$$

Evaluating Flange on Nozzle [SH. OUT NOZ. S2]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(42560) + 4 \cdot \text{abs}(19580) \cdot 647.700 \leq \pi \cdot 647.700^3 \cdot ((219.101 - 193.709) + .300 \cdot 219.101)$$

$$= 0.73169E+06 \leq 0.79323E+06 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle SH. OUT NOZ. S2 [Pmax]:
 $= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$
 $= \min(219.101, 0.300 \cdot 219.101 + 219.101 - 731689.5 / (\pi \cdot 647.700^3))$
 $= \min(219.101, 200.779)$
 $= 200.779 \text{ bars}$

Evaluating Flange on Nozzle [TS. IN NOZ. T1]:
 $= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$
 $= 16(19191) + 4 \cdot \text{abs}(13053) \cdot 431.800 \leq \pi \cdot 431.800^3 [(238.123 - 155.487) + 0.300 \cdot 238.123]$
 $= 0.32960E+06 \leq 0.39740E+06 \text{ Check Passes}$

Maximum Pressure for Flange on Nozzle TS. IN NOZ. T1 [Pmax]:
 $= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$
 $= \min(238.123, 0.300 \cdot 238.123 + 238.123 - 329602.2 / (\pi \cdot 431.800^3))$
 $= \min(238.123, 181.773)$
 $= 181.773 \text{ bars}$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Large Diameter Flange Pressure Ratings are per ASME B16.47 2025 Edition

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-22
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	[V] 180 Km/hr
Surface Roughness Category	D: Flat, unobstructed
Type of Surface	Moderately Smooth
Base Elevation	300 mm.
Percent Wind for Hydrotest	33.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm.
Distance Upwind of Crest Lh	0 mm.
Distance from Crest to the Vessel x	0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot gV \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.19 \cdot 0.98) / (1 + 1.7 \cdot 3.4 \cdot 0.19)))$
 $= \min(0.85, 0.92)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	33.000 Hz
Natural Frequency of Vessel (Empty)	33.000 Hz
Natural Frequency of Vessel (Test)	33.000 Hz

User Entered Importance Factor is	1.000
Force Coefficient	[Cf] 0.619
Structure Height to Diameter ratio	10.495

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Zg]$:
 Exposure Category: D from Table 26.11-1
 $\alpha = 11.5 : Zg = 589788 \text{ mm.}$

Effective Height $[z]$:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 1300 + 300 = 1600 \text{ mm.}$
 $= 5.25 \text{ ft. Imperial Units}$

Velocity Pressure coefficient evaluated at height z $[Kz]$:
 Because $z (5.25 \text{ ft.}) < 15 \text{ ft.}$
 $= 2.41(15 / Zg)^{2/\alpha}$
 $= 2.41(15/1935)^{2/11.5}$
 $= 1.04$

Type of Hill: No Hill

Wind Directionality Factor $[Kd]$:
 $= 1, \text{ per Table 26.6-1}$

As there is No Hill Present: $[Kzt]$:
 $K1 = 0, K2 = 0, K3 = 0$

Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0 \cdot 0 \cdot 0)^2$$

$$= 1$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 1.04 \cdot 1 \cdot 1 \cdot 1 \cdot 112^2)$$

$$= 33.1 \text{ psf } [162 \text{ Kgs/m}^2]$$

Force on the first element [F]:

$$= qz \cdot G \cdot Cf \cdot \text{WindArea}$$

$$= 33.1 \cdot 0.85 \cdot 0.62 \cdot 6.14$$

$$= 107 \text{ lbs. } [48.6 \text{ Kgf}]$$

Element	Hgt (z) mm.	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
CH. COVER-1	1600.0	0.000	0.000	0.000	1.035	1.000	161.849
CH. FLG-1	1600.0	0.000	0.000	0.000	1.035	1.000	161.849
CH. SHELL-1	1600.0	0.000	0.000	0.000	1.035	1.000	161.849
MAIN SHELL	1600.0	0.000	0.000	0.000	1.035	1.000	161.849
CH. SHELL-2	1600.0	0.000	0.000	0.000	1.035	1.000	161.849
CH. FLG-2	1600.0	0.000	0.000	0.000	1.035	1.000	161.849
CH. COVER-2	1600.0	0.000	0.000	0.000	1.035	1.000	161.849

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation mm.	Pressure Kgs/m ²	Force Kgf
datasheet wei	0.00	1600.00	161.85	0.00

Wind Load Calculation:

From	To	Wind Height mm.	Wind Diameter mm.	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	1600	1965.6	5700.24	161.849	53.4318
20	30	1600	1314	5531.94	161.849	51.8543
30	40	1600	1491.6	19390.8	161.849	181.762
40	50	1600	1626	147868	161.849	1386.06
50	60	1600	1491.6	19390.8	161.849	181.762
60	70	1600	1314	5531.94	161.849	51.8543
70	80	1600	1965.6	5700.24	161.849	53.4318

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code		ASCE/SEI 7-22
Importance Factor		1.500
Force Modification Factor R		3.000
Site Class		D
Component Elevation Ratio	z/h	0.000
Amplification Factor	Ap	0.000
Force Factor		0.000
Consider Vertical Acceleration		Yes
Minimum Acceleration Multiplier		0.000
User Value of Sds (used if > 0)		0.000
User Value of Sd1 (used if > 0)		0.000
Component resonance duct. fact.	[CAR]	0.000
Component strength factor	[Rpo]	0.000
Overstrength factor	[Omega0]	0.000
Response Acceleration Factor	[SMS]	0.326
Response Acceleration Factor	[SM1]	0.139
Building Importance Factor	[Ie]	1.000
Site Latitude		0.000 deg
Site Longitude		0.000 deg

Seismic Analysis Results:

$$SDS = 2/3 * Sms = 2/3 * 0.33 = 0.22$$

$$SD1 = 2/3 * Sm1 = 2/3 * 0.14 = 0.093$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 = $C_t \cdot h_{nx}$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 = $0.020(6.950.75)$
 = 0.086 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 = $(1/\text{Natural Frequency}) = (1/33)$
 = 0.030

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 = Minimum Value of $(1.7 \cdot 0.086, 0.03)$ per 12.8.2
 = 0.030

As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:
 = $0.3 \cdot Sds \cdot W \cdot I$
 = $0.3 \cdot 0.22 \cdot 88082 \cdot 1.5$
 = 8614.398 Kgf

Vertical load per 12.4-4, [Ev]:
 = $0.2 \cdot Sds \cdot W$
 = $0.2 \cdot 0.22 \cdot 88082$
 = 3828.6 Kgf

Earthquake Load Calculation:

From	To	Earthquake Height mm.	Earthquake Weight Kgf	Element Ope Load Kgf
10	20	819	9786.86	1289.68
20	30	547.5	9786.86	862.148
30	40	547.5	9786.86	862.148
40	Sad1	547.5	9786.86	862.148
Sad1	50	547.5	9786.86	862.148
40	50	547.5	9786.86	862.148



50	60	547.5	9786.86	862.148
60	70	547.5	9786.86	862.148
70	80	819	9786.86	1289.68

The Earthquake Weight does not include the weight of the saddles.

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Shop/Field Installation Options :

Insulation is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	6761.176 mm.
Center of Gravity of the Liquid	6761.351 mm.
Center of Gravity of the Insulation	6761.175 mm.
Center of Gravity of the Nozzles	6762.764 mm.
Center of Gravity of the Added Weights (Operating)	6761.175 mm.
Center of Gravity of the Added Weights (Empty)	6761.175 mm.
Center of Gravity of the Tubesheet(s)	6761.175 mm.
Center of Gravity of the Tubes	6761.175 mm.
Center of Gravity of the Baffles	7023.675 mm.
Center of Gravity of Bare Shell New and Cold	6761.959 mm.
Center of Gravity of Bare Shell Corroded	6761.974 mm.
Vessel CG in the Operating Condition	6763.868 mm.
Vessel CG in the Fabricated (Shop/Empty) Condition	6764.172 mm.
Vessel CG in the Test Condition	6763.832 mm.

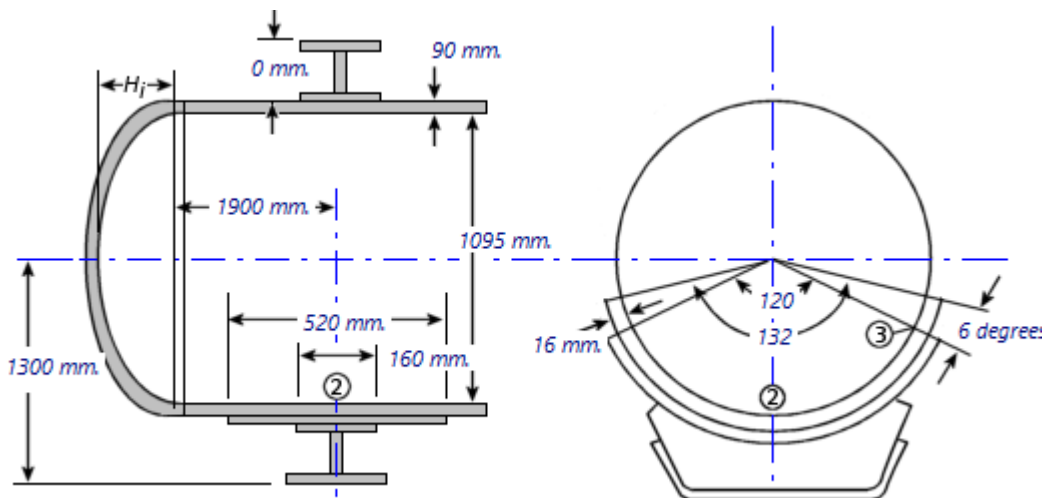
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Input and Calculated Values:

Vessel Mean Radius	Rm	594.00	mm.
Shell Thickness used in this Case	t	90.000	mm.
Stiffened Vessel Length per 4.15.6	L	9094.00	mm.
Distance from Saddle to Vessel tangent	a1 or a	1900.00	mm.
Saddle Width	b1 or b	160.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	520.00	mm.
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	16.0	mm.
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1300.00	mm.
Coefficient of Friction	mu	0.60	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case	60750.69	Kgf
Pressure used in Analysis	P	191.040 bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	62.79	138.00
Long. Stress at Bottom of Midspan	67.65	138.00
Long. Stress at Top of Saddles	87.10	138.00
Long. Stress at Bottom of Saddles	53.09	138.00
Tangential Shear in Shell	7.86	110.40
Circ. Stress at Horn of Saddle	7.26	172.50
Circ. Compressive Stress in Shell	0.85	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:



$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (1960/2 + 0) \cdot 1300/1132$$

$$= 3376.9 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(327, 0) \cdot 1300/5294$$

$$= 80.4 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(8614, 0) \cdot 1300/5294$$

$$= 2115.4 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (8614/2 + 0) \cdot 1300/1132$$

$$= 14840.7 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 45910 + \max(80.4, 3377, 2115, 14841)$$

$$= 60750.7 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1587 (0 + \pi/4(1.28 \cdot 1.2)^2)$$

$$= 3209.902 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	61125.77	Kgf
Transverse Shear Load	4307.20	Kgf
Longitudinal Shear Load	8614.40	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot Rm)$$

$$= \min(2(1095/2 + 90 + 16) \sin(120/2), 2 \cdot 594)$$

$$= 1131.895 \text{ mm.}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $Rm/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$$

$$= -60751 \cdot 1900 [1 - (1 - 1900/9094 + (594^2 - 0^2)/(2 \cdot 1900 \cdot 9094)) / (1 + (4 \cdot 0)/(3 \cdot 9094))]$$

$$= -22937.8 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 ((1 + 2(Rm^2 - hm^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 60751 \cdot 9094/4 ((1 + 2(594^2 - 0^2)/(9094^2)) / (1 + (4 \cdot 0)/(3 \cdot 9094)) - 4 \cdot 1900/9094)$$

$$= 23869.4 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 t)$$

$$= 191 \cdot 594/(2 \cdot 87) - 23869/(\pi \cdot 594^2 \cdot 87)$$

$$= 62.79 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t)$$

$$= 191 \cdot 594 / (2 \cdot 87) + 23869 / (\pi \cdot 594^2 \cdot 87)$$

$$= 67.65 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 191 \cdot 594 / (2 \cdot 87) - 22938 / (0.11 \cdot \pi \cdot 594^2 \cdot 87)$$

$$= 87.10 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 191 \cdot 594 / (2 \cdot 87) + 22938 / (0.19 \cdot \pi \cdot 594^2 \cdot 87)$$

$$= 53.09 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 60751 \cdot (9094 - 2 \cdot 1900) / (9094 + (4 \cdot 0 / 3))$$

$$= 35365.5 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 35366 / (594 \cdot 87)$$

$$= 7.86 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(594 \cdot 87)}$$

$$= 177.316 \text{ mm.}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(160 + 1.56 \cdot \sqrt{(594 \cdot 87)}, 2 \cdot 1900)$$

$$= 514.63 \text{ mm.}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r / S, 1)$$

$$= \min(138 / 138, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 60751 \cdot 0.1 / (515(87 + 1 \cdot 16))$$

$$= -0.85 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -60751 / (4(87 + 1 \cdot 16)515) -$$

$$3 \cdot 0.053 \cdot 60751 / (2(87 + 1 \cdot 16)^2)$$

$$= -7.26 \text{ N./mm}^2$$

Distance between Saddle Supports [Ls]:

$$= 5294.0 \text{ mm.}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot L_s(\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.12750 \text{E-04} \cdot 5294(200 - 21.1)$$

$$= 12.075 \text{ mm.}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1180.0000	mm.
Baseplate Thickness	Bpthk	32.0000	mm.
Baseplate Width	Bpwid	200.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	16.0000	mm.
Web Thickness	Webtk	16.0000	mm.
Web Location	Webloc	Side	
Saddle Design Temperature		200.0	°C
Saddle Yield Stress	Sy	225.0	N./mm ²
Height of Web at Center	Hw,c	306.0	mm.
Friction Coefficient	mu	0.600	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 $= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$
 $= ((1.56 \cdot \sqrt{(550 \cdot 87)} + 520)^3) / 12$
 $= 4726.947 \text{ cm}^4$

Resolved Inertia for the Shell [I,shell]:
 $= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$
 $= 4727 + 749 (128 - 43.5)^2$
 $= 58787.812 \text{ cm}^4$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	861.4	87.0	43.5	749.4	3259966.5	0.588E+05
Wearplate	520.0	16.0	95.0	83.2	790399.9	948.
Web	16.0	614.5	410.2	98.3	4033577.5	0.109E+06
BasePlate	200.0	32.0	733.5	64.0	4694400.0	0.234E+06
Totals	...	...	...	994.9	12778344.0	0.403E+06

Distance to Centroid [C1]:
 $= AY / A$
 $= 12778344 / 995$
 $= 128.434 \text{ mm.}$

Angle [β]:
 $= 180 - \text{Saddle Angle} / 2$
 $= 180 - 120 / 2$
 $= 120.0$

Saddle Splitting Coefficient [K1]:
 $= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$
 $= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$
 $= 0.2035$

Saddle Splitting Force [Fh]:
 $= K1 \cdot Q$
 $= 0.2 \cdot 60751$
 $= 12364.0820 \text{ Kgf}$

Tension Stress, St = (Fh/As) = 4.9386 N./mm²
 Allowed Stress, Sa = 0.6 * Yield Str = 134.9988 N./mm²

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$
 $= 1300 - 550 \cdot \sin(120/2) / 1.05$
 $= 844.740 \text{ mm.}$

Bending Moment, M = Fh * d = 10444.6426 Kg-m.

Bending Stress, Sb = (M * C1 / I) = 3.2633 N./mm²
 Allowed Stress, Sa = 2/3 * Yield Str = 149.9987 N./mm²

Minimum Thickness of Baseplate per Moss:

$= \sqrt[3]{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$
 $= \sqrt[3]{(3(60751 + 375)200 / (2 \cdot 1180 \cdot 150))}$
 $= 31.875 \text{ mm.}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 \cdot \cos(90 - 120/2) (548 + 90 + 16)$
 $= 1131.895 \text{ mm.}$

Distance between Ribs [e]:
 $= \text{Web Length} / (\text{Nr ribs} - 1)$
 $= 1132 / (4 - 1)$
 $= 377.298 \text{ mm.}$

Baseplate Pressure Area [Ap]:

$$= e \cdot Bpwid / 2$$

$$= 377 \cdot 200 / 2$$

$$= 377.298 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth)$$

$$= 61126 / (1180 \cdot 200)$$

$$= 25.901 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 377 \cdot 25.9$$

$$= 9772.312 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= Rib \text{ Area} + Web \text{ Area}$$

$$= 23 + 30.2$$

$$= 53.224 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 9772 / 53.2$$

$$= 18.006 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	16.0	144.0	88.0	23.0	202752.0	872.
Web	188.6	16.0	8.0	30.2	24147.1	368.
Totals	...	...	...	53.2	226899.1	0.124E+04

Rib dimension [D]:

$$= Saddle \text{ Width} - Web \text{ Thickness}$$

$$= 160 - 16$$

$$= 144.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 226899 / 53.2$$

$$= 42.631 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(ytot, Saddle \text{ Width} - ytot)$$

$$= \max(42.6, 160 - 42.6)$$

$$= 117.369 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{ (Total \text{ Inertia} / Total \text{ Area}) }$$

$$= \sqrt{ (1241 / 53.2) }$$

$$= 48.284 \text{ mm.}$$

Length of Outer Rib [L]:

$$= Saddle \text{ Height} - \cos(\theta / 2) (radius + shlthk + wpdthk) - bpthk$$

$$= 1300 - \cos(120 / 2) (548 + 90 + 16) - 32$$

$$= 941.250 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{ (2 \cdot \pi^2 \cdot Elastic \text{ Modulus} / Yield \text{ Stress}) }$$

$$= \sqrt{ (2 \cdot \pi^2 \cdot 192580992 / 225) }$$

$$= 129.985$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 941 / 48.3$$

$$= 19.494$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{pln}) \cdot e \cdot L / 2$$

$$= 8614 / (2 \cdot 1180) \cdot 377 \cdot 941 / 2$$

$$= 648.159 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($19.5 < 130$) per AISC E2-1 [Sca]:

$$= (1 - (Klr)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (Klr) / (8 \cdot Cc) - (Klr^3) / (8 \cdot Cc^3))$$

$$= (1 - (19.5)^2 / (2 \cdot 130^2)) 225 /$$

$$(5/3 + 3 \cdot (19.5) / (8 \cdot 130) - (19.5^3) / (8 \cdot 130^3))$$

$$= 129.2 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 18 / 129 + (648 \cdot 117 / 12408103) / 150$$

$$= 0.540$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	144.0	88.0	23.0	202752.0	0.117E+04
Web	377.3	16.0	8.0	60.4	48294.2	308.
Totals	...	...	...	83.4	251046.2	0.148E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 251046 / 83.4$$

$$= 30.099 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(y_{tot}, \text{abs}(\text{Saddle Width} - y_{tot}))$$

$$= \max(30.1, \text{abs}(160 - 30.1))$$

$$= 129.901 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{ (Ro + W_{pdthk})^2 - (Pitch/2)^2 } - B_{pthk}$$

$$= 1300 - \sqrt{ (654 + 16)^2 - (377/2)^2 } - 32$$

$$= 642.321 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (1478 / 83.4) }$$

$$= 42.099 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 642 / 42.1$$

$$= 15.257$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 8614 / (2 \cdot 1180)$$

$$= 3.650 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 3.65 \cdot 377 \cdot 642$$

$$= 884.624 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 885 / 113798$$

$$= 76.233 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($15.3 < 130$) per AISC E2-1 [Sca]:

$$= (1 - (Klr)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (Klr) / (8 \cdot Cc) - (Klr^3) / (8 \cdot Cc^3))$$

$$= (1 - (15.3)^2 / (2 \cdot 130^2)) 225 /$$

$$(5/3 + 3 \cdot (15.3) / (8 \cdot 130) - (15.3^3) / (8 \cdot 130^3))$$

= 130.6 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= $Sc/Sca + (M_{base,c} \cdot C1/I) / Sba$
 = $22.4/131 + (885 \cdot 130/1478) / 150$
 = 0.679

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	3
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-36
Bolt Allowable Stress	Stba	150.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	85.0 mm.
Nominal Bolt Diameter	Bnd	27.0000 mm.
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	3.1284 cm ²
Saddle Load Q0 (Weight)	Q0	46285.1 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	2115.4 Kg
Maximum Transverse Force	Ft	4307.2 Kg
Maximum Longitudinal Force	F1	8614.4 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_{b,l}]:

= $F1 / (Bolt\ Area \cdot Number\ of\ Bolts)$
 = $8614 / (3.13 \cdot 3)$
 = 90.0 N./mm². Must be less than 100.0 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [τ_{b,t}]:

= $Ft / (Bolt\ Area \cdot Number\ of\ Bolts)$
 = $4307 / (3.13 \cdot 3)$
 = 45.0 N./mm². Must be less than 100.0 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

= 0.0 (Q0 > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:

= $F1 / (BoltShearAllowable \cdot Nbolts)$
 = $8614 / (100 \cdot 3)$
 = 2.8160 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

= $B \cdot Ft + Sum\ of\ X\ Moments$
 = $1300 \cdot 4307 + 0$
 = 5599.47 Kg-m.

Eccentricity (e):

= $Rmom / Q0$
 = $5599/46285$
 = 120.98 mm. < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:

= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:

= $\max[Bltarearl, Bltarears, Bltareart]$
 = $\max[0, 2.82, 0]$
 = 2.8160 cm²

MDMT Calculations for: FIXED SADDLE, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 16, c = 3 mm., E* = 1

Thickness Ratio = $tr * E / (tg - c) = 0.97$, Temp. Reduction = 2 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
 Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit -31 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -32 °C

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	594.00	mm.
Shell Thickness used in this Case	t	90.000	mm.
Stiffened Vessel Length per 4.15.6	L	9094.00	mm.
Distance from Saddle to Vessel tangent	a1 or a	1820.00	mm.
Saddle Width	b1 or b	160.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	520.00	mm.
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	16.0	mm.
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1300.00	mm.
Coefficient of Friction	mu	0.10	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case 60841.05 Kgf
 Pressure used in Analysis P 191.040 bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	62.30	138.00
Long. Stress at Bottom of Midspan	68.15	138.00
Long. Stress at Top of Saddles	85.23	138.00
Long. Stress at Bottom of Saddles	54.13	138.00
Tangential Shear in Shell	8.11	110.40
Circ. Stress at Horn of Saddle	7.27	172.50
Circ. Compressive Stress in Shell	0.86	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= Ftr(Ft/Num \text{ of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (1960/2 + 0) \cdot 1300/1132$
 $= 3376.9 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(Fl, wind; \text{Sum of X Forces}) \cdot B / Ls$
 $= \max(327, 0) \cdot 1300/5294$
 $= 80.4 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(Fl, eq; \text{Sum of X Forces}) \cdot B / Ls$
 $= \max(8614, 0) \cdot 1300/5294$
 $= 2115.4 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= Ftr(Ft/Num \text{ of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (8614/2 + 0) \cdot 1300/1132$

= 14840.7 Kgf

Load Combination Results for Q + Wind or Seismic [Q]:

= Saddle Load + max(Fwl, Fwt, Fsl, Fst)
 = 46000 + max(80.4, 3377, 2115, 14841)
 = 60841.0 Kgf

Longitudinal Wind Force [FI]:

= WindScalar • WindPress(Platform Area + ($\pi/4$ (OD • WindDiaMult)²))
 = 1.1 • 1587 (0 + $\pi/4$ (1.28 • 1.2)²)
 = 3209.902 N

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	61216.12	Kgf
Transverse Shear Load	4307.20	Kgf
Longitudinal Shear Load	8614.40	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= min(2(ShellID/2 + t + WearPadThickness)sin($\theta/2$), 2•Rm)
 = min(2(1095/2 + 90 + 16)sin(120/2), 2•594)
 = 1131.895 mm.

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

= -Q•a [1 - (1 - a/L + (Rm²-hm²)/(2a•L))/(1+(4hm)/3L)]
 = -60841•1820[1 - (1 - 1820/9094 + (594²-0²)/(2•1820•9094))/(1+(4•0)/(3•9094))]
 = -20980.9 Kg-m.

Moment per Equation 4.15.2 [M2]:

= Q•L/4((1+2(Rm²-hm²)/(L²))/(1+(4hm²)/(3L)) - 4a/L)
 = 60841•9094/4((1+2(594²-0²)/(9094²))/(1+(4•0)/(3•9094)) - 4•1820/9094)
 = 28772.3 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

= P • Rm/(2t) - M2/(π •Rm²t)
 = 191 • 594/(2•87) - 28772/(π •594²•87)
 = 62.30 N./mm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

= P • Rm/(2t) + M2/(π • Rm² • t)
 = 191 • 594/(2 • 87) + 28772/(π • 594² • 87)
 = 68.15 N./mm²

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_3]:

= P • Rm/(2t) - M1/(K1• π •Rm²t)
 = 191•594/(2•87) - 20981/(0.11• π •594²•87)
 = 85.23 N./mm²

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_4]:

= P • Rm/(2t) + M1/(K1* • π • Rm² • t)
 = 191•594/(2•87) + 20981/(0.19• π •594²•87)
 = 54.13 N./mm²

Maximum Shear Force in the Saddle (4.15.3) [T]:

= Q(L-2a)/(L+(4•hm/3))
 = 60841 (9094 - 2 • 1820)/(9094 + (4 • 0/3))
 = 36488.6 Kgf

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 36489 / (594 \cdot 87)$$

$$= 8.11 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(594 \cdot 87)}$$

$$= 177.316 \text{ mm.}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(160 + 1.56 \cdot \sqrt{(594 \cdot 87)}, 2 \cdot 1820)$$

$$= 514.63 \text{ mm.}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r/S, 1)$$

$$= \min(138/138, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 60841 \cdot 0.1 / (515(87 + 1 \cdot 16))$$

$$= -0.86 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -60841 / (4(87 + 1 \cdot 16)515) -$$

$$3 \cdot 0.053 \cdot 60841 / (2(87 + 1 \cdot 16)^2)$$

$$= -7.27 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1180.0000	mm.
Baseplate Thickness	Bpthk	32.0000	mm.
Baseplate Width	Bpwid	200.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	16.0000	mm.
Web Thickness	Webtk	16.0000	mm.
Web Location	Webloc	Side	
Saddle Design Temperature		200.0	°C
Saddle Yield Stress	Sy	225.0	N./mm ²
Height of Web at Center	Hw,c	306.0	mm.
Friction Coefficient	mu	0.100	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(550 \cdot 87)} + 520)87^3) / 12$$

$$= 4726.947 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 4727 + 749(128 - 43.5)^2$$

$$= 58787.812 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ² (2) cm ⁴
Shell	861.4	87.0	43.5	749.4	3259966.5	0.588E+05
Wearplate	520.0	16.0	95.0	83.2	790399.9	948.
Web	16.0	614.5	410.2	98.3	4033577.5	0.109E+06
BasePlate	200.0	32.0	733.5	64.0	4694400.0	0.234E+06
Totals	...	...	...	994.9	12778344.0	0.403E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 12778344/995$$

$$= 128.434 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 60841$$

$$= 12382.4727 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh/As) = 4.9459 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 134.9988 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1300 - 550 \cdot \sin(120/2) / 1.05$$

$$= 844.740 \text{ mm.}$$

$$\text{Bending Moment, } M = Fh \cdot d = 10460.1777 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 3.2681 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 149.9987 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$$

$$= \sqrt{(3(60841 + 375) \cdot 200) / (2 \cdot 1180 \cdot 150)}$$

$$= 31.899 \text{ mm.}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (548 + 90 + 16)$$

$$= 1131.895 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nrings} - 1)$$

$$= 1132 / (4 - 1)$$

$$= 377.298 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 377 \cdot 200/2$$

$$= 377.298 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 61216 / (1180 \cdot 200)$$

$$= 25.939 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 377 \cdot 25.9$$

$$= 9786.757 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 23 + 30.2$$

$$= 53.224 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/Ar$$

$$= 9787/53.2$$

= 18.033 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	16.0	144.0	88.0	23.0	202752.0	872.
Web	188.6	16.0	8.0	30.2	24147.1	368.
Totals	...	...	...	53.2	226899.1	0.124E+04

Rib dimension [D]:

= Saddle Width - Web Thickness
 = 160 - 16
 = 144.000 mm.

Distance to Centroid from Datum [ytot]:

= AY / A
 = 226899/53.2
 = 42.631 mm.

Distance to Centroid [C1]:

= max(ytot, Saddle Width - ytot)
 = max(42.6, 160 - 42.6)
 = 117.369 mm.

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{1241/53.2}$
 = 48.284 mm.

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 1300 - cos(120/2)(548 + 90 + 16) - 32
 = 941.250 mm.

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 = $\sqrt{2 \cdot \pi^2 \cdot 192580992/225}$
 = 129.985

Slenderness ratio [KL/r]:

= KL/r
 = 1 • 941/48.3
 = 19.494

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2
 = 8614 / (2 • 1180) • 377 • 941/2
 = 648.159 Kg-m.

Compressive Allowable, KL/r < Cc (19.5 < 130) per AISC E2-1 [Sca]:

= (1 - (KL/r)² / (2 • Cc²)) Fy / (5/3 + 3 • (KL/r) / (8 • Cc) - (KL/r)³ / (8 • Cc³))
 = (1 - (19.5)² / (2 • 130²)) 225 /
 (5/3 + 3 • (19.5) / (8 • 130) - (19.5)³ / (8 • 130³))
 = 129.2 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I) / Sba
 = 18/129 + (648 • 117/12408103) / 150
 = 0.540

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	144.0	88.0	23.0	202752.0	0.117E+04
Web	377.3	16.0	8.0	60.4	48294.2	308.



Totals	...	...	...	83.4	251046.2	0.148E+04
--------	-----	-----	-----	------	----------	-----------

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 251046/83.4$$

$$= 30.099 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(ytot, \text{abs}(\text{Saddle Width} - ytot))$$

$$= \max(30.1, \text{abs}(160 - 30.1))$$

$$= 129.901 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(\text{Ro} + \text{Wpdthk})^2 - (\text{Pitch}/2)^2} - \text{Bpthk}$$

$$= 1300 - \sqrt{(654 + 16)^2 - (377/2)^2} - 32$$

$$= 642.321 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(1478/83.4)}$$

$$= 42.099 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 642/42.1$$

$$= 15.257$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 8614 / (2 \cdot 1180)$$

$$= 3.650 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 3.65 \cdot 377 \cdot 642$$

$$= 884.624 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 885/113798$$

$$= 76.233 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($15.3 < 130$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (15.3)^2 / (2 \cdot 130^2)) 225 / (5/3 + 3 \cdot (15.3) / (8 \cdot 130) - (15.3^3) / (8 \cdot 130^3))$$

$$= 130.6 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 22.4/131 + (885 \cdot 130/1478) / 150$$

$$= 0.680$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	3
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-36
Bolt Allowable Stress	Stba	150.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	85.0 mm.
Nominal Bolt Diameter	Bnd	27.0000 mm.
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	3.1284 cm ²
Saddle Load Q0 (Weight)	Q0	46375.5 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	2115.4 Kg
Maximum Transverse Force	Ft	4307.2 Kg
Maximum Longitudinal Force	Fl	8614.4 Kg

Saddle Bolted to Steel Foundation

No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:
 $= F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 8614 / (3.13 \cdot 3)$
 $= 90.0 \text{ N./mm}^2$. Must be less than 100.0 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:
 $= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 4307 / (3.13 \cdot 3)$
 $= 45.0 \text{ N./mm}^2$. Must be less than 100.0 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 $= 0.0$ (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 $= F_l / (\text{BoltShearAllowable} \cdot \text{Nbols})$
 $= 8614 / (100 \cdot 3)$
 $= 2.8160 \text{ cm}^2$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 $= B \cdot F_t + \text{Sum of X Moments}$
 $= 1300 \cdot 4307 + 0$
 $= 5599.47 \text{ Kg-m.}$

Eccentricity (e):
 $= Rmom / QO$
 $= 5599 / 46375$
 $= 120.74 \text{ mm.}$ < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 $= 0$ (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 $= \max[Bltarearl, Bltarears, Bltareart]$
 $= \max[0, 2.82, 0]$
 $= 2.8160 \text{ cm}^2$

MDMT Calculations for: SLIDING SADDLE, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 16, c = 3 mm., E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.97$, Temp. Reduction = 2 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-31 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-32 °C

Review notes about nozzle loadings and supports in the Warnings report.

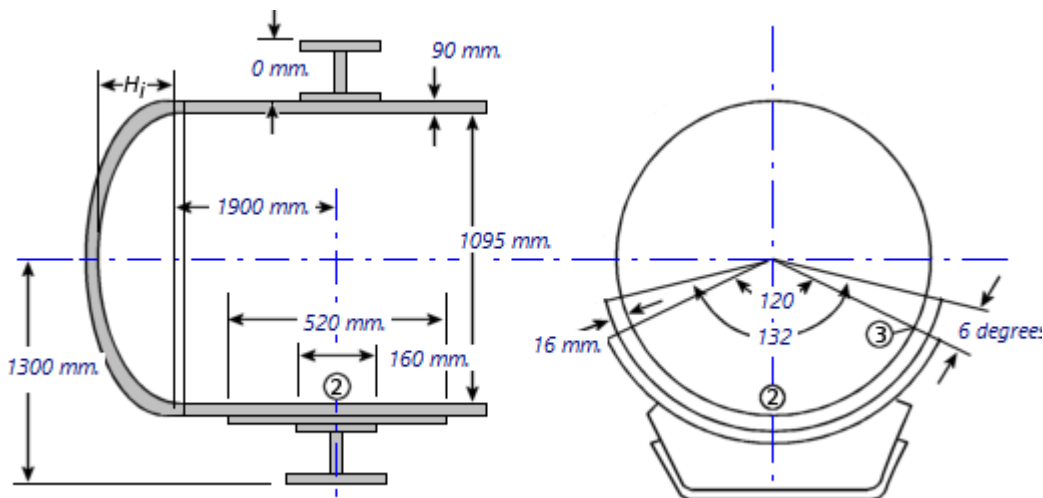
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Input and Calculated Values:

Vessel Mean Radius	Rm	594.00	mm.
Shell Thickness used in this Case	t	90.000	mm.
Stiffened Vessel Length per 4.15.6	L	9094.00	mm.
Distance from Saddle to Vessel tangent	a1 or a	1900.00	mm.
Saddle Width	b1 or b	160.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	520.00	mm.
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	16.0	mm.
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1300.00	mm.
Coefficient of Friction	mu	0.60	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces	P	45787.32	Kgf
Pressure used in Analysis		251.880	bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	84.16	234.00
Long. Stress at Bottom of Midspan	87.82	234.00
Long. Stress at Top of Saddles	102.48	234.00
Long. Stress at Bottom of Saddles	76.85	234.00
Tangential Shear in Shell	5.92	138.58
Circ. Stress at Horn of Saddle	5.47	350.99
Circ. Compressive Stress in Shell	0.64	234.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:



$$= Ftr(Ft/Num of Saddles + Z Force Load) \cdot B / E$$

$$= 3 (647/2 + 0) \cdot 1300/1132$$

$$= 1114.4 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(108, 0) \cdot 1300/5294$$

$$= 26.5 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$$

$$= 44673 + \max(26.5, 1114, 0, 0)$$

$$= 45787.3 \text{ Kgf}$$

Longitudinal Wind Force [FI]:

$$= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1587 (0 + \pi/4 (1.28 \cdot 1.2)^2)$$

$$= 3209.902 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	46162.39	Kgf
Transverse Shear Load	323.43	Kgf
Longitudinal Shear Load	108.02	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(1095/2 + 90 + 16) \sin(120/2), 2 \cdot 594)$$

$$= 1131.895 \text{ mm.}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$$

$$= -45787 \cdot 1900 [1 - (1 - 1900/9094 + (594^2 - 0^2)/(2 \cdot 1900 \cdot 9094)) / (1 + (4 \cdot 0)/(3 \cdot 9094))]$$

$$= -17288.1 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 45787 \cdot 9094/4 ((1 + 2(594^2 - 0^2)/(9094^2)) / (1 + (4 \cdot 0)/(3 \cdot 9094)) - 4 \cdot 1900/9094)$$

$$= 17990.2 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t)$$

$$= 252 \cdot 594/(2 \cdot 87) - 17990/(\pi \cdot 594^2 \cdot 87)$$

$$= 84.16 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$$

$$= 252 \cdot 594/(2 \cdot 87) + 17990/(\pi \cdot 594^2 \cdot 87)$$

$$= 87.82 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 t)$$

$$= 252 \cdot 594/(2 \cdot 87) - 17288/(0.11 \cdot \pi \cdot 594^2 \cdot 87)$$

$$= 102.48 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m/(2t) + M1/(K1^* \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 252 \cdot 594 / (2 \cdot 87) + (-17288 / (0.19 \cdot \pi \cdot 594^2 \cdot 87))$$

$$= 76.85 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a)/(L+(4 \cdot hm/3))$$

$$= 45787 (9094 - 2 \cdot 1900) / (9094 + (4 \cdot 0/3))$$

$$= 26654.7 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 26655 / (594 \cdot 87)$$

$$= 5.92 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(594 \cdot 87)}$$

$$= 177.316 \text{ mm.}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(160 + 1.56 \cdot \sqrt{(594 \cdot 87)}, 2 \cdot 1900)$$

$$= 514.63 \text{ mm.}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 45787 \cdot 0.1 / (515(87 + 1 \cdot 16))$$

$$= -0.64 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -45787 / (4(87 + 1 \cdot 16)515) -$$

$$3 \cdot 0.053 \cdot 45787 / (2(87 + 1 \cdot 16)^2)$$

$$= -5.47 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1180.0000	mm.
Baseplate Thickness	Bpthk	32.0000	mm.
Baseplate Width	Bpwid	200.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	16.0000	mm.
Web Thickness	Webtk	16.0000	mm.
Web Location	Webloc	Side	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	306.0	mm.
Friction Coefficient	mu	0.600	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(550 \cdot 87)} + 520)87^3) / 12$$

$$= 4726.947 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 4727 + 749(128 - 43.5)^2$$

$$= 58787.812 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ² (2) cm ⁴
Shell	861.4	87.0	43.5	749.4	3259966.5	0.588E+05
Wearplate	520.0	16.0	95.0	83.2	790399.9	948.
Web	16.0	614.5	410.2	98.3	4033577.5	0.109E+06
BasePlate	200.0	32.0	733.5	64.0	4694400.0	0.234E+06
Totals	...	...	...	994.9	12778344.0	0.403E+06



Distance to Centroid [C1]:

$$= AY / A$$

$$= 12778344/995$$

$$= 128.434 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 45787$$

$$= 9318.7109 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh/As) = 3.7222 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 1300 - 550 \cdot \sin(120/2) / 1.05$$

$$= 844.740 \text{ mm.}$$

$$\text{Bending Moment, } M = Fh \cdot d = 7872.0444 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 2.4595 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$$

$$= \sqrt{(3(45787 + 375) \cdot 200) / (2 \cdot 1180 \cdot 173))}$$

$$= 25.769 \text{ mm.}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (548 + 90 + 16)$$

$$= 1131.895 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 1132 / (4 - 1)$$

$$= 377.298 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 377 \cdot 200/2$$

$$= 377.298 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 46162 / (1180 \cdot 200)$$

$$= 19.560 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 377 \cdot 19.6$$

$$= 7380.083 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 23 + 30.2$$

$$= 53.224 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/Ar$$

$$= 7380/53.2$$

$$= 13.598 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	16.0	144.0	88.0	23.0	202752.0	872.
Web	188.6	16.0	8.0	30.2	24147.1	368.
Totals	...	...	...	53.2	226899.1	0.124E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 160 - 16$$

$$= 144.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 226899/53.2$$

$$= 42.631 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(\text{ytot}, \text{Saddle Width} - \text{ytot})$$

$$= \max(42.6, 160 - 42.6)$$

$$= 117.369 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(1241/53.2)}$$

$$= 48.284 \text{ mm.}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2)(\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1300 - \cos(120/2)(548 + 90 + 16) - 32$$

$$= 941.250 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 192580992/260)}$$

$$= 120.921$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 941/48.3$$

$$= 19.494$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 108 / (2 \cdot 1180) \cdot 377 \cdot 941/2$$

$$= 8.127 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($19.5 < 121$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (19.5)^2 / (2 \cdot 121^2)) 260 /$$

$$(5/3 + 3 \cdot (19.5) / (8 \cdot 121) - (19.5^3) / (8 \cdot 121^3))$$

$$= 148.6 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 13.6/149 + (8.13 \cdot 117/12408103) / 173$$

$$= 0.096$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
--	---	---	---	---	----	----

Rib	16.0	144.0	88.0	23.0	202752.0	0.117E+04
Web	377.3	16.0	8.0	60.4	48294.2	308.
Totals	...	...	...	83.4	251046.2	0.148E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 251046/83.4$$

$$= 30.099 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(ytot, \text{abs}(\text{Saddle Width} - ytot))$$

$$= \max(30.1, \text{abs}(160 - 30.1))$$

$$= 129.901 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{ (Ro + Wpdthk)^2 - (Pitch/2)^2 } - Bpthk$$

$$= 1300 - \sqrt{ (654 + 16)^2 - (377/2)^2 } - 32$$

$$= 642.321 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{ (1478/83.4)}$$

$$= 42.099 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 642/42.1$$

$$= 15.257$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 108 / (2 \cdot 1180)$$

$$= 0.046 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.046 \cdot 377 \cdot 642$$

$$= 11.092 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 11.1/113798$$

$$= 0.956 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($15.3 < 121$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (15.3)^2 / (2 \cdot 121^2)) 260 / (5/3 + 3 \cdot (15.3) / (8 \cdot 121) - (15.3^3) / (8 \cdot 121^3))$$

$$= 150.5 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 16.9/151 + (11.1 \cdot 130/1478) / 173$$

$$= 0.118$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	3
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Material Specification		SA-36
Bolt Allowable Stress	Stba	114.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	85.0 mm.
Nominal Bolt Diameter	Bnd	27.0000 mm.
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	3.1284 cm ²
Saddle Load Q0 (Weight)	Q0	45048.0 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	26.5 Kg



Maximum Transverse Force	Ft	323.4 Kg
Maximum Longitudinal Force	Fl	108.0 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:
 = $F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 = $108 / (3.13 \cdot 3)$
 = 1.1 N./mm². Must be less than 68.4 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:
 = $F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 = $323 / (3.13 \cdot 3)$
 = 3.4 N./mm². Must be less than 68.4 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 = 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 = $F_l / (\text{BoltShearAllowable} \cdot \text{Nbols})$
 = $108 / (68.4 \cdot 3)$
 = 0.0516 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 = $B \cdot F_t + \text{Sum of X Moments}$
 = $1300 \cdot 323 + 0$
 = 420.46 Kg-m.

Eccentricity (e):
 = R_{mom} / QO
 = 420/45048
 = 9.33 mm. < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 = $\max[Bltarearl, Bltarears, Bltareart]$
 = $\max[0, 0.052, 0]$
 = 0.0516 cm²

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	594.00	mm.
Shell Thickness used in this Case	t	90.000	mm.
Stiffened Vessel Length per 4.15.6	L	9094.00	mm.
Distance from Saddle to Vessel tangent	a1 or a	1820.00	mm.
Saddle Width	b1 or b	160.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	520.00	mm.
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	16.0	mm.
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1300.00	mm.
Coefficient of Friction	mu	0.10	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces 45877.81 Kgf
 Pressure used in Analysis P 251.880 bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	83.79	234.00
Long. Stress at Bottom of Midspan	88.20	234.00
Long. Stress at Top of Saddles	101.08	234.00
Long. Stress at Bottom of Saddles	77.63	234.00
Tangential Shear in Shell	6.11	138.58
Circ. Stress at Horn of Saddle	5.48	350.99
Circ. Compressive Stress in Shell	0.65	234.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (647/2 + 0) \cdot 1300/1132$
 $= 1114.4 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(108, 0) \cdot 1300/5294$
 $= 26.5 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 44763 + \max(26.5, 1114, 0, 0)$
 $= 45877.8 \text{ Kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1587 (0 + \pi/4 (1.28 \cdot 1.2)^2)$
 $= 3209.902 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 46252.89 Kgf
 Transverse Shear Load 323.43 Kgf
 Longitudinal Shear Load 108.02 Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$
 $= \min(2(1095/2 + 90 + 16) \sin(120/2), 2 \cdot 594)$
 $= 1131.895 \text{ mm.}$

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011
 K5 = 0.7603 K6 = 0.0529 K7 = 0.0529 K8 = 0.3405
 K9 = 0.2711 K10 = 0.0581 K1* = 0.1923 K6p = 0.0434
 K7p = 0.0434

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$
 $= -45878 \cdot 1820 [1 - (1 - 1820/9094 + (594^2 - 0^2)/(2 \cdot 1820 \cdot 9094))/(1 + (4 \cdot 0)/(3 \cdot 9094))]$
 $= -15820.9 \text{ Kg-m.}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$

$$= 45878 \cdot 9094 / 4 \cdot ((1 + 2(594^2 - 0^2) / (9094^2)) / (1 + (4 \cdot 0) / (3 \cdot 9094))) - 4 \cdot 1820 / 9094$$

$$= 21696.0 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 252 \cdot 594 / (2 \cdot 87) - 21696 / (\pi \cdot 594^2 \cdot 87)$$

$$= 83.79 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 252 \cdot 594 / (2 \cdot 87) + 21696 / (\pi \cdot 594^2 \cdot 87)$$

$$= 88.20 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 252 \cdot 594 / (2 \cdot 87) - 15821 / (0.11 \cdot \pi \cdot 594^2 \cdot 87)$$

$$= 101.08 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 252 \cdot 594 / (2 \cdot 87) + 15821 / (0.19 \cdot \pi \cdot 594^2 \cdot 87)$$

$$= 77.63 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3))$$

$$= 45878 \cdot (9094 - 2 \cdot 1820) / (9094 + (4 \cdot 0 / 3))$$

$$= 27514.6 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 27515 / (594 \cdot 87)$$

$$= 6.11 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(594 \cdot 87)}$$

$$= 177.316 \text{ mm.}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(160 + 1.56 \cdot \sqrt{(594 \cdot 87)}, 2 \cdot 1820)$$

$$= 514.63 \text{ mm.}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 45878 \cdot 0.1 / (515(87 + 1 \cdot 16))$$

$$= -0.65 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -45878 / (4(87 + 1 \cdot 16)515) - 3 \cdot 0.053 \cdot 45878 / (2(87 + 1 \cdot 16)^2)$$

$$= -5.48 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1180.0000	mm.
Baseplate Thickness	Bpthk	32.0000	mm.
Baseplate Width	Bpwid	200.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	16.0000	mm.
Web Thickness	Webtk	16.0000	mm.
Web Location	Webloc	Side	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	306.0	mm.
Friction Coefficient	mu	0.100	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth}) t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(550 \cdot 87)} + 520) 87^3) / 12$$

$$= 4726.947 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 4727 + 749 (128 - 43.5)^2$$

$$= 58787.812 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ² cm ⁴
Shell	861.4	87.0	43.5	749.4	3259966.5	0.588E+05
Wearplate	520.0	16.0	95.0	83.2	790399.9	948.
Web	16.0	614.5	410.2	98.3	4033577.5	0.109E+06
BasePlate	200.0	32.0	733.5	64.0	4694400.0	0.234E+06
Totals	...	...	...	994.9	12778344.0	0.403E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 12778344 / 995$$

$$= 128.434 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 45878$$

$$= 9337.1289 \text{ Kgf}$$

Tension Stress, St = (Fh/As) = 3.7295 N./mm²
 Allowed Stress, Sa = 0.6 * Yield Str = 155.9972 N./mm²

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 1300 - 550 \cdot \sin(120/2) / 1.05$$

$$= 844.740 \text{ mm.}$$

Bending Moment, M = Fh * d = 7887.6030 Kg-m.

Bending Stress, Sb = (M * C1 / I) = 2.4644 N./mm²
 Allowed Stress, Sa = 2/3 * Yield Str = 173.3302 N./mm²

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(45878 + 375) 200 / (2 \cdot 1180 \cdot 173)}$$

$$= 25.794 \text{ mm.}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (548 + 90 + 16)$$

$$= 1131.895 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 1132 / (4 - 1)$$

= 377.298 mm.

Baseplate Pressure Area [Ap]:

= e • Bpwid / 2
 = 377 • 200/2
 = 377.298 cm²

Bearing Pressure [Bp]:

= (Q + SaddleWt) / (BasePlateLength • BasePlateWidth)
 = 46253/(1180 • 200)
 = 19.599 Kgf/cm²

Axial Load [P]:

= Ap • Bp
 = 377 • 19.6
 = 7394.551 Kgf

Area of the Rib and Web [Ar]:

= Rib Area + Web Area
 = 23 + 30.2
 = 53.224 cm²

Compressive Stress [Sc]:

= P/Ar
 = 7395/53.2
 = 13.625 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	16.0	144.0	88.0	23.0	202752.0	872.
Web	188.6	16.0	8.0	30.2	24147.1	368.
Totals	...	...	...	53.2	226899.1	0.124E+04

Rib dimension [D]:

= Saddle Width - Web Thickness
 = 160 - 16
 = 144.000 mm.

Distance to Centroid from Datum [ytot]:

= AY / A
 = 226899/53.2
 = 42.631 mm.

Distance to Centroid [C1]:

= max(ytot, Saddle Width - ytot)
 = max(42.6, 160 - 42.6)
 = 117.369 mm.

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{1241/53.2}$
 = 48.284 mm.

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 1300 - cos(120/2)(548 + 90 + 16) - 32
 = 941.250 mm.

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 = $\sqrt{2 \cdot \pi^2 \cdot 192580992/260}$
 = 120.921

Slenderness ratio [KL/r]:

= KL/r
 = 1 • 941/48.3

= 19.494

Bending Moment [Rm]:

$$= F_l / (2 \cdot B_{pln}) \cdot e \cdot L / 2$$

$$= 108 / (2 \cdot 1180) \cdot 377 \cdot 941 / 2$$

$$= 8.127 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($19.5 < 121$) per AISC E2-1 [Sca]:

$$= (1 - (Kl/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (Kl/r) / (8 \cdot C_c) - (Kl/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (19.5)^2 / (2 \cdot 121^2)) 260 /$$

$$(5/3 + 3 \cdot (19.5) / (8 \cdot 121) - (19.5^3) / (8 \cdot 121^3))$$

$$= 148.6 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 13.6 / 149 + (8.13 \cdot 117 / 12408103) / 173$$

$$= 0.096$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	144.0	88.0	23.0	202752.0	0.117E+04
Web	377.3	16.0	8.0	60.4	48294.2	308.
Totals	...	...	...	83.4	251046.2	0.148E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 251046 / 83.4$$

$$= 30.099 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(y_{tot}, \text{abs}(\text{Saddle Width} - y_{tot}))$$

$$= \max(30.1, \text{abs}(160 - 30.1))$$

$$= 129.901 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{ (R_o + W_{pdthk})^2 - (Pitch/2)^2 } - B_{pthk}$$

$$= 1300 - \sqrt{ (654 + 16)^2 - (377/2)^2 } - 32$$

$$= 642.321 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ 1478 / 83.4 }$$

$$= 42.099 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 642 / 42.1$$

$$= 15.257$$

Unit Force [Force,u]:

$$= F_l / (2 \cdot \text{Baseplate Length})$$

$$= 108 / (2 \cdot 1180)$$

$$= 0.046 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.046 \cdot 377 \cdot 642$$

$$= 11.092 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 11.1 / 113798$$

$$= 0.956 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($15.3 < 121$) per AISC E2-1 [Sca]:

$$= (1 - (Kl/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (Kl/r) / (8 \cdot C_c) - (Kl/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (15.3)^2 / (2 \cdot 121^2)) 260 /$$

$$\begin{aligned} & (5/3 + 3 \cdot (15.3)/(8 \cdot 121) - (15.3^3)/(8 \cdot 121^3)) \\ & = 150.5 \text{ N./mm}^2 \end{aligned}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$\begin{aligned} & = S_c/S_{ca} + (M_{base,c} \cdot C1/I)/S_{ba} \\ & = 16.9/151 + (11.1 \cdot 130/1478)/173 \\ & = 0.118 \end{aligned}$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	3
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Material Specification		SA-36
Bolt Allowable Stress	Stba	114.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	85.0 mm.
Nominal Bolt Diameter	Bnd	27.0000 mm.
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	3.1284 cm ²
Saddle Load Q0 (Weight)	Q0	45138.5 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	26.5 Kg
Maximum Transverse Force	Ft	323.4 Kg
Maximum Longitudinal Force	Fl	108.0 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$\begin{aligned} & = F_l / (\text{Bolt Area} \cdot \text{Number of Bolts}) \\ & = 108 / (3.13 \cdot 3) \\ & = 1.1 \text{ N./mm}^2. \text{ Must be less than } 68.4 \text{ N./mm}^2. \end{aligned}$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$\begin{aligned} & = F_t / (\text{Bolt Area} \cdot \text{Number of Bolts}) \\ & = 323 / (3.13 \cdot 3) \\ & = 3.4 \text{ N./mm}^2. \text{ Must be less than } 68.4 \text{ N./mm}^2. \end{aligned}$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} Q_0 > Q_L \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$\begin{aligned} & = F_l / (\text{BoltShearAllowable} \cdot \text{Nbolts}) \\ & = 108 / (68.4 \cdot 3) \\ & = 0.0516 \text{ cm}^2 \end{aligned}$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$\begin{aligned} & = B \cdot F_t + \text{Sum of X Moments} \\ & = 1300 \cdot 323 + 0 \\ & = 420.46 \text{ Kg-m.} \end{aligned}$$

Eccentricity (e):

$$\begin{aligned} & = R_{mom} / Q_0 \\ & = 420/45139 \\ & = 9.31 \text{ mm.} < B_{plen}/6 \text{ --> No Uplift in Transverse direction} \end{aligned}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$\begin{aligned} & = \max[Bltarearl, Bltarears, Bltareart] \\ & = \max[0, 0.052, 0] \\ & = 0.0516 \text{ cm}^2 \end{aligned}$$

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: TS OUT NOZ. T2

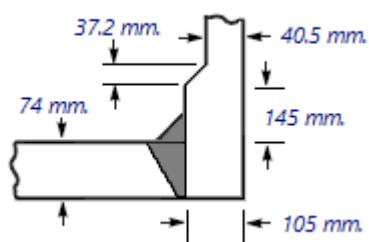
From: 30

Pressure for Reinforcement Calculations	P	147.000	bars
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1095.00	mm.
Shell Finished (Minimum) Thickness	t	74.0000	mm.
Shell Internal Corrosion Allowance	c	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		1264.17	mm.
User Entered Minimum Design Metal Temperature		-28.89	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		16.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	40.4900	mm.
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	145.0000	mm.
Height of Beveled Transition	L'	37.2448	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		105.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		60.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		1500	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: TS OUT NOZ. T2

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 16.000 in.
 Actual Thickness Used in Calculation 1.594 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 147 \cdot 550 / (138 \cdot 1 - 0.6 \cdot 147)$
 $= 62.6491$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 147 \cdot 206 / (138 \cdot 1 - 0.6 \cdot 147)$
 $= 23.4664$ mm.

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 64.5 / 0.5773$
 $= 111.7346$ mm.

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

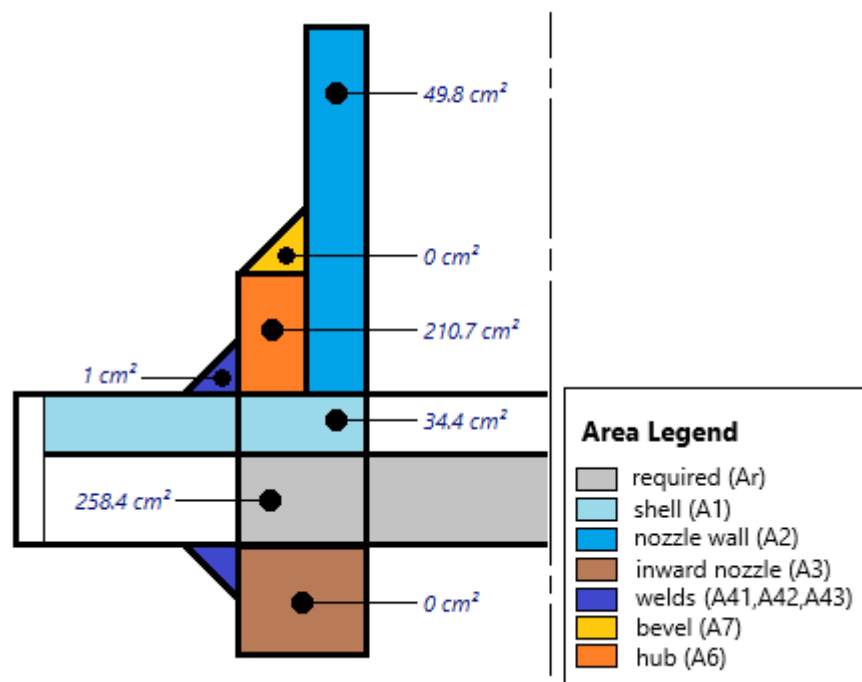
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	824.8000	mm.
Parallel to Vessel Wall, opening length	d	412.4000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	177.5000	mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	258.365	NA	NA
Area in Shell	A1	34.439	NA	NA
Area in Nozzle Wall	A2	49.784	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds A41+A42+A43		1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	210.716	NA	NA
TOTAL AREA AVAILABLE	Atot	295.939	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (412 \cdot 62.6 \cdot 1 + 2 \cdot 102 \cdot 62.6 \cdot 1 \cdot (1 - 1))$
 $= 258.365 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 412(1 \cdot 71 - 1 \cdot 62.6) - 2 \cdot 102$
 $(1 \cdot 71 - 1 \cdot 62.6) \cdot (1 - 1)$
 $= 34.439 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 178)(37.5 - 23.5)1$
 $= 49.784 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10^2 \cdot 1 + (0)^2 \cdot 1$
 $= 1.000 \text{ cm}^2$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, Hubht)) \cdot (Hubtk - t_n) \cdot fr2$$

$$= (2 \cdot \min(178, 300, 145)) \cdot (105 - 40.5) \cdot 1$$

$$= 210.716 \text{ cm}^2 \text{ Includes Bevel Area of } 23.637 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 26.4664 mm.
Wall Thickness per UG16(b),	tr16b = 4.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 65.6491 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 65.6491 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm.
Wall Thickness per table UG-45	tb3 = 11.3312 mm.

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.3, \max(65.6, 4.5)]$$

$$= 11.3312 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(26.5, 11.3)$$

$$= 26.4664 \text{ mm.}$$

Available Nozzle Neck Thickness = 40.4900 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 68.6,	Allowable	: 138.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 276.4 N./mm ²	Passed
Occasional	: 37.1,	Allowable	: 183.5 N./mm ²	Passed
Shear	: 21.9,	Allowable	: 96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 40.5, tr = 23.5, c = 3 mm., E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.63$, Temp. Reduction = 21 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	31 °C
Min Metal Temp. at Required thickness (UCS 66.1)	11 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 74, tr = 62.6, c = 3 mm., E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.88$, Temp. Reduction = 7 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	25 °C
Min Metal Temp. at Required thickness (UCS 66.1)	19 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 19 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-42 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $147.00 / 255.10 = 0.576$

Weld Size Calculations, Description: TS OUT NOZ. T2

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.

Results Per UW-16.1:

Required Thickness	Actual Thickness
--------------------	------------------



Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

$$\begin{aligned} \text{Weld Load [W]:} \\ &= \max(0, (A-A1+2 \cdot t_n \cdot f r_1 \cdot (E1 \cdot t - t_r)) \cdot S_v) \\ &= \max(0, (258 - 34.4 + 2 \cdot 102 \cdot 1 \cdot \\ &\quad (1 \cdot 71 - 62.6)) \cdot 138) \\ &= 339070.72 \text{ Kgf} \end{aligned}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

$$\begin{aligned} \text{Weld Load [W1]:} \\ &= (A2+A4-(W_i \cdot \text{Can}/.707)^2 \cdot f r_2) \cdot S_v \\ &= (260 + 1 - 0 \cdot 1) \cdot 138 \\ &= 367970.84 \text{ Kgf} \end{aligned}$$

$$\begin{aligned} \text{Weld Load [W2]:} \\ &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f r_1)) \cdot S_v \\ &= (260 + 0 + 1 + (145)) \cdot 138 \\ &= 571783.50 \text{ Kgf} \end{aligned}$$

$$\begin{aligned} \text{Weld Load [W3]:} \\ &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f r_1)) \cdot S \\ &= (260 + 0 + 1 + 0 + (145)) \cdot 138 \\ &= 571783.50 \text{ Kgf} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

$$\begin{aligned} \text{Shear, Outward Nozzle Weld [Sonw]:} \\ &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.14/2.0) \cdot 616 \cdot 10 \cdot 0.49 \cdot 138 \\ &= 66761. \text{ Kgf} \end{aligned}$$

$$\begin{aligned} \text{Shear, Nozzle Wall [Snw]:} \\ &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n \\ &= (3.14 \cdot 257) \cdot (105 - 3) \cdot 0.7 \cdot 138 \\ &= 811823. \text{ Kgf} \end{aligned}$$

$$\begin{aligned} \text{Tension, Shell Groove Weld [Tngw]:} \\ &= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng} \\ &= (3.14/2.0) \cdot 616 \cdot (60 - 0) \cdot 0.74 \cdot 138 \\ &= 604934. \text{ Kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

$$\begin{aligned} \text{Failure Path [Path 1-1]:} \\ &= (\text{SONW} + \text{SNW}) = (66761 + 811823) = 878584 \text{ Kgf} \end{aligned}$$

$$\begin{aligned} \text{Failure Path [Path 2-2]:} \\ &= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ &= (66761 + 0 + 604934 + 0) = 671695 \text{ Kgf} \end{aligned}$$

$$\begin{aligned} \text{Failure Path [Path 3-3]:} \\ &= (\text{Sonw} + \text{Tngw} + \text{Sinw}) \\ &= (66761 + 604934 + 0) = 671695 \text{ Kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 878584 Kgf, must exceed W = 339070 Kgf	or W1 = 367970 Kgf
Path 2-2 = 671694 Kgf, must exceed W = 339070 Kgf	or W2 = 571783 Kgf
Path 3-3 = 671694 Kgf, must exceed W = 339070 Kgf	or W3 = 571783 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 156 bars

The Drop for this Nozzle is : 94.9859 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 468.9858 mm.

Input Echo, WRC297 Item 1, Description: TS OUT NOZ. T2:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	1095.000	mm.
Shell Thickness	74.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm.
Nozzle Diameter	406.400	mm.
Nozzle Thickness	105.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-350 LF2	
Nozzle Cold S.I. Allowable	SNmc	138.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	138.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSstained loads.*

Design Internal Pressure	Dp	147.00	bars
Radial Load	P	-13053.20	Kgf
Circumferential Shear	Vc	9789.91	Kgf
Longitudinal Shear	Vl	13053.20	Kgf
Circumferential Moment	Mc	13570.10	Kg-m.
Longitudinal Moment	Ml	13570.10	Kg-m.
Torsional Moment	Mt	15657.81	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dv/Tv (16.797) is < 20
 check the Limitations of WRC bulletin 297.*

*Warning - The ratio, Dn/Tn (5.870) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$
 $= 616.4 + 2 \cdot 1.65 \cdot \sqrt{586.0(74.0 - 3.0)}$
 $= 1289.519 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1172.000 mm.
Nozzle Outside Diameter	(d) =	616.400 mm.
Vessel Thickness used	(T) =	71.000 mm.
Nozzle Thickness used	(t) =	102.000 mm.
T / t	=	0.696
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.137$$

Nr/P	=	0.074
Mr/P	=	0.128



M0/P = 0.039
 N0/P = 0.099
 MrD/Mc = 0.267
 NrDL/Mc = 0.117
 M0d/Mc = 0.087
 N0DL/Mc = 0.065
 MrD/Ml = 0.110
 NrDL/Ml = 0.066
 M0D/Ml = 0.037
 N0DL/Ml = 0.123

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	5	-5	5	-5
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-5	-5	5	5
Outplane Bending (ML)	-9	9	9	-9
Normal Pressure Stress	107	121	107	121
Outplane Stress Summary	100	122	130	114

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	19	-19	19	-19
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-2	-2	2	2
Inplane Bending (ML)	-28	28	28	-28
Inplane Pressure Stress	53	53	53	53
Inplane Stress Summary	43	61	106	10

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3
Shear Stress Summary	5	5	2	2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	101	123	130	114

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	5	-5	5	-5
Outplane Membrane (Mc)	-2	-2	2	2
Outplane Bending (Mc)	-22	22	22	-22
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0



Normal Pressure Stress	53	53	53	53
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Outplane Stress Summary	36	69	87	30
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	19	-19	19	-19
Inplane Membrane (Mc)	-5	-5	5	5
Inplane Bending (Mc)	-68	68	68	-68
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	107	121	107	121

Inplane Stress Summary	54	167	202	40
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3

Shear Stress Summary	1	1	5	5
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Two * Max Shear Stress	54	167	202	42
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-5	-5	5	5
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	28	46	28	46

Hoop Stress Summary	25	43	36	53
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	6	-6	6	-6
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-7	-7	7	7
Axial Bending (ML)	-9	9	9	-9
Axial Pressure Stress	14	14	14	14

Axial Stress Summary	4	10	38	5
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
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Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	2	2	2	2

Shear Stress Summary	3	3	1	1
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	26	43	39	54
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-2	-2	2	2
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	28	46	28	46

Hoop Stress Summary	28	45	33	51
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	6	-6	6	-6
Axial Membrane (Mc)	-7	-7	7	7
Axial Bending (Mc)	-25	25	25	-25
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	14	14	14	14

Axial Stress Summary	-11	27	55	-10
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	2	2	2	2

Shear Stress Summary	1	1	3	3
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	39	46	55	62
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WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	107	121	107	121	107	121	107	121
Circ. Pl (SUS)	-3	-3	7	7	-4	-4	6	6
Circ. Q (SUS)	-4	4	14	-14	-49	49	87	-87

Long. Pm (SUS)	53	53	53	53	53	53	53	53
Long. Pl (SUS)	-1	-1	3	3	0	0	4	4
Long. Q (SUS)	-9	9	47	-47	-17	17	27	-27
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	107.0	121.0	107.0	121.0	107.0	121.0	107.0	121.0
Pm+Pl (SUS)	104.0	118.0	114.0	128.0	103.0	117.0	113.0	127.0
Pm+Pl+Q (Total)	100.3	122.3	128.2	114.0	54.2	166.0	200.1	41.4

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	121.00	138.00	Passed
Pm+Pl (SUS)	128.01	207.00	Passed
Pm+Pl+Q (TOTAL)	200.14	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	28	46	28	46	28	46	28	46
Circ. Pl (SUS)	-3	-3	7	7	0	0	4	4
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	14	14	14	14	14	14	14	14
Long. Pl (SUS)	-7	-7	7	7	-7	-7	7	7
Long. Q (SUS)	-3	3	15	-15	-19	19	31	-31
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	-1	-1	1	1
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	28.0	46.0	28.0	46.0	28.0	46.0	28.0	46.0
Pm+Pl (SUS)	25.0	43.0	35.0	53.0	28.0	46.0	32.1	50.0
Pm+Pl+Q (Total)	25.2	43.1	37.6	53.1	40.0	46.0	52.4	60.3

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	46.00	138.00	Passed
Pm+Pl (SUS)	53.00	207.00	Passed
Pm+Pl+Q (TOTAL)	60.30	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: DRAIN NOZ. D1

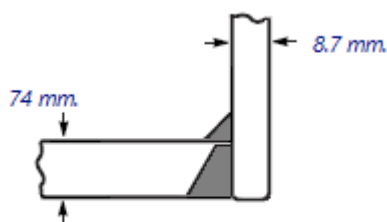
From: 30

Pressure for Reinforcement Calculations	P	147.094	bars
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1095.00	mm.
Shell Finished (Minimum) Thickness	t	74.0000	mm.
Shell Internal Corrosion Allowance	c	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		1314.17	mm.
User Entered Minimum Design Metal Temperature		-28.89	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		-90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Flange Material		SA-350 LF2 +IT	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	60.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		1500	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: DRAIN NOZ. D1

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.687	in.
Actual Thickness Used in Calculation	0.344	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 147 \cdot 550 / (138 \cdot 1 - 0.6 \cdot 147)$
 $= 62.6917$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 147 \cdot 24.4 / (138 \cdot 1 - 0.6 \cdot 147)$
 $= 2.7815$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	202.3250	mm.
Parallel to Vessel Wall	Rn+tn+tt	101.1625	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	14.3440	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: DRAIN NOZ. D1.
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 5.7815	mm.
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 65.6917	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 65.6917	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm.
Wall Thickness per table UG-45	tb3	= 6.4200	mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.42, \max(65.7, 4.5)]$
 $= 6.4200$ mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(5.78, 6.42)$
 $= 6.4200$ mm.

Available Nozzle Neck Thickness = $0.875 \cdot 8.738 = 7.645$ mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 7.65, tr = 2.78, c = 3 mm., $E^* = 1$
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.6$, Temp. Reduction = 23 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Weld Size Calculations, Description: DRAIN NOZ. D1

Intermediate Calc. for nozzle/shell Welds	Tmin	5.7376	mm.
---	------	--------	-----

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.0163 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o$ mm.

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

The Drop for this Nozzle is : 0.8315 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 374.8315 mm.





Input, Nozzle Desc: SH. IN NOZ. S1

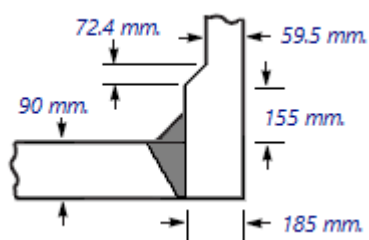
From: 40

Pressure for Reinforcement Calculations	P	191.000	bars
Temperature for Internal Pressure	Temp	200	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1095.00	mm.
Shell Finished (Minimum) Thickness	t	90.0000	mm.
Shell Internal Corrosion Allowance	c	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		3114.18	mm.
User Entered Minimum Design Metal Temperature		-28.89	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	59.5400	mm.
Flange Material		SA-350 LF2 +IT	
Hub Height of Integral Nozzle	h	155.0000	mm.
Height of Beveled Transition	L'	72.4343	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		185.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		80.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		1500	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: SH. IN NOZ. S1

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 24.000 in.
Actual Thickness Used in Calculation 2.344 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $191 \cdot 550 / (138 \cdot 1 - 0.6 \cdot 191)$
= 83.0996 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $191 \cdot 308 / (138 \cdot 1 - 0.6 \cdot 191)$
= 46.4633 mm.

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= $125 / 0.5773$
= 217.3032 mm.

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

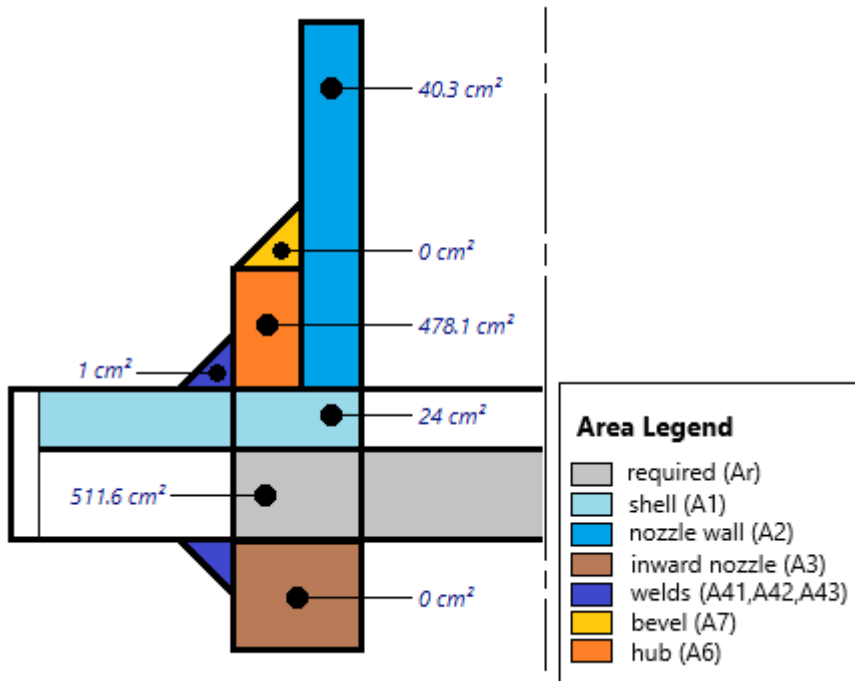
UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1231.2001	mm.
Parallel to Vessel Wall, opening length	d	615.6000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	217.5000	mm.

Weld Strength Reduction Factor [fr1]:
= min(1, S_n / S_v)
= min(1, $138 / 138$)
= 1.000

Weld Strength Reduction Factor [fr2]:
= min(1, S_n / S_v)
= min(1, $138 / 138$)
= 1.000

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(1, 1)
= 1.000



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	511.561	NA	NA
Area in Shell	A1	24.011	NA	NA
Area in Nozzle Wall	A2	40.307	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	478.093	NA	NA
TOTAL AREA AVAILABLE	Atot	543.410	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (616 \cdot 83.1 \cdot 1 + 2 \cdot 182 \cdot 83.1 \cdot 1 \cdot (1 - 1))$
 $= 511.561 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 616(1 \cdot 87 - 1 \cdot 83.1) - 2 \cdot 182$
 $(1 \cdot 87 - 1 \cdot 83.1) \cdot (1 - 1)$
 $= 24.011 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot ho)(tn - trn)fr2$
 $= (2 \cdot 200)(56.5 - 46.5)1$
 $= 40.307 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10^2 \cdot 1 + (0)^2 \cdot 1$
 $= 1.000 \text{ cm}^2$

Area Available in the Hub Section [A6]:
 $= (2 \cdot \min(T_{lp}, h_o, Hubht)) \cdot (Hubtk - t_n) \cdot fr2$
 $= (2 \cdot \min(200, 200, 155)) \cdot (185 - 59.5) \cdot 1$
 $= 478.093 \text{ cm}^2$ Includes Bevel Area of 89.167 cm^2

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 49.4633 \text{ mm.}$
 Wall Thickness per UG16(b), $tr16b = 4.5000 \text{ mm.}$
 Wall Thickness, shell/head, internal pressure $trb1 = 86.0996 \text{ mm.}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 86.0996 \text{ mm.}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 4.5000 \text{ mm.}$
 Wall Thickness per table UG-45 $tb3 = 11.3312 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[11.3, \max(86.1, 4.5)]$
 $= 11.3312 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(49.5, 11.3)$
 $= 49.4633 \text{ mm.}$

Available Nozzle Neck Thickness = 59.5400 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	68.7,	Allowable	:	138.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	276.3 N./mm ²	Passed
Occasional	:	47.6,	Allowable	:	183.5 N./mm ²	Passed
Shear	:	14.7,	Allowable	:	96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification $-46 \text{ }^{\circ}\text{C}$

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $t_g = 90$, $t_r = 83.1$, $c = 3 \text{ mm.}$, $E* = 1$
 Thickness Ratio $= tr \cdot E* / (t_g - c) = 0.96$, Temp. Reduction $= 2 \text{ }^{\circ}\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve B	$29 \text{ }^{\circ}\text{C}$
Min Metal Temp. at Required thickness (UCS 66.1)	$27 \text{ }^{\circ}\text{C}$

Gov. MDMT of the nozzle to shell joint welded assembly : $27 \text{ }^{\circ}\text{C}$

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification	$-46 \text{ }^{\circ}\text{C}$
Flange MDMT with Temp reduction per UCS-66(i)(2)	$-60 \text{ }^{\circ}\text{C}$

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
 Design Pressure/Ambient Rating $= 191.00/255.10 = 0.749$

Intermediate Hub Nozzle Calculations:

The Diameter Limit is less than the Hub OD !

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:

$$= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$$

$$= 97.4/0.5773$$

$$= 168.6326 \text{ mm.}$$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 923.4000 mm.
 Parallel to Vessel Wall, opening length d 461.7000 mm.
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 217.5000 mm.

Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (cm²)

AREA AVAILABLE, A1 to A6		No Pad	With Pad
Area Required [Internal Pressure]	AR	341.058	341.058
Area Available in Shell	A1	12.005	12.005
Area Available in Nozzle Wall	A2	40.307	40.307
Area Available in Inward Nozzle	A3	0.000	0.000
Area Available in Welds	A4	-0.000	-0.000
Area Available in Pad	A5	0.000	0.000
Area Available in Hub	A6	371.012	371.012
TOTAL AREA AVAILABLE	Atot	423.324	423.324

The area available without a pad is Sufficient.

The Nozzle does not fall within the scope of the diameter requirements of App. 1-7(b). Therefore, the Membrane and Bending stress checks are not required.

Weld Size Calculations, Description: SH. IN NOZ. S1

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (512 - 24 + 2 \cdot 182 \cdot 1 \cdot (1 \cdot 87 - 83.1)) \cdot 138)$$

$$= 706037.88 \text{ Kg}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A4-(W_i \cdot \text{Can}/.707)^2 \cdot f_r2) \cdot S_v$$

$$= (411 + 1 - 0 \cdot 1) \cdot 138$$

$$= 580196.62 \text{ Kg}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v$$

$$= (411 + 0 + 1 + (317)) \cdot 138$$

$$= 1025815.12 \text{ Kg}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$$

$$= (411 + 0 + 1 + 0 + (317)) \cdot 138$$

$$= 1025815.12 \text{ Kg}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 980 \cdot 10 \cdot 0.49 \cdot 138$$

= 106098. Kgf

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 399) \cdot (185 - 3) \cdot 0.7 \cdot 138$$

$$= 2246038. Kgf$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 980 \cdot (80 - 0) \cdot 0.74 \cdot 138$$

$$= 1281837. Kgf$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (106098 + 2246037) = 2352135 Kgf$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (106098 + 0 + 1281837 + 0) = 1387935 Kgf$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (106098 + 1281837 + 0) = 1387935 Kgf$$

Summary of Failure Path Calculations:

Path 1-1 = 2352135 Kgf, must exceed W = 706037 Kgf or W1 = 580196 Kgf
 Path 2-2 = 1387935 Kgf, must exceed W = 706037 Kgf or W2 = 1025815 Kgf
 Path 3-3 = 1387935 Kgf, must exceed W = 706037 Kgf or W3 = 1025815 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 196 bars

The Drop for this Nozzle is : 302.8528 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 592.8528 mm.

Warning:

Review Appendix 1-7 (b)(3) for notes regarding externally applied loads to large nozzles and U-2(g).

Input Echo, WRC297 Item 1, Description: SH. IN NOZ. S1:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	1095.000	mm.
Shell Thickness	90.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm.
Nozzle Diameter	609.600	mm.
Nozzle Thickness	185.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-350 LF2	
Nozzle Cold S.I. Allowable	SNmc	138.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	138.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUstained loads.

Design Internal Pressure	Dp	191.00	bars
Radial Load	P	-19579.92	Kgf
Circumferential Shear	Vc	14684.89	Kgf
Longitudinal Shear	Vl	19579.92	Kgf
Circumferential Moment	Mc	30532.79	Kg-m.
Longitudinal Moment	Ml	30532.79	Kg-m.
Torsional Moment	Mt	35230.17	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dv/Tv (14.167) is < 20
 check the Limitations of WRC bulletin 297.*
*Warning - The ratio, Dn/Dv (.768) is > 0.5
 check the Limitations of WRC bulletin 297.*
*Warning - The ratio, Dn/Tn (5.295) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - c_a))}$
 $= 979.6 + 2 \cdot 1.65 \cdot \sqrt{(594.0(90.0 - 3.0))}$
 $= 1729.782 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1188.000 mm.
Nozzle Outside Diameter	(d) =	979.600 mm.
Vessel Thickness used	(T) =	87.000 mm.
Nozzle Thickness used	(t) =	182.000 mm.
T / t	=	0.478
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.047$$

Nr/P	=	0.060
Mr/P	=	0.102
M0/P	=	0.034
N0/P	=	0.084
MrD/Mc	=	0.260
NrDL/Mc	=	0.100
M0d/Mc	=	0.078
N0DL/Mc	=	0.086
MrD/Ml	=	0.093
NrDL/Ml	=	0.047
M0D/Ml	=	0.025
N0DL/Ml	=	0.091

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	5	-5	5	-5
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-3	-3	3	3
Outplane Bending (Ml)	-6	6	6	-6
Normal Pressure Stress	112	131	112	131
Outplane Stress Summary	109	130	129	125

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
--------------------	----	----	----	----



(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	15	-15	15	-15
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-22	22	22	-22
Inplane Pressure Stress	56	56	56	56
Inplane Stress Summary	48	62	97	21

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	109	130	129	125

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	5	-5	5	-5
Outplane Membrane (Mc)	-3	-3	3	3
Outplane Bending (Mc)	-18	18	18	-18
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	56	56	56	56
Outplane Stress Summary	40	68	85	37

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	15	-15	15	-15
Inplane Membrane (Mc)	-4	-4	4	4
Inplane Bending (Mc)	-63	63	63	-63
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	112	131	112	131
Inplane Stress Summary	61	176	196	58

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	2	2	2	2



Shear Stress Summary 1 1 4 4

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	61	176	196	58

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the	Top	Top	Bottom	Bottom
hoop direction)	Outside	Inside	Outside	Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-3	-3	3	3
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	29	52	29	52

Hoop Stress Summary 28 51 35 58

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom
pipe cross-section)	Outside	Inside	Outside	Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	2	-2	2	-2
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-3	-3	3	3
Axial Bending (ML)	-3	3	3	-3
Axial Pressure Stress	14	14	14	14

Axial Stress Summary 10 12 25 12

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1

Shear Stress Summary 1 1 0 0

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside
Two * Max Shear Stress	28	51	35	58

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-3	-3	3	3
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	29	52	29	52

Hoop Stress Summary 28 51 35 58

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	2	-2	2	-2
Axial Membrane (Mc)	-3	-3	3	3
Axial Bending (Mc)	-11	11	11	-11
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	14	14	14	14
Axial Stress Summary	2	20	33	5

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1
Shear Stress Summary	0	0	1	1

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	28	51	36	58

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	112	131	112	131	112	131	112	131
Circ. Pl (SUS)	-1	-1	5	5	-3	-3	5	5
Circ. Q (SUS)	-1	1	11	-11	-48	48	78	-78
Long. Pm (SUS)	56	56	56	56	56	56	56	56
Long. Pl (SUS)	0	0	2	2	-1	-1	5	5
Long. Q (SUS)	-7	7	37	-37	-13	13	23	-23
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	112.0	131.0	112.0	131.0	112.0	131.0	112.0	131.0
Pm+Pl (SUS)	111.0	130.0	117.0	136.0	109.0	128.0	117.0	136.0
Pm+Pl+Q (Total)	110.1	131.1	128.0	125.0	61.1	176.0	195.1	58.4

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	131.00	138.00	Passed
Pm+Pl (SUS)	136.01	207.00	Passed
Pm+Pl+Q (TOTAL)	195.08	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:



Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	29	52	29	52	29	52	29	52
Circ. Pl (SUS)	-1	-1	5	5	-1	-1	5	5
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	14	14	14	14	14	14	14	14
Long. Pl (SUS)	-3	-3	3	3	-3	-3	3	3
Long. Q (SUS)	-1	1	5	-5	-9	9	13	-13
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	29.0	52.0	29.0	52.0	29.0	52.0	29.0	52.0
Pm+Pl (SUS)	28.0	51.0	34.0	57.0	28.0	51.0	34.0	57.0
Pm+Pl+Q (Total)	28.1	51.0	34.1	57.0	28.0	51.0	34.2	57.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	52.00	138.00	Passed
Pm+Pl (SUS)	57.00	207.00	Passed
Pm+Pl+Q (TOTAL)	57.02	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: SH. OUT NOZ. S2

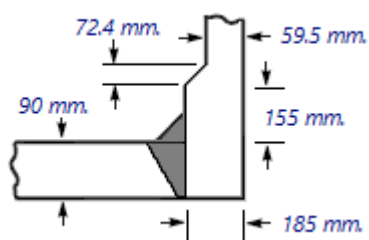
From: 40

Pressure for Reinforcement Calculations	P	191.094	bars
Temperature for Internal Pressure	Temp	200	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1095.00	mm.
Shell Finished (Minimum) Thickness	t	90.0000	mm.
Shell Internal Corrosion Allowance	c	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		10408.18	mm.
User Entered Minimum Design Metal Temperature		-28.89	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		-90.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	59.5400	mm.
Flange Material		SA-350 LF2 +IT	
Hub Height of Integral Nozzle	h	155.0000	mm.
Height of Beveled Transition	L'	72.4343	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		185.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		80.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		1500	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: SH. OUT NOZ. S2

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 24.000 in.
 Actual Thickness Used in Calculation 2.344 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 191 \cdot 550 / (138 \cdot 1 - 0.6 \cdot 191)$
 $= 83.1440$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 191 \cdot 308 / (138 \cdot 1 - 0.6 \cdot 191)$
 $= 46.4882$ mm.

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 125 / 0.5773$
 $= 217.3032$ mm.

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

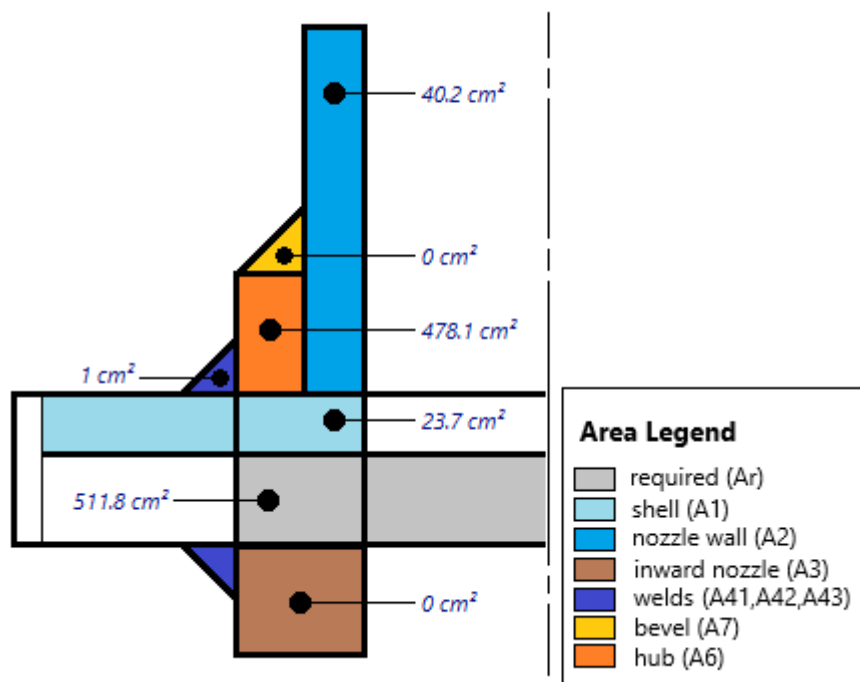
UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1231.2001	mm.
Parallel to Vessel Wall, opening length	d	615.6000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	217.5000	mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	511.835	NA	NA
Area in Shell	A1	23.737	NA	NA
Area in Nozzle Wall	A2	40.207	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds A41+A42+A43		1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	478.093	NA	NA
TOTAL AREA AVAILABLE	Atot	543.037	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (616 \cdot 83.1 \cdot 1 + 2 \cdot 182 \cdot 83.1 \cdot 1 \cdot (1 - 1))$
 $= 511.835 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 616(1 \cdot 87 - 1 \cdot 83.1) - 2 \cdot 182$
 $(1 \cdot 87 - 1 \cdot 83.1) \cdot (1 - 1)$
 $= 23.737 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot ho)(tn - trn)fr2$
 $= (2 \cdot 200)(56.5 - 46.5)1$
 $= 40.207 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10^2 \cdot 1 + (0)^2 \cdot 1$
 $= 1.000 \text{ cm}^2$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lp}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot \text{fr2}$$

$$= (2 \cdot \min(200, 200, 155)) \cdot (185 - 59.5) \cdot 1$$

$$= 478.093 \text{ cm}^2 \text{ Includes Bevel Area of } 89.167 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 49.4882 mm.
Wall Thickness per UG16(b),	tr16b = 4.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 86.1440 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 86.1440 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm.
Wall Thickness per table UG-45	tb3 = 11.3312 mm.

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.3, \max(86.1, 4.5)]$$

$$= 11.3312 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(49.5, 11.3)$$

$$= 49.4882 \text{ mm.}$$

Available Nozzle Neck Thickness = 59.5400 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 68.4,	Allowable	: 138.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 276.6 N./mm ²	Passed
Occasional	: 47.6,	Allowable	: 183.5 N./mm ²	Passed
Shear	: 14.7,	Allowable	: 96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 90, tr = 83.1, c = 3 mm., E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.96, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	27 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 27 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification	-46 °C
Flange MDMT with Temp reduction per UCS-66(i)(2)	-59 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :

$$\text{Design Pressure/Ambient Rating} = 191.09/255.10 = 0.749$$

Intermediate Hub Nozzle Calculations:

The Diameter Limit is less than the Hub OD !

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:

$$= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$$

$$= 97.4/0.5773$$

$$= 168.6326 \text{ mm.}$$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 923.4000 mm.
 Parallel to Vessel Wall, opening length d 461.7000 mm.
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 217.5000 mm.

Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (cm²)

AREA AVAILABLE, A1 to A6		No Pad	With Pad
Area Required [Internal Pressure]	AR	341.240	341.240
Area Available in Shell	A1	11.869	11.869
Area Available in Nozzle Wall	A2	40.207	40.207
Area Available in Inward Nozzle	A3	0.000	0.000
Area Available in Welds	A4	-0.000	-0.000
Area Available in Pad	A5	0.000	0.000
Area Available in Hub	A6	371.012	371.012
TOTAL AREA AVAILABLE	Atot	423.088	423.088

The area available without a pad is Sufficient.

The Nozzle does not fall within the scope of the diameter requirements of App. 1-7(b). Therefore, the Membrane and Bending stress checks are not required.

Weld Size Calculations, Description: SH. OUT NOZ. S2

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (512 - 23.7 + 2 \cdot 182 \cdot 1 \cdot (1 \cdot 87 - 83.1)) \cdot 138)$$

$$= 706580.00 \text{ Kg}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A4-(W_i \cdot \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (411 + 1 - 0 \cdot 1) \cdot 138$$

$$= 580056.81 \text{ Kg}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (411 + 0 + 1 + (317)) \cdot 138$$

$$= 1025675.31 \text{ Kg}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (411 + 0 + 1 + 0 + (317)) \cdot 138$$

$$= 1025675.31 \text{ Kg}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 980 \cdot 10 \cdot 0.49 \cdot 138$$

= 106098. Kgf

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo}) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 399) \cdot (185 - 3) \cdot 0.7 \cdot 138$$

$$= 2246038. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 980 \cdot (80 - 0) \cdot 0.74 \cdot 138$$

$$= 1281837. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (106098 + 2246037) = 2352135 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (106098 + 0 + 1281837 + 0) = 1387935 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (106098 + 1281837 + 0) = 1387935 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 2352135 Kgf, must exceed W = 706580 Kgf or W1 = 580056 Kgf
 Path 2-2 = 1387935 Kgf, must exceed W = 706580 Kgf or W2 = 1025675 Kgf
 Path 3-3 = 1387935 Kgf, must exceed W = 706580 Kgf or W3 = 1025675 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 196 bars

The Drop for this Nozzle is : 302.8528 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 592.8528 mm.

Warning:

Review Appendix 1-7 (b)(3) for notes regarding externally applied loads to large nozzles and U-2(g).

Input Echo, WRC297 Item 1, Description: SH. OUT NOZ. S:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	1095.000	mm.
Shell Thickness	90.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm.
Nozzle Diameter	609.600	mm.
Nozzle Thickness	185.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-350 LF2	
Nozzle Cold S.I. Allowable	SNmc	138.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	138.00 N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUstained loads.

Design Internal Pressure	Dp	191.09	bars
Radial Load	P	-19579.92	Kgf
Circumferential Shear	Vc	14684.89	Kgf
Longitudinal Shear	Vl	19579.92	Kgf
Circumferential Moment	Mc	30094.50	Kg-m.
Longitudinal Moment	Ml	30094.50	Kg-m.
Torsional Moment	Mt	35230.17	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dv/Tv (14.167) is < 20
 check the Limitations of WRC bulletin 297.*
*Warning - The ratio, Dn/Dv (.768) is > 0.5
 check the Limitations of WRC bulletin 297.*
*Warning - The ratio, Dn/Tn (5.295) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 979.6 + 2 \cdot 1.65 \cdot \sqrt{(594.0(90.0 - 3.0))}$
 $= 1729.782 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1188.000 mm.
Nozzle Outside Diameter	(d) =	979.600 mm.
Vessel Thickness used	(T) =	87.000 mm.
Nozzle Thickness used	(t) =	182.000 mm.
T / t	=	0.478
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.047$$

Nr/P	=	0.060
Mr/P	=	0.102
M0/P	=	0.034
N0/P	=	0.084
MrD/Mc	=	0.260
NrDL/Mc	=	0.100
M0d/Mc	=	0.078
N0DL/Mc	=	0.086
MrD/Ml	=	0.093
NrDL/Ml	=	0.047
M0D/Ml	=	0.025
N0DL/Ml	=	0.091

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	5	-5	5	-5
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-3	-3	3	3
Outplane Bending (Ml)	-5	5	5	-5
Normal Pressure Stress	112	131	112	131
Outplane Stress Summary	109	130	128	125

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
--------------------	----	----	----	----



(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	15	-15	15	-15
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-22	22	22	-22
Inplane Pressure Stress	56	56	56	56
Inplane Stress Summary	48	62	97	21

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	109	130	129	125

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	5	-5	5	-5
Outplane Membrane (Mc)	-3	-3	3	3
Outplane Bending (Mc)	-18	18	18	-18
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	56	56	56	56
Outplane Stress Summary	41	68	85	37

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	15	-15	15	-15
Inplane Membrane (Mc)	-3	-3	3	3
Inplane Bending (Mc)	-62	62	62	-62
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	112	131	112	131
Inplane Stress Summary	62	175	195	58

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	2	2	2	2



Shear Stress Summary 1 1 4 4

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	62	175	195	59

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-3	-3	3	3
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	29	52	29	52

Hoop Stress Summary 28 51 35 58

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	2	-2	2	-2
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-3	-3	3	3
Axial Bending (ML)	-3	3	3	-3
Axial Pressure Stress	14	14	14	14

Axial Stress Summary 10 12 25 12

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1

Shear Stress Summary 1 1 0 0

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	28	51	35	58

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-3	-3	3	3
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	29	52	29	52

Hoop Stress Summary 28 51 35 58

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	2	-2	2	-2
Axial Membrane (Mc)	-3	-3	3	3
Axial Bending (Mc)	-11	11	11	-11
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	14	14	14	14
Axial Stress Summary	2	20	33	5

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1
Shear Stress Summary	0	0	1	1

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	28	51	36	58

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	112	131	112	131	112	131	112	131
Circ. Pl (SUS)	-1	-1	5	5	-2	-2	4	4
Circ. Q (SUS)	0	0	10	-10	-47	47	77	-77
Long. Pm (SUS)	56	56	56	56	56	56	56	56
Long. Pl (SUS)	0	0	2	2	-1	-1	5	5
Long. Q (SUS)	-7	7	37	-37	-13	13	23	-23
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	112.0	131.0	112.0	131.0	112.0	131.0	112.0	131.0
Pm+Pl (SUS)	111.0	130.0	117.0	136.0	110.0	129.0	116.0	135.0
Pm+Pl+Q (Total)	111.1	130.1	127.0	126.0	63.0	176.0	193.1	58.4

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	131.00	138.00	Passed
Pm+Pl (SUS)	136.01	207.00	Passed
Pm+Pl+Q (TOTAL)	193.08	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	29	52	29	52	29	52	29	52
Circ. Pl (SUS)	-1	-1	5	5	-1	-1	5	5
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	14	14	14	14	14	14	14	14
Long. Pl (SUS)	-3	-3	3	3	-3	-3	3	3
Long. Q (SUS)	-1	1	5	-5	-9	9	13	-13
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	29.0	52.0	29.0	52.0	29.0	52.0	29.0	52.0
Pm+Pl (SUS)	28.0	51.0	34.0	57.0	28.0	51.0	34.0	57.0
Pm+Pl+Q (Total)	28.1	51.0	34.1	57.0	28.0	51.0	34.2	57.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	52.00	138.00	Passed
Pm+Pl (SUS)	57.00	207.00	Passed
Pm+Pl+Q (TOTAL)	57.02	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: TS. IN NOZ. T1

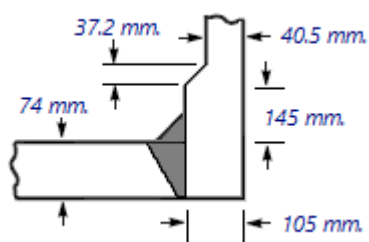
From: 50

Pressure for Reinforcement Calculations	P	147.094	bars
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1095.00	mm.
Shell Finished (Minimum) Thickness	t	74.0000	mm.
Shell Internal Corrosion Allowance	c	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		12258.17	mm.
User Entered Minimum Design Metal Temperature		-28.89	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		-90.00	deg
Diameter		16.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	40.4900	mm.
Flange Material		SA-350 LF2 +IT	
Hub Height of Integral Nozzle	h	145.0000	mm.
Height of Beveled Transition	L'	37.2448	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		105.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		60.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		1500	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: TS. IN NOZ. T1

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 16.000 in.
 Actual Thickness Used in Calculation 1.594 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 147 \cdot 550 / (138 \cdot 1 - 0.6 \cdot 147)$
 $= 62.6917$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 147 \cdot 206 / (138 \cdot 1 - 0.6 \cdot 147)$
 $= 23.4823$ mm.

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 64.5 / 0.5773$
 $= 111.7346$ mm.

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

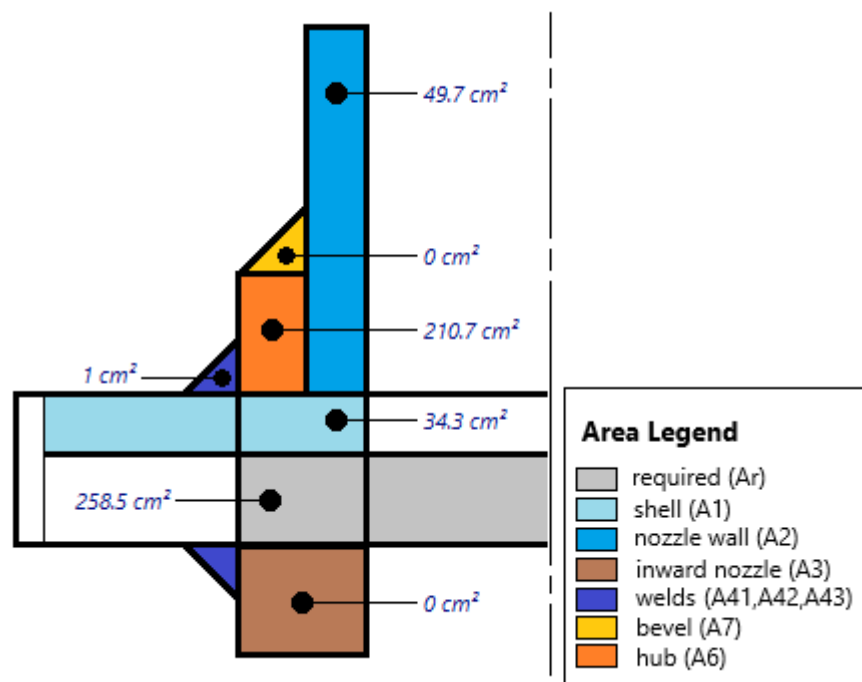
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	824.8000	mm.
Parallel to Vessel Wall, opening length	d	412.4000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	177.5000	mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	258.541	NA	NA
Area in Shell	A1	34.263	NA	NA
Area in Nozzle Wall	A2	49.727	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	210.716	NA	NA
TOTAL AREA AVAILABLE	Atot	295.706	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (412 \cdot 62.7 \cdot 1 + 2 \cdot 102 \cdot 62.7 \cdot 1 \cdot (1 - 1))$$

$$= 258.541 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 412(1 \cdot 71 - 1 \cdot 62.7) - 2 \cdot 102(1 \cdot 71 - 1 \cdot 62.7) \cdot (1 - 1)$$

$$= 34.263 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 178)(37.5 - 23.5)1$$

$$= 49.727 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 1.000 \text{ cm}^2$$



Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, Hubht)) \cdot (Hubtk - t_n) \cdot fr2$$

$$= (2 \cdot \min(178, 300, 145)) \cdot (105 - 40.5) \cdot 1$$

$$= 210.716 \text{ cm}^2 \text{ Includes Bevel Area of } 23.637 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 26.4823 \text{ mm.}$
 Wall Thickness per UG16(b), $tr16b = 4.5000 \text{ mm.}$
 Wall Thickness, shell/head, internal pressure $trb1 = 65.6917 \text{ mm.}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 65.6917 \text{ mm.}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 4.5000 \text{ mm.}$
 Wall Thickness per table UG-45 $tb3 = 11.3312 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[11.3, \max(65.7, 4.5)]$
 $= 11.3312 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(26.5, 11.3)$
 $= 26.4823 \text{ mm.}$

Available Nozzle Neck Thickness = 40.4900 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	68.6,	Allowable	:	138.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	276.4 N./mm ²	Passed
Occasional	:	37.1,	Allowable	:	183.5 N./mm ²	Passed
Shear	:	21.9,	Allowable	:	96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, $t_g = 74$, $t_r = 62.7$, $c = 3 \text{ mm.}$, $E* = 1$
 Thickness Ratio = $t_r \cdot E* / (t_g - c) = 0.88$, Temp. Reduction = 7 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 25 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 19 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 19 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification -46 °C
 Flange MDMT with Temp reduction per UCS-66(i)(2) -70 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
 Design Pressure/Ambient Rating = $147.09 / 255.10 = 0.577$

Weld Size Calculations, Description: TS. IN NOZ. T1

Intermediate Calc. for nozzle/shell Welds $T_{min} \quad 19.0000 \text{ mm.}$

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld $6.0000 = \text{Min per Code } 7.0700 = 0.7 \cdot W_o \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (259 - 34.3 + 2 \cdot 102 \cdot 1 \cdot (1 \cdot 71 - 62.7)) \cdot 138)$$

$$= 339443.09 \text{ Kgf}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (260 + 1 - 0 \cdot 1) \cdot 138$$

$$= 367891.09 \text{ Kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (260 + 0 + 1 + (145)) \cdot 138$$

$$= 571703.69 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (260 + 0 + 1 + 0 + (145)) \cdot 138$$

$$= 571703.69 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 616 \cdot 10 \cdot 0.49 \cdot 138$$

$$= 66761. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 257) \cdot (105 - 3) \cdot 0.7 \cdot 138$$

$$= 811823. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{g_{nv}} \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 616 \cdot (60 - 0) \cdot 0.74 \cdot 138$$

$$= 604934. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (66761 + 811823) = 878584 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (66761 + 0 + 604934 + 0) = 671695 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$

$$= (66761 + 604934 + 0) = 671695 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 878584 Kgf, must exceed W = 339443 Kgf or W1 = 367891 Kgf
 Path 2-2 = 671694 Kgf, must exceed W = 339443 Kgf or W2 = 571703 Kgf
 Path 3-3 = 671694 Kgf, must exceed W = 339443 Kgf or W3 = 571703 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 155 bars

The Drop for this Nozzle is : 94.9859 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 468.9858 mm.

Input Echo, WRC297 Item 1, Description: TS. IN NOZ. T1:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	1095.000	mm.
Shell Thickness	74.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm.
Nozzle Diameter	406.400	mm.
Nozzle Thickness	105.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-350 LF2	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUStained loads.*

Design Internal Pressure	Dp	147.09	bars
Radial Load	P	-13053.20	Kgf
Circumferential Shear	Vc	9789.91	Kgf
Longitudinal Shear	Vl	13053.20	Kgf
Circumferential Moment	Mc	13570.10	Kg-m.
Longitudinal Moment	Ml	13570.10	Kg-m.
Torsional Moment	Mt	15657.81	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dv/Tv (16.797) is < 20
 check the Limitations of WRC bulletin 297.*

*Warning - The ratio, Dn/Tn (5.870) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 616.4 + 2 \cdot 1.65 \cdot \sqrt{(586.0(74.0 - 3.0))}$
 $= 1289.519 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1172.000	mm.
Nozzle Outside Diameter	(d) =	616.400	mm.
Vessel Thickness used	(T) =	71.000	mm.
Nozzle Thickness used	(t) =	102.000	mm.
T / t	=	0.696	
d / t	=	10.000	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.137$$

Nr/P	=	0.074
Mr/P	=	0.128
M0/P	=	0.039



NØ/P = 0.099
 MrD/Mc = 0.267
 NrDL/Mc = 0.117
 MØd/Mc = 0.087
 NØDL/Mc = 0.065
 MrD/Ml = 0.110
 NrDL/Ml = 0.066
 MØD/Ml = 0.037
 NØDL/Ml = 0.123

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses Normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	5	-5	5	-5
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-5	-5	5	5
Outplane Bending (ML)	-9	9	9	-9
Normal Pressure Stress	107	121	107	121
Outplane Stress Summary	100	122	130	114

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	19	-19	19	-19
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-2	-2	2	2
Inplane Bending (ML)	-28	28	28	-28
Inplane Pressure Stress	53	53	53	53
Inplane Stress Summary	43	61	106	10

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3
Shear Stress Summary	5	5	2	2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	101	123	130	114

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	5	-5	5	-5
Outplane Membrane (Mc)	-2	-2	2	2
Outplane Bending (Mc)	-22	22	22	-22
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	53	53	53	53



Outplane Stress Summary	36	69	87	30
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	19	-19	19	-19
Inplane Membrane (Mc)	-5	-5	5	5
Inplane Bending (Mc)	-68	68	68	-68
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	107	121	107	121

Inplane Stress Summary	54	167	202	40
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3

Shear Stress Summary	1	1	5	5
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	55	167	202	42
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the	Top	Top	Bottom	Bottom
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-5	-5	5	5
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	28	46	28	46

Hoop Stress Summary	25	43	36	54
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	6	-6	6	-6
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-7	-7	7	7
Axial Bending (ML)	-9	9	9	-9
Axial Pressure Stress	14	14	14	14

Axial Stress Summary	4	10	38	5
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside



Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	2	2	2	2
Shear Stress Summary	3	3	1	1

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	26	43	39	54

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-2	-2	2	2
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	28	46	28	46
Hoop Stress Summary	28	45	33	51

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	6	-6	6	-6
Axial Membrane (Mc)	-7	-7	7	7
Axial Bending (Mc)	-25	25	25	-25
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	14	14	14	14
Axial Stress Summary	-11	27	55	-10

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	2	2	2	2
Shear Stress Summary	1	1	3	3

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	39	46	55	62

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	107	121	107	121	107	121	107	121
Circ. Pl (SUS)	-3	-3	7	7	-4	-4	6	6
Circ. Q (SUS)	-4	4	14	-14	-49	49	87	-87



Long. Pm (SUS)	53	53	53	53	53	53	53	53
Long. Pl (SUS)	-1	-1	3	3	0	0	4	4
Long. Q (SUS)	-9	9	47	-47	-17	17	27	-27
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	107.0	121.0	107.0	121.0	107.0	121.0	107.0	121.0
Pm+Pl (SUS)	104.0	118.0	114.0	128.0	103.0	117.0	113.0	127.0
Pm+Pl+Q (Total)	100.3	122.3	128.2	114.0	54.2	166.0	200.1	41.4

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	121.00	138.00	Passed
Pm+Pl (SUS)	128.01	207.00	Passed
Pm+Pl+Q (TOTAL)	200.14	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	28	46	28	46	28	46	28	46
Circ. Pl (SUS)	-3	-3	7	7	0	0	4	4
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	14	14	14	14	14	14	14	14
Long. Pl (SUS)	-7	-7	7	7	-7	-7	7	7
Long. Q (SUS)	-3	3	15	-15	-19	19	31	-31
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	-1	-1	1	1
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	28.0	46.0	28.0	46.0	28.0	46.0	28.0	46.0
Pm+Pl (SUS)	25.0	43.0	35.0	53.0	28.0	46.0	32.1	50.0
Pm+Pl+Q (Total)	25.2	43.1	37.6	53.1	40.0	46.0	52.4	60.3

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	46.00	138.00	Passed
Pm+Pl (SUS)	53.00	207.00	Passed
Pm+Pl+Q (TOTAL)	60.30	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: VENT NOZ. V1

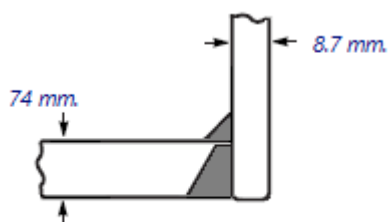
From: 50

Pressure for Reinforcement Calculations	P	147.000	bars
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1095.00	mm.
Shell Finished (Minimum) Thickness	t	74.0000	mm.
Shell Internal Corrosion Allowance	c	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		12108.18	mm.
User Entered Minimum Design Metal Temperature		-28.89	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Flange Material		SA-350 LF2 +IT	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	8.7376	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		1500	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: VENT NOZ. V1

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.687	in.
Actual Thickness Used in Calculation	0.344	in.



Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 147 \cdot 550 / (138 \cdot 1 - 0.6 \cdot 147)$
 $= 62.6491 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 147 \cdot 24.4 / (138 \cdot 1 - 0.6 \cdot 147)$
 $= 2.7797 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	202.3250	mm.
Parallel to Vessel Wall	Rn+tn+tt	101.1625	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	14.3440	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: VENT NOZ. V1.
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 5.7797	mm.
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 65.6491	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 65.6491	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm.
Wall Thickness per table UG-45	tb3	= 6.4200	mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.42, \max(65.6, 4.5)]$
 $= 6.4200 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(5.78, 6.42)$
 $= 6.4200 \text{ mm.}$

Available Nozzle Neck Thickness = $0.875 \cdot 8.738 = 7.645 \text{ mm.} \rightarrow \text{OK}$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 7.65, tr = 2.78, c = 3 mm., $E^* = 1$
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.6$, Temp. Reduction = 23 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Weld Size Calculations, Description: VENT NOZ. V1

Intermediate Calc. for nozzle/shell Welds	Tmin	5.7376	mm.
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.0163 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o \text{ mm.}$

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

The Drop for this Nozzle is : 0.8315 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 374.8315 mm.





Input Echo, Tubesheet Number 1, Description: FIXED TS

Shell Data:

Main Shell Description: MAIN SHELL

Shell Maximum Design Pressure	Psd,max	191.00 bars
Shell Maximum Operating Pressure	Psox,max	191.00 bars
Shell Minimum Operating Pressure	Psox,min	0.00 bars
Shell Thickness	ts	90.0000 mm.
Shell Internal Corrosion Allowance	cas	3.0000 mm.
Shell External Corrosion Allowance	casext	0.0000 mm.
Inside Diameter of Shell	Ds	1095.000 mm.
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	200.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: CH. SHELL-1

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	147.00 bars
Channel Maximum Operating Pressure	Ptox,max	147.00 bars
Channel Minimum Operating Pressure	Ptox,min	0.00 bars
Channel Thickness	tc	74.0000 mm.
Channel Internal Corrosion Allowance	cac	3.0000 mm.
Channel External Corrosion Allowance	cacext	0.0000 mm.
Inside Diameter of Channel	Dc	1095.000 mm.
Channel Design Temperature	TEMPC	85.00 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	468
Tube Wall Thickness	et	3.4100 mm.
Tube Outside Diameter	D	25.4000 mm.
Total Straight Tube Length	Lt	9494.00 mm.
Straight Tube Length (bet. inner tubsht faces) L		9094.00 mm.
Design Temperature of the Tubes		200.00 °C
Tube Material		SA-179
Tube Material UNS Number		K01200
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		92.40 N./mm ²
Tube Allowable Stress At Ambient		92.40 N./mm ²
Tube Yield Stress At design Temperature	Syt	154.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	40.3900 mm.
Tube Layout Pattern		Triangular

Fillet Weld Leg	af	3.0000 mm.
Groove Weld Leg	ag	3.0000 mm.
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No

Radius to Outermost Tube Hole Center	ro	506.085 mm.
Largest Center-to-Center Tube Distance	Ul	0.0000 mm.
Length of Expanded Portion of Tube	ltx	51.0000 mm.
Tube-side pass partition groove depth	hg	0.0000 mm.
Area of Tube Layout	Ap	5976.1 cm ²
Perimeter of Tube Layout	Cp	4005.8 mm.



Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf A

Tubesheet Design Metal Temperature T 200.00 °C
 Tubesheet Material SA-350 LF2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number K03011
 Tubesheet Allowable Stress at Temperature S 138.00 N./mm²
 Tubesheet Allowable Stress at Ambient Tt 138.00 N./mm²
 Thickness of Tubesheet h 200.0000 mm.
 Tubesheet Corr. Allowance (Shell side) Cats 3.0000 mm.
 Tubesheet Corr. Allowance (Channel side) Catc 3.0000 mm.
 Tubesheet Outside Diameter A 1275.000 mm.

Area of the Untubed Lanes AL 0.0 cm²

Additional Data for Fixed/Floating Tubesheet Exchangers:

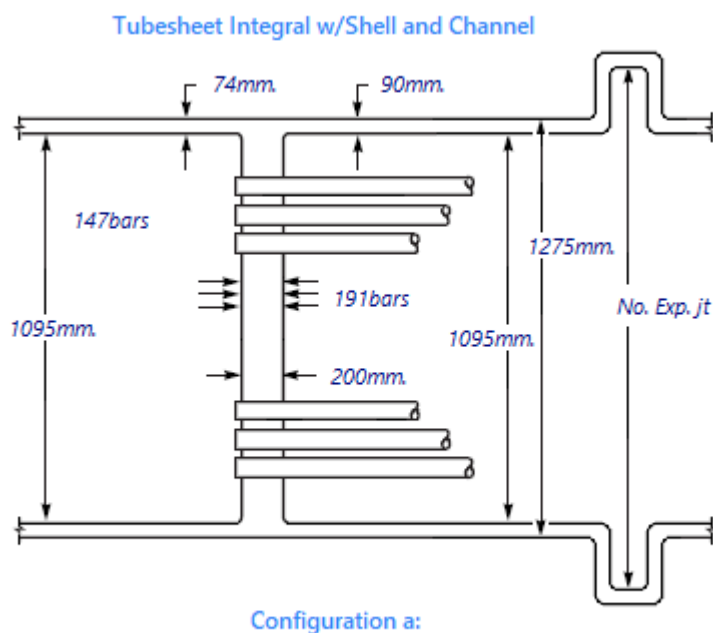
Unsupported Tube Span under consideration l 1608.000 mm.
 Tube End condition corresponding to Span (l) k 1.00

Ignore Radial Thermal Exp. effects (4.18.8/4.18.9) Yes
 Perform Differential Pressure Design No
 Run Multiple Load Cases Yes

Mean Shell Metal Temp. along Shell len. Tsm 75.62 °C
 Mean Tube Metal Temp. along Tube length Ttm 57.65 °C
 Junction Stress Reduction option Perform Plastic Calculation

Tubesheet Integral with Both
 Tubesheet Extended as Flange No

Is Exchanger in Creep range (skip EP, Use 3S for Sps) No



ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1



Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	75.6 °C	0.0000119442 /°C
Elastic Mod. at Design Temperature	200.0 °C	0.19258E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	75.6 °C	0.19958E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
 Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Design Temperature	85.0 °C	0.19904E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
 Tubes - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	57.7 °C	0.0000118278 /°C
Elastic Mod. at Design Temperature	200.0 °C	0.19258E+09 KPa.
Elastic Mod. at Metal Temp. along Len	57.7 °C	0.20061E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	200.0 °C	0.19258E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
 TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	200.0 °C	0.19258E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.
 Please insure these properties are consistent with the
 type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (147 \cdot (25.4/2 - 0)) / (92.4 \cdot 1 + 0.4 \cdot 147)$
 $= 1.8997 + 0.0000 = 1.8997 \text{ mm.}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-1 at 200.00 °C
 Elastic Modulus for Material 193602528.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
3.4100	25.40	9094.00	7.45	50.0000	0.0198260	93.30
EMAWP = $(2.167 / (D/T) - 0.0833) \cdot B = 194 \text{ bars}$						

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
3.3762	25.40	9094.00	7.52	50.0000	0.0194353	93.30
EMAWP = $(2.167 / (D/T) - 0.0833) \cdot B = 191 \text{ bars}$						

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	3.3762 mm.
Allowable Internal Pressure at Corroded thickness	277.93 bars
Required Internal Design Pressure	147.00 bars
Allowable External Pressure at Corroded thickness	193.71 bars
Required External Design Pressure	191.00 bars
Required Thickness due to Shell Side pressure	3.3762 mm.

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)



Results for ASME Fixed Tubesheet Calculations for Configuration a:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.8.4 Step 1:

Tube expansion depth ratio $[\rho]$:

$$= l_{tx} / h \text{ (modified for corrosion if present)}$$

$$= 51/200$$

$$= 0.2550 \text{ (must be } 0 \leq \rho \leq 1 \text{)}$$

Effective tube hole diameter $[d^*]$:

$$= \max(dt - 2 \cdot tt(EtT/E) \cdot (StT/S) \cdot (\rho); dt - 2 \cdot tt)$$

$$= \max(25.4 - 2 \cdot 3.41 (192580832/192580832) \cdot (92.4/138) \cdot (0.25); 25.4 - 2 \cdot 3.41)$$

$$= 24.2356 \text{ mm.}$$

Equivalent outer tube limit circle diameter $[Do]$:

$$= 2 \cdot r_o + dt$$

$$= 2 \cdot 506 + 25.4$$

$$= 1037.570 \text{ mm.}$$

Determine the basic ligament efficiency for shear $[\mu]$:

$$= (p - dt) / p$$

$$= (40.4 - 25.4) / 40.4$$

$$= 0.371$$

Equivalent outer tube limit radius $[ao]$:

$$= Do / 2$$

$$= 1038/2$$

$$= 518.7850 \text{ mm.}$$

Effective tube pitch $[p^*]$:

$$= p / \sqrt{ (1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot Do \cdot p) / (\pi \cdot Do^2)) }$$

$$= 40.4 / \sqrt{ (1 - 4 \cdot \min(0 \cdot 100 ; 4 \cdot 1038 \cdot 40.4) / (\pi \cdot 1038^2)) }$$

$$= 40.3900 \text{ mm.}$$

Effective ligament efficiency for bending $[\mu^*]$:

$$= (p^* - d^*) / p^*$$

$$= (40.4 - 24.2) / 40.4$$

$$= 0.4000$$

Ratio $[ps]$:

$$= as / ao$$

$$= 548/519$$

$$= 1.0554$$

Ratio $[pc]$:

$$= ac / ao$$

$$= 548/519$$

$$= 1.0554$$

Compute parameter $[xt]$:

$$= 1 - Nt((dt - 2 \cdot tt) / (2 \cdot ao))^2$$

$$= 1 - 468 ((25.4 - 2 \cdot 3.41) / (2 \cdot 519))^2$$

$$= 0.8499$$

Determine parameter $[xs]$:

$$= 1 - Nt(dt / (2 \cdot ao))^2$$

$$= 1 - 468 (25.4 / (2 \cdot 519))^2$$

$$= 0.7195$$

Determine the value $[h'g]$:

$$= \max((hg - CATC); 0) \text{ (For pressure only cases)}$$

$$= \max((0 - 0); 0)$$

$$= 0.000 \text{ mm.}$$

ASME Section VIII-2 Part 4.18.8.4 Step 2:

Axial Shell Stiffness $[Ks]$:

$$\begin{aligned}
 &= \pi \cdot ts (Ds + ts) Es / L \\
 &= \pi \cdot 90 (1095 + 90) 192580832 / 9094 \\
 &= 723544.7500 \text{ Kgf/mm.}
 \end{aligned}$$

Axial Tube Stiffness [Kt]:

$$\begin{aligned}
 &= \pi \cdot tt (Dt - tt) Et / L \\
 &= \pi \cdot 3.41 (25.4 - 3.41) 192580832 / 9094 \\
 &= 508.7263 \text{ Kgf/mm.}
 \end{aligned}$$

Stiffness Factor [Ks,t]:

$$\begin{aligned}
 &= Ks / (Nt \cdot Kt) \\
 &= 723545 / (468 \cdot 509) \\
 &= 3.03903
 \end{aligned}$$

Rigidity Ratio [J]:

$$\begin{aligned}
 &= 1 / (1 + Ks/Kj) \\
 &= 1 / (1 + 723545/0) \\
 &= 1.0000 (= 1 \text{ if No Exp. Jt.})
 \end{aligned}$$

Shell coefficient [β_s]:

$$\begin{aligned}
 &= ((12(1 - \nu_s^2))^{1/4}) / ((Ds + ts)ts) \\
 &= ((12(1 - 0.3^2))^{1/4}) / ((1095 + 90)90) \\
 &= 0.0056 \text{ 1/mm.}
 \end{aligned}$$

Shell coefficient [ks]:

$$\begin{aligned}
 &= \beta_s \cdot Es \cdot ts^3 / (6(1 - \nu_s^2)) \\
 &= 0.0056 \cdot 192580832 \cdot 90^3 / (6(1 - 0.3^2)) \\
 &= 1431275136.0000 \text{ bars} \cdot \text{mm.}^2
 \end{aligned}$$

Shell coefficient [λ_s]:

$$\begin{aligned}
 &= 6(Ds + ts) \cdot ks / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2) \\
 &= 6(1095 + 90) \cdot 1431275136 / 200^3 \cdot (1 + 200 \cdot 0.0056 + (200 \cdot 0.0056)^2 / 2) \\
 &= 3476476.7500 \text{ bars}
 \end{aligned}$$

Shell coefficient [δ_s]:

$$\begin{aligned}
 &= Ds^2 / (4 \cdot Es \cdot ts) \cdot (1 - Ds \cdot \nu_s / (2(Ds + ts))) \\
 &= 1095^2 / (4 \cdot 192580832 \cdot 90) \cdot (1 - 1095 \cdot 0.3 / (2(1095 + 90))) \\
 &= 0.14897E-01 \text{ mm./N./mm}^2
 \end{aligned}$$

Channel coefficient [β_c]:

$$\begin{aligned}
 &= ((12(1 - \nu_c^2))^{1/4}) / ((Dc + tc)tc) \\
 &= ((12(1 - 0.3^2))^{1/4}) / ((1095 + 74)74) \\
 &= 0.0062 \text{ 1/mm.}
 \end{aligned}$$

Channel coefficient [kc]:

$$\begin{aligned}
 &= \beta_c \cdot Ec \cdot tc^3 / (6(1 - \nu_c^2)) \\
 &= 0.0062 \cdot 199042256 \cdot 74^3 / (6(1 - 0.3^2)) \\
 &= 913019904.0000 \text{ bars} \cdot \text{mm.}^2
 \end{aligned}$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$\text{CqDelta} = \text{CqTheta} = \text{CmTheta} = \text{CmDelta} = 1$$

Channel coefficient [λ_c]:

$$\begin{aligned}
 &= 6(Dc + tc / \cos(\theta)) \cdot kc / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (Cm\Delta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot Cq\Delta) \\
 &= 6(1095 + 74 / \cos(0)) \cdot 913019904 / 200^3 \cdot (1 + 200 \cdot 0.0062 \cdot (1 + 1) / 2 + (200 \cdot 0.0062)^2 / 2 \cdot 1) \\
 &= 2401573.5 \text{ bars}
 \end{aligned}$$

Channel coefficient [δ_c]:

$$\begin{aligned}
 &= Dc^2 / (4 \cdot Ec \cdot Tc) \cdot (1 - Dc \cdot \nu_c / (2(Dc + tc))) \\
 &= 1095^2 / (4 \cdot 199042256 \cdot 74) \cdot (1 - 1095 \cdot 0.3 / (2(1095 + 74))) \\
 &= 0.17491E-01 \text{ mm./N./mm}^2
 \end{aligned}$$

Channel coefficient [zetac]:

$$\text{zetac} = 0$$

ASME Section VIII-2 Part 4.18.8.4 Step 3:

$$h/p = 4.951721; \mu^* = 0.399961; E^* = 79601176 \text{ KPa.}$$

$$E^*/E = 0.413339 ; v^* = 0.310763$$

Note: As h/p (4.952) is > 2 , data values for $h/p = 2$ were used.

Tube Bundle Stiffness Factor [Xa]:

$$= ((24(1 - v^{*2})Nt \cdot Et \cdot tt(dt - tt)ao^2) / (E^* \cdot L \cdot H^3))^{1/4}$$

$$= ((24(1 - 0.31^2)468 \cdot 192580832 \cdot 3.41$$

$$(25.4 - 3.41)519^2) / (79601176 \cdot 9094 \cdot 200^3))^{1/4}$$

$$= 1.6154$$

Values from Table 4.18.3

$$Zd = 0.347565 ; Zv = 0.178075 ; Zm = 0.724769$$

$$Za = 0.577057E+00 ; Zw = 0.178075$$

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Diameter ratio [K]:

$$= A / Do$$

$$= 1275/1038$$

$$= 1.2288$$

Compute coefficient [F]:

$$= (1 - v^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.31) / 79601176 \cdot (3476476 + 2401573 +$$

$$192580832 \cdot \ln(1.23))$$

$$= 5.4332$$

Compute parameter [ϕ]:

$$= (1 + v^*) F$$

$$= (1 + 0.31) 5.43$$

$$= 7.1216$$

Compute parameter [Q1]:

$$= (ps - 1 - \phi \cdot Zv) / (1 + \phi \cdot Zm)$$

$$= (1.06 - 1 - 7.12 \cdot 0.18) / (1 + 7.12 \cdot 0.72)$$

$$= -0.196839$$

Compute Parameter [Qz1]:

$$= (Zd + Q1 \cdot Zw) / 2 \cdot Xa^4$$

$$= (0.35 + -0.2 \cdot 0.18) / 2 \cdot 1.62^4$$

$$= 1.0641$$

Compute parameter [Qz2]:

$$= (Zv + Q1 \cdot Zm) / 2 \cdot Xa^4$$

$$= (0.18 + -0.2 \cdot 0.72) / 2 \cdot 1.62^4$$

$$= 0.1206$$

Compute parameter [U]:

$$= (Zw + (ps - 1) Zm) Xa^4 / (1 + \phi \cdot Zm)$$

$$= (0.18 + (1.06 - 1) 0.72) 1.62^4 / (1 + 7.12 \cdot 0.72)$$

$$= 0.2412$$

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor [γ_b]:

$$= 0 \text{ (config a)}$$

Compute Parameter [γ]:

$$= 0.000 \text{ mm. (For Pressure only cases)}$$

Calculate parameter [ω_s]:

$$= ps \cdot ks \cdot \beta_s \cdot \delta S (1 + h \cdot \beta_s)$$

$$= 1.06 \cdot 1431275136 \cdot 0.0056 \cdot 0.014896646 (1 + 200 \cdot 0.0056)$$

$$= 26470.7539 \text{ mm.}^2$$

Calculate parameter [ω^*s]:

$$= ao^2(ps^2 - 1) \cdot (ps - 1) / 4 - \omega_s$$

$$= 519^2(1.06^2 - 1) \cdot (1.06 - 1) / 4 - 26471$$

$$= -26047.0664 \text{ mm.}^2$$

Parameter [wc]:

$$= \rho_c \cdot k_c [\delta C (C_m \theta + \beta C + C_q \Delta \cdot h \cdot \beta C^2) - \zeta c (C_m \theta + h \cdot C_q \theta + h \cdot \beta C / 2)]$$

$$= 1.06 \cdot 913019904 [0.017490795 (1 + 0.0062 \cdot 1 \cdot 200) - 0.000 (1/200 + 1 \cdot 0.0062/2)]$$

$$= 23293.8066 \text{ mm.}^2$$

Parameter [ω*c]:

$$= a_o^2 [(\rho_c^2 + 1) \cdot (\rho_c - 1) / 4 - (\rho_s - 1) / 2] - \omega_c$$

$$= 519^2 [(1.06^2 + 1) \cdot (1.06 - 1) / 4 - (1.06 - 1) / 2] - 23294$$

$$= -22870.1191 \text{ mm.}^2$$

Pressure [P*S]:

$$= 0 \text{ For Pressure only cases or Configurations d,e,f,A,B,C,D}$$

Pressure [P*C]:

$$= 0 \text{ For Pressure only cases or Configurations b,c,d,B,C,D}$$

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Pressure [P's]:

$$= P_s \{ x_s + 2(1 - x_s) v_t + [2/Kst \cdot (D_s/D_o)^2] v_s - [(\rho_s^2 - 1) / (J \cdot Kst)] - [(1 - J) / (2J \cdot Kst)] [(D_j^2 - (D_s)^2) / D_o^2] \}$$

$$= 191 \{ 0.72 + 2(1 - 0.72) 0.3 + [2/3.04 \cdot (1095/1038)^2] 0.3 - [(1.06^2 - 1) / (1 \cdot 3.04)] - [(1 - 1) / (2 \cdot 1 \cdot 3.04)] [(0^2 - (1095)^2) / 1038^2] \}$$

$$= 204.4218 \text{ bars}$$

Pressure [P't]:

$$= [x_t + 2(1 - x_t) v_t + 1 / (J \cdot Kst)] P_t$$

$$= [0.85 + 2(1 - 0.85) 0.3 + 1 / (1 \cdot 3.04)] 147$$

$$= 186.5464 \text{ bars}$$

Pressure [P_γ]:

$$= N_t \cdot K_t \cdot \gamma / (\pi \cdot a_o^2)$$

$$= 468 \cdot 509 \cdot 0 / (3.14 \cdot 519^2)$$

$$= 0.0000 \text{ bars}$$

Pressure [P_w]:

$$= -\gamma b \cdot U \cdot W^* / (2 \cdot \pi \cdot a_o^2)$$

$$= -0 \cdot 0.24 \cdot 0 / (2 \cdot \pi \cdot 519^2)$$

$$= -0.0000 \text{ bars}$$

Calculate the pressure [Prim]:

$$= - (U / a_o^2) \cdot (\omega^* s \cdot P_s - \omega^* c \cdot P_t)$$

$$= - (0.24 / 519^2) \cdot (-26047 \cdot 191 - -22870 \cdot 147)$$

$$= 1.4454 \text{ bars}$$

Calculate the pressure [P_ω]:

$$= (U / a_o^2) \cdot (\omega^* s \cdot P^* s - \omega^* c \cdot P^* c)$$

$$= (0.24 / 519^2) \cdot (26471 \cdot 0 - 23294 \cdot 0)$$

$$= 0.0000 \text{ bars}$$

Determine the effective pressure [Pe]:

$$= J \cdot Kst / (1 + J \cdot Kst (Qz1 + (\rho_s - 1) Qz2)) \cdot (P^* s - P^* t + P_\gamma + P_\omega + P_w + P_{\text{Prim}})$$

$$= 0.1000E+01 \cdot 3.04 / (1 + 1 \cdot 3.04 (1.06 + (1.06 - 1) 0.12)) \cdot (204 - 187 + 0 + 0 + -0 + 1.45)$$

$$= 13.8021 \text{ bars}$$

ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor [Q2]:

$$= [((\omega^* s \cdot P_s - \omega^* c \cdot P_t) - (\omega^* s \cdot P^* s - \omega^* c \cdot P^* c)) + W^* \cdot \gamma b / (2 \cdot \pi)] / (1 + \phi \cdot Z_m)$$

$$= [((-26047 \cdot 191 - -22870 \cdot 147) - (26471 \cdot 0 - 23294 \cdot 0)) \cdot 0.01 + 0 \cdot 0 / (2 \cdot \pi)] / (1 + 7.12 \cdot 0.72)$$

$$= -2669.7104 \text{ Kg}$$

Calculate factor [Q3]:
 $= Q1 + 2 \cdot Q2 / (Pe \cdot ao^2)$
 $= -0.2 + 2 \cdot -2670 / (13.8 \cdot 519^2)$
 $= -0.33779$

Fm Value from Table 4.18.3 = 0.168896

Tubesheet Bending Stress - original thickness [σ]:
 $= (1.5 \cdot Fm / \mu \cdot) \cdot (2 \cdot ao / (H - h'g))^2 \cdot Pe$
 $= (1.5 \cdot 0.17 / 0.4) \cdot (2 \cdot 519 / (200 - 0))^2 \cdot 13.8$
 $= 23.5310 \text{ N./mm}^2$

Allowable Tubesheet Bending Stress [σ allowed]:
 $= 1.5 \cdot S = 1.5 \cdot 138 = 207 \text{ N./mm}^2$

Tubesheet Bending Stress - final thickness [σ]:
 $= (1.5 \cdot Fm / \mu \cdot) \cdot (2 \cdot ao / (h - h'g))^2 \cdot Pe$
 $= (1.5 \cdot 0.043 / 0.4) \cdot (2 \cdot 519 / (21.2 - 0))^2 \cdot 5.3$
 $= 206.9870 \text{ N./mm}^2$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:
 $= h + Cats + Catc$
 $= 21.2 + 0 + 0$
 $= 21.1805 \text{ mm.}$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [τ limit]:
 $= (2 \cdot \mu \cdot h / ao) \cdot \min(0.8S; 0.533Sy)$
 $= (2 \cdot 0.37 \cdot 200 / 519) \cdot 110$
 $= 31.5909 \text{ N./mm}^2$

Shear Stress is not required to be computed; [Pe] <= τ limit

Tubesheet Shear Stress is probably low, use the following req. thk:

Tubesheet thickness (Incl. Corr.) = 3.8100 mm.
 Tubesheet Shear Stress = 17.5442 N./mm²

Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd]:
 $= \max(HreqB; HreqS)$
 $= \max(21.2; 3.81)$
 $= 21.1805 \text{ mm.}$

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The F_{tmin} and F_{tmax} Coefficients from Table 4.18.4:
 F_{tmin} = 0.98, F_{tmax} = 1.01

First Extreme Tube Axial Stress from among all the tubes [σ 1]:
 $= ((Ps \cdot xs - Pt \cdot xt) - Pe \cdot F_{tmin}) / (Xt - Xs)$
 $= ((191 \cdot 0.72 - 147 \cdot 0.85) - (13.8) \cdot 0.98) / (0.85 - 0.72)$
 $= -0.7789 \text{ N./mm}^2$

Second Extreme value of Tube Axial Stress from among all the tubes [σ 2]:
 $= ((Ps \cdot xs - Pt \cdot xt) - Pe \cdot F_{tmax}) / (Xt - Xs)$
 $= ((191 \cdot 0.72 - 147 \cdot 0.85) - (13.8) \cdot 1.01) / (0.85 - 0.72)$
 $= -1.0998 \text{ N./mm}^2$

Maximum Tube Axial Stress [σ ,max]:
 $= \max(|\sigma_{t,1}|; |\sigma_{t,2}|) = 1.1 \text{ N./mm}^2$

Allowable Tube Stress [σ ,A]:
 $= Sot = 92.3983 \text{ N./mm}^2$

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:

$$= \max((3.25 - 0.25(Z_d + Q_3 \cdot Z_w)X_{a^4}); 1.25)$$

$$= \max((3.25 - 0.25(0.35 + -0.34 \cdot 0.18)1.62^4); 1.25)$$

$$= 2.0000 \text{ (Should be } \leq 2 \text{)}$$

Determine the factor [rt]:

$$= (\sqrt{ dt^2 + (dt - 2 \cdot tt)^2 }) / 4$$

$$= (\sqrt{ 25.4^2 + (25.4 - 2 \cdot 3.41)^2 }) / 4$$

$$= 7.8676 \text{ mm.}$$

Determine the factor [Ct]:

$$= \sqrt{ (2 \cdot \pi^2 \cdot Et / S_{yt})}$$

$$= \sqrt{ (2 \cdot \pi^2 \cdot 192580832 / 154)}$$

$$= 157.1179$$

Determine the factor [Ft]:

$$= k \cdot L / r$$

$$= 1 \cdot 1608 / 7.87$$

$$= 204.3835$$

Buckling Allowable Stress [Stb]:

$$= (1 / F_s) \cdot (\pi^2 \cdot Et / Ft^2) (Ct \leq Ft)$$

$$= (1 / 2) \cdot (\pi^2 \cdot 192580832 / 204^2)$$

$$= 22.752 \text{ N./mm}^2 \text{ (Never greater than } S_{ot} \text{)}$$

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:

$$= \sigma_{t,max} \cdot \text{Tube Area}$$

$$= 1.1 \cdot 2.36$$

$$= 26.42 \text{ Kgf}$$

Tube Metal Area [At]:

$$= \pi (d_o - t) t$$

$$= \pi (25.4 - 3.41) 3.41$$

$$= 2.356 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 3.41 (25.4 - 3.41) 92.4$$

$$= 2219.550 \text{ Kgf}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > Ft \text{)}$$

$$= 0.85 \cdot \pi \cdot 3 (25.4 + 0.67 \cdot 3) 92.4$$

$$= 2068.8767 \text{ Kgf}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 92.4 / 92.4$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > Ft \text{)}$$

$$= 0.55 \cdot \pi \cdot 3 (25.4 + 0.67 \cdot 3) 92.4$$

$$= 1338.6848 \text{ Kgf}$$

Minimum Length of Weld [ar]:

$$= 2 (\sqrt{ (0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d } - 0.75 \cdot d_o)$$

$$= 2 (\sqrt{ (0.75 \cdot 25.4)^2 + 1.07 \cdot 3.41 \cdot (25.4 - 3.41) 1 \cdot 1 } - 0.75 \cdot 25.4)$$

$$= 4.002 \text{ mm.}$$

Minimum Required Fillet Weld Leg [afr]: 2.0460 mm.
 Minimum Required Groove Weld Leg [agr]: 2.0460 mm.

Tube-Tubesheet Jt allowable, 2220 is >= tube strength 2220 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:
 $= S_s \cdot E_{sw} = 138 \cdot 1 = 138 \text{ N./mm}^2$

Axial Membrane Stress in Shell $[\sigma_s, m]$:
 $= a_o^2 / ((D_s + t_s) t_s) \cdot [P_e + (\rho s^2 - 1)(P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s)$
 $= 519^2 / ((1095 + 90) 90) \cdot [13.8 + (1.06^2 - 1)(191 - 147)] + 548^2 \cdot 147 / ((1095 + 90) 90)$
 $= 46.0656 \text{ N./mm}^2$

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Note:
 For a given Shell thickness of 90 mm., the minimum Shell length adjacent to the tubesheet should be 565.068 mm.

Shell Membrane Stress due to Joint Interaction $[\sigma_s, m]$:
 $= a_o^2 / ((D_s + t_s) t_s) [P_e + (\rho s^2 - 1)(P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s)$
 $= 519^2 / ((1095 + 90) 90) [13.8 + (1.06^2 - 1)(191 - 147)] + 548^2 \cdot 147 / ((1095 + 90) 90)$
 $= 46.0656 \text{ N./mm}^2$

Shell Bending Stress due to Joint Interaction $[\sigma_s, b]$:
 $= 6 \cdot k_s / t_s^2 \{ \beta_s [\delta_s \cdot P_s + a_s^2 \cdot P \cdot s / (E_s \cdot t_s)] + 6(1 - \nu^2) / (E^*) (a_o / h)^3 (1 + h \cdot \beta_s / 2) [P_e (Z_v + Z_m \cdot Q_1) + 2 / a_o^2 \cdot Z_m \cdot Q_2] \}$
 $= 6 \cdot 1431275136 / 90^2 \{ 0.0056 [0.014896646 \cdot 191 + 548^2 \cdot 0 / (17.33E+09)] + 6(1 - 0.31^2) / (79601176) (519 / 200)^3 (1 + 200 \cdot 0.0056 / 2) [13.8 (0.18 + 0.72 \cdot -0.2) + 2 / 519^2 \cdot 0.72 \cdot -2670] \}$
 $= 149.8625 \text{ N./mm}^2$

Shell Stress Summation vs. Allowable
 $:: |\sigma_s, m| + |\sigma_s, b| \leq 1.5 \cdot S_s$
 $:: |46.1| + |150| \leq 207 \text{ N./mm}^2$
 $:: 196, \text{ must be } \leq 207 \text{ N./mm}^2$

Note:
 For a given Channel thickness of 74 mm., the minimum Channel length adjacent to the tubesheet should be 512.384 mm.

Channel Membrane Stress due to Joint Interaction $[\sigma_c, m]$:
 $= a_c^2 \cdot P_t / ((D_c + t_c) t_c \cdot \cos(\theta))$
 $= 548^2 \cdot 147 / ((1095 + 74) 74 \cdot \cos(0))$
 $= 50.941 \text{ N./mm}^2$

Channel Bending Stress due to Joint Interaction $[\sigma_c, b]$:
 $= 6 \cdot k_c / t_c^2 \cdot \{ (\delta_c \cdot C_m \Delta \cdot \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t - 6(1 - \nu^2) / E^* \cdot (a_o / h)^3 \cdot (C_m \theta + h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot [P_e (Z_v + Z_m \cdot Q_1) + 2 / a_o^2 \cdot Z_m \cdot Q_2] \}$
 $= 6 \cdot 913019904 / 74^2 \cdot \{ (0.017490795 \cdot 1 \cdot 0.0062 - 0.000 \cdot 1) \cdot 147 - 6(1 - 0.31^2) / 79601176 \cdot (519 / 200)^3 \cdot (1 + 200 \cdot 0.0062 / 2 \cdot 1) \cdot [13.8 (0.18 + 0.72 \cdot -0.2) + 2 / 519^2 \cdot 0.72 \cdot -2670] \}$
 $= 176.7170 \text{ N./mm}^2$

Channel Stress Summation vs. Allowable
 $:: |\sigma_c, m| + |\sigma_c, b| \leq 1.5 \cdot S_c$
 $:: |50.9| + |177| \leq 207 \text{ N./mm}^2$
 $:: 228 \text{ must be } \leq 207 \text{ N./mm}^2 \text{ (See Elast-Plast iteration below)}$

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
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Tubesheet Bend. Stress	23.5	<=	207.0 N./mm ²	Pass
Tubesheet Shear Stress	2.8	<=	110.4 N./mm ²	Pass
Maximum Tube Stress	1.1	<=	92.4 N./mm ²	Pass
Minimum Tube Stress (Buckling)	-1.1	<=	-22.8 N./mm ²	Pass
Maximum Force on any one Tube	26.4	<=	2219.6 Kgf	Pass
Axial Membrane Stress in Shell	46.1	<=	138.0 N./mm ²	Pass
Shell Stress (jt. inter.)	195.9	<=	207.0 N./mm ²	Pass
Channel Stress (jt. inter.)	227.7	<=	207.0 N./mm ²	See below

Additional Results for the 2nd Elastic-Plastic Iteration.



These calculations are not intended to be used in the creep range. Check Part 4.18 in Section VIII Division 2.

Compute Allowable [Sstars]:
 = min(Sys, SPS,s/2)
 = min(225, 450/2) = 225 N./mm²

Compute Allowable [S*c]:
 = min(Syc, SPS,c/2)
 = min(242, 484/2) = 242 N./mm²

Compute Factor [facts]:
 = min([1.4 - 0.4* abs(σs,b) / Sstars], 1)
 = min([1.4 - 0.4* abs(150)/225], 1) = 1

Compute Factor [factc]:
 = min([1.4 - 0.4* abs(σc,b) / Sstarc], 1)
 = min([1.4 - 0.4* abs(177)/242], 1) = 1

Note: Since facts & factc are both >= 1, the design Passes.

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Shell Stress (jt. inter.)	195.9 <=	207.0 N./mm ²	Pass
Channel Stress (jt. inter.)	227.7 <=	484.0 N./mm ²	Pass

Thickness Results for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm.)	Required	Actual	P/F
Tubesheet Thickness :	21.180	200.000	Ok
Tube-Tubesheet Fillet Weld Leg :	2.046	3.000	Ok
Tube-Tubesheet Groove Weld Leg :	2.046	3.000	Ok

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 14 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend Allwd	Stresses Shear Allwd	Case Type	Pass/ Fail
D1uc	171.859	...	155	207	27	110 Ps+Pt-Th D1 Ok
D2uc	184.594	...	179	207	30	110 Ps+Pt-Th D2 Ok
D3uc	21.180	...	24	207	3	110 Ps+Pt-Th D3 Ok
O1uc	131.552	...	194	426	35	110 Ps+Pt+Th O1 Ok
O2uc	100.843	...	140	426	22	110 Ps+Pt+Th O2 Ok
O3uc	17.019	...	34	426	5	110 Ps+Pt+Th O3 Ok
O4uc	32.276	...	40	426	8	110 Ps+Pt+Th O4 Ok
D1c	176.784	...	162	207	28	110 Ps+Pt-Th-c D1 Ok
D2c	189.296	...	186	207	31	110 Ps+Pt-Th-c D2 Ok
D3c	26.081	...	24	207	3	110 Ps+Pt-Th-c D3 Ok



01c	136.658	...	203	426	36	110	Ps+Pt+Th-c	01	Ok
02c	106.420	...	146	426	23	110	Ps+Pt+Th-c	02	Ok
03c	23.286	...	37	426	5	110	Ps+Pt+Th-c	03	Ok
04c	37.951	...	42	426	8	110	Ps+Pt+Th-c	04	Ok

Max: 189.2957 ... mm. 0.900 0.326 (Str. Ratio)

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[(+/-)Th]:

With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:

Operating Load Cases using the Maximum and Minimum Operating Pressures and
 Operating Temperatures

Shell Axial Membrane Stress Summary:

Case#	Shell Stresses				:	Shell Band Stress				:	Pass Fail
	Ten	Allwd	Cmp	Allwd		Ten	Allwd	Cmp	Allwd		
D1uc	3	137	...	...	...	...	...	...	...	...	Ok
D2uc	43	137	...	...	...	...	...	...	...	...	Ok
D3uc	46	137	...	...	...	...	...	...	...	...	Ok
O1uc	7	450	-7	-118	...	...	...	...	...	...	Ok
O2uc	33	450	...	...	...	...	...	...	...	...	Ok
O3uc	36	450	...	...	...	...	...	...	...	...	Ok
O4uc	10	450	-10	-118	...	...	...	...	...	...	Ok
D1c	3	137	...	...	...	...	...	...	...	...	Ok
D2c	45	137	...	...	...	...	...	...	...	...	Ok
D3c	48	137	...	...	...	...	...	...	...	...	Ok
O1c	7	450	-7	-117	...	...	...	...	...	...	Ok
O2c	35	450	...	...	...	...	...	...	...	...	Ok
O3c	38	450	...	...	...	...	...	...	...	...	Ok
O4c	10	450	-10	-117	...	...	...	...	...	...	Ok
Max RATIO	0.348				0.086	...				...	

Tube, Shell and Channel Stress Summary:

Case#	Tube Stresses				Tube Loads		Shell Stress		Channel Stress		Pass Fail
	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	
D1uc	13	92	...	-22	308	2219	39	206	246	484	Ok
D2uc	14	92	-14	-22	326	2219	229	450	18	206	Ok
D3uc	1	92	-1	-22	26	2219	196	206	228	484	Ok
O1uc	45	184	...	...	1077	4439	71	450	272	484	Ok
O2uc	27	184	...	...	638	4439	192	450	9	484	Ok
O3uc	31	184	...	...	751	4439	159	450	254	484	Ok
O4uc	32	184	...	...	770	4439	37	450	27	484	Ok
D1c	13	92	-2	-22	311	2219	46	206	263	484	Ok
D2c	13	92	-13	-22	318	2219	244	450	23	206	Ok
D3c	1	92	-1	-22	15	2219	204	206	239	484	Ok
O1c	45	184	...	...	1076	4439	80	450	292	484	Ok
O2c	28	184	...	...	664	4439	204	450	6	484	Ok
O3c	32	184	...	...	758	4439	164	450	268	484	Ok

04c 32 184 ... 764 4439 40 450 29 484 Ok

Max RATIO 0.243 0.597 0.243 0.985 0.602

Summary of Thickness Comparisons for 14 Load Cases:

Thickness (mm.)	Required	Actual	P/F
Tubesheet thickness :	189.296	200.000	Ok
Tube thickness :	3.376	3.410	Ok
Tube-tubesheet fillet weld leg :	2.046	3.000	Ok
Tube-tubesheet groove weld leg :	2.046	3.000	Ok

Min Shell length of thk, (90.000) adj. to tubesheet: 565.068 mm.

Min Channel length of thk, (74.000) adj. to tubesheet: 512.384 mm.

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Compresses by : -1.992 mm.
 Due to Pressure Shell Compresses by : -0.933 mm.
 Due to Pressure + Thermal Shell Compresses by : -2.925 mm.

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	187.852	1.000	212.127	1.000
Tubesheet Shear Stress	538.448	1.000	687.245	1.000
Tube Tensile Stress	1048.69	1.000	1298.96	1.000
Tube Compressive Stress	2191.14	1.000	319.850	1.000
Tube-Tubesheet Joint load	1048.67	1.000	1298.96	1.000
Shell Stress (Axial, Junction)	187.852	1.000	193.709	0.549
Tubesheet-Channel Junction	187.852	1.000	193.709	0.114
Tube Pressure Stress	277.928	1.000	193.709	1.000
Minimum MAWP	187.852		193.709	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	196.510	1.000	221.554	1.000
Tubesheet Shear Stress	596.290	1.000	703.067	1.000
Tube Tensile Stress	1060.45	1.000	1298.92	1.000
Tube Compressive Stress	12960.98	1.000	336.648	1.000
Tube-Tubesheet Joint load	1060.45	1.000	1298.91	1.000
Shell Stress (Axial, Junction)	196.510	0.257	198.975	0.460
Tubesheet-Channel Junction	196.510	0.629	198.975	0.089
Tube Pressure Stress	277.928	1.000	198.975	1.000
Minimum MAPnc	196.510		198.975	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Summary of Exchanger Pressure and Thermal Displacements: (mm.)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.1275	0.1276	...	...	...	...
2	-0.9323	-0.9329	...	...	...	...
3	-0.8048	-0.8053	...	...	...	...
4	...	...	...	...	...	...
5	0.1275	0.1276	-1.9906	-1.9919	...	...
6	-0.9323	-0.9329	-1.9906	-1.9919	...	...



7	-0.8048	-0.8053	-1.9906	-1.9919	...	...
8	...	...	-1.9906	-1.9919	...	...

Tubesheet MDMT Calculations:

Shell Side MDMT calculation:

Governing thickness on the shell side per figure UCS-66.3 (e):

$$= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{shell thickness}))$$

$$= \max(200/4, \min(200, 90))$$

$$= 90.000 \text{ mm.}$$

MDMT reduction ratio per UCS 66 (b)(1)(b):

$$= \min(\text{ps}/\text{Shellside MAPnc}, 1)$$

$$= \min(191/199, 1)$$

$$= 0.960$$

Thickness Ratio = 0.96, Temperature Reduction per Fig. UCS 66.1 = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	29 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	13 °C
Min Metal Temp. at Required thickness (UCS 66.1)	10 °C

Channel Side MDMT calculation:

Governing Channel side thickness per figure UCS-66.3 (e):

$$= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{channel thickness}))$$

$$= \max(200/4, \min(200, 74))$$

$$= 74.000 \text{ mm.}$$

MDMT reduction ratio per UCS 66 (b)(1)(b):

$$= \min(\text{pt}/\text{Tubeside MAPnc}, 1)$$

$$= \min(147/197, 1)$$

$$= 0.748$$

Thickness Ratio = 0.75, Temperature Reduction per Fig. UCS 66.1 = 14 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	25 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	9 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-5 °C

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